Statistics

Lecture notes, Chapters 1–12

Based on *General Statistics*, 3rd ed. (Hanbit Academy)

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Chapter 1. Generating Data

1.1 What is statistics?

Motivation. Opinion polls, drug trials and quality checks all try to say something about a large group after looking at only a small part of it. Statistics is the science of doing this in a way whose reliability can itself be assessed.

Table 1-1 shows the result of a poll taken before a mayoral election.

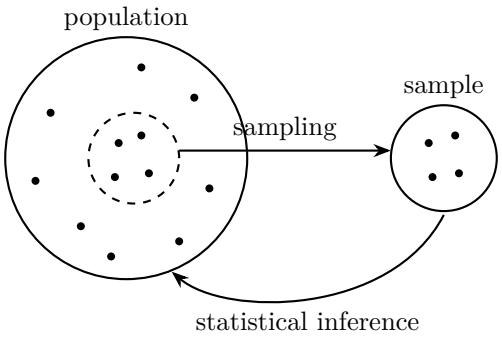
Candidate	Support
John Smith	42%
Emily Johnson	37%
Michael Lee	15%
Sarah Davis	6%

Such a poll raises three tasks. First, *whom* to survey: how many voters, chosen how, so that they represent all voters (including by age and occupation). Second, how to *organize and summarize* the data and draw conclusions about all voters—e.g. does Smith also lead Johnson in the whole electorate? Third, since the conclusion may be wrong, how to *assess its reliability*. **Statistics** is the science that collects, organizes and interprets numerical information (**data**) so as to reach rational conclusions with a measurable degree of reliability.

Usually only a tiny part of the units of interest can be observed: surveying everyone is too costly or impossible (one cannot record the sleep of every smartphone user in the world, nor give a new drug to every patient). So we collect data on a part and reason about the whole.

Definition (Population and sample)

Each individual unit making up the whole is a **sampling unit**; the value describing the feature of interest of a unit (support for a candidate, sleep quality, drug effect) is a **characteristic**. The collection of the characteristic values of *all* sampling units is the **population**; the collection of values actually observed is the **sample**.
A population made of finitely many units is a **finite population** (all voters of a city). A population with infinitely many units, e.g. all present and future smartphone users, is an **infinite population**; it is a conceptual, not a physically existing, object.



A brief history. The word "statistics" comes from *State Arithmetics*: rulers took censuses and valued land to levy taxes. In the 17th–18th centuries physics and astronomy demanded accurate measurement, and astronomers and surveyors met **variability**—repeated measurements of the

same object differ—which stimulated statistical methods. In the 19th century statistics spread to agriculture and biology (heredity, comparison of crop varieties and fertilizers), relying on data from experiments and observation; in the 20th century it became a unified science. Today it is used in epidemic forecasting, polling, medical data, quality control, autonomous-vehicle safety, AI evaluation, genetics, social-media analysis and climate research.

1.2 Simple random sampling

Motivation. Inference from a part to the whole is only trustworthy if the part represents the whole. The basic tool for guaranteeing this is to let chance, not people, pick the sample.

If a survey of the online-learning satisfaction of all Korean university students only covers students near the surveyor's home, the population is all students but the sample is not representative; if students in that region differ, the conclusion is wrong.

Illustration 1-1: a biased product sample

A factory makes thin steel plates for precision instruments. To test strength, ten $5\,\mathrm{cm}^2$ pieces are cut, but the inspector avoids visible defects and cuts only from good parts. The test then says the plates are excellent, although the sample does not reflect actual quality—customer complaints are likely.

Illustration 1-2: voluntary response

A singing contest let viewers vote by web, text or phone. Of 5 million viewers only 10% (500,000) voted; contestant A won with 50%. A later random survey of 500 viewers found only 10% preferred A and 90% preferred B: the voters were mostly A's fans. A sample made only of people who choose to respond is a **voluntary response sample**; it can badly distort the population's opinion.

Both illustrations show samples that misrepresent the population. A good sample gives every unit an equal chance of selection, which removes the surveyor's subjective intervention (Illustration 1-1) and the respondents' self-selection (Illustration 1-2), the sources of structural error. This requires careful *sample design*.

Definition 1-1 (Simple random sampling)

A sampling method designed so that, from a finite population, every subset consisting of n sampling units has the same probability of being selected as the sample is called **simple random sampling**.

If the population has size N, there are $\binom{N}{n}$ subsets of size n, so each has selection probability $1/\binom{N}{n}$. Concretely: pick the first unit with probability $1/N$; after k units have been chosen ($k = 1, \ldots, n-1$), pick one of the remaining $N-k$ units with probability $1/(N-k)$. This is **sampling without replacement**: a chosen unit is not returned before the next draw. In **sampling with replacement** the chosen unit is returned, so each draw picks one of the N units with probability $1/N$ (*simple random sampling with replacement*); by contrast, *simple random sampling without replacement* picks with probabilities $\frac{1}{N}, \frac{1}{N-1}, \ldots, \frac{1}{N-n+1}$.

Proposition (Properties of simple random sampling)

Simple random sampling (without replacement) of size n from a population of size N satisfies
(1) every possible sample of size n has the same chance of being selected;
(2) every unit of the population has the same chance of being selected, namely n/N.

Proof. [Supplement]

Step 1. With the sequential scheme, a particular ordered list of n distinct units is drawn with probability $\frac{1}{N} \cdot \frac{1}{N-1} \cdots \frac{1}{N-n+1}$. A given subset arises from $n!$ orderings, so its probability is

$$\frac{n!}{N(N-1)\cdots(N-n+1)} = \frac{n!\,(N-n)!}{N!} = \frac{1}{\binom{N}{n}},$$

the same for every subset; this is (1).

Step 2. A fixed unit belongs to $\binom{N-1}{n-1}$ of the $\binom{N}{n}$ equally likely subsets, so it is selected with probability $\binom{N-1}{n-1}/\binom{N}{n} = n/N$. ■

Systematic sampling (choose a random start, then every k-th unit) satisfies (2) but not (1). Statistical sample designs are based on **probability sampling**, in which the selection probability of each unit is explicitly defined; other designs are treated in Chapter 12.

1.3 Cautions in sample surveys

Motivation. Random selection removes the surveyor's bias, but a real survey can still go wrong in several practical ways. Recognizing them is as important as the sampling formula.

(1) Incomplete frame. One needs an accurate, complete list (frame) of the population's units. Homeless people, prisoners and dormitory residents may be missing from a household list; households without phones are missing from a telephone poll. If the omitted units differ from the rest, the sample suffers **bias**.

(2) Non-response. Selected people may refuse or be unreachable; non-response rates often reach 30%, and are higher in cities. Replacing non-respondents by extra respondents does not help if respondents and non-respondents hold different views.

(3) Response and interviewer effects. For sensitive questions ("Have you ever downloaded illegally?") people answer untruthfully, so the frequency is underestimated. An interviewer's tone, attitude or sex also matters; "Most people exercise three times a week; how often do you?" invites over-reporting. Interviewers must be trained and supervised.

(4) Question wording. The wording strongly steers answers. A question such as "Environmental regulation places an excessive burden on small firms, costing jobs and hindering growth. Do you favour relaxing it?" induces support for deregulation.

Reliable surveys therefore need a statistical sample design *and* careful questionnaires, a complete frame, trained interviewers and management of non-response.

1.4 Statistical experiments

Motivation. Surveys observe units as they are; experiments deliberately act on units and watch the response. This section fixes the vocabulary of experiments.

An experiment applies a manipulation to people, animals or objects and studies how they respond.

Definition 1-2 (Experimental unit and treatment)

The individual on which an experiment is performed is an **experimental unit**; applying a specific experimental environment or condition to an experimental unit is a **treatment**.

Definition 1-3 (Response variable, factor, factor level)

In a statistical experiment, a variable expressing the experimental environment or condition is a **factor**, and the variable expressing the reaction to it is the **response variable**. The values a factor takes are the **levels** of the factor.

Illustration 1-3: vaccine dose

The same dose of a vaccine is injected into 25 people and the blood concentration 30 minutes later is the response variable. The experiment has one factor (vaccine) at one level; with three different doses it would have one factor at three levels.

Example 1-1 (Diabetes treatment)

To study the effect of a new diabetes treatment on blood-sugar control, half of 18,000 participating patients take the new treatment daily and the other half the existing one. After six months the group on the new treatment has a lower mean blood sugar. Identify the experimental units, factor, levels and response variable.
Solution. Experimental units: the 18,000 diabetic patients. Factor: type of treatment (the experimental condition). Levels: two—new treatment and existing treatment. Response variable: blood sugar measured after six months.

Illustration 1-4: two factors

A textbook author asks whether practice problems work best at the start or at the end of a chapter, depending on the problem type. Six versions of the textbook are made and students get the same study time, homework and final test.

Position \ Type	Simple fact	Computation	Concept
End of chapter	Group 1	Group 2	Group 3
Start of chapter	Group 4	Group 5	Group 6

Students are the experimental units and the test score is the response. There are two factors, position (2 levels) and type (3 levels); their $2 \times 3 = 6$ combinations are the six treatments.

1.5 Comparative experiments and randomization

Motivation. A single-treatment experiment cannot separate the treatment's effect from everything else that happens to the units. Comparison with a control group and random assignment are the two remedies.

Illustration 1-5: gastric freezing

Gastric freezing—a patient swallows a balloon through which a coolant circulates for an hour—was claimed to reduce acid and relieve ulcer pain. The original study was

$$\text{observe pain} \;\rightarrow\; \text{gastric freezing} \;\rightarrow\; \text{observe pain.}$$

Patients improved, and the method was used for years. The improvement, however, came from trust in the doctor and expectation of the treatment: the **placebo effect**, a positive response to any treatment, which affects not only subjective symptoms but even physiological responses such as acid secretion. A later controlled study split 160 ulcer patients into 82 who received gastric freezing and 78 who received a saline solution: 34% of the former and 38% of the latter improved.

The first study used one treatment only and was misleading. In the second, both groups received the placebo-type influence, so the difference between them is the effect of gastric freezing itself—which turned out to be nil. The group receiving the treatment is the **treatment group**; the group receiving the dummy treatment is the **control group**. Controlling extraneous factors through a control group is the core of experimental design.

First principle of experimental design

Minimize the effect of extraneous factors other than the factor of interest.

A control group alone is not enough. If the doctor knows who is truly treated, this can influence the assessment of pain; so neither patient nor experimenter should know the assignment: a **double-blind experiment**. Moreover, the two groups must be alike with respect to extraneous factors. If one group is more affected by an extraneous factor, the difference between groups mixes the treatment effect with that factor: this is **confounding**.

Illustration 1-6: confounded study

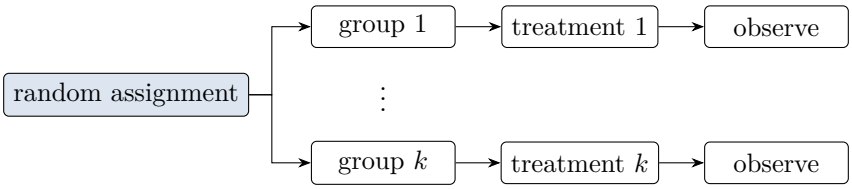
After toxic chemicals leaked near an industrial area, an agency reported genetic damage in residents, and the government decided on compensation. A review panel found the study had not controlled age or living habits, so damage could not be attributed to the chemicals. Genetic damage varies with age, sex and environment, and its assessment is subjective; residents should have been compared with a control group similar in age, sex and region, measured identically, and assessed blind.

In general there are too many extraneous factors to equalize them by hand, and an experimenter assigning patients by judgement (e.g. giving the new cancer drug to milder cases) can unintentionally bias the result. Hence units should be assigned purely by chance.

Illustration 1-7: random assignment

A food company evaluates a new instant food on 30 newly weaned male rats: 10 (treatment group) get the new food and 20 (control) the existing food, and weight gain after 4 weeks is measured. The 10 treated rats are chosen from the 30 by simple random sampling.

Each rat has the same probability $1/3$ of being treated. Assigning experimental units by chance so that each has the same chance of each treatment is **randomization** (introduced by R. A. Fisher at Rothamsted).



Completely randomized design (Figures 1-2 and 1-3; $k = 2$ is the randomized comparative experiment)

Second principle of experimental design (randomization)

When deciding which treatment each experimental unit receives, give every unit the same chance of receiving any particular treatment.

When randomized experiments were proposed in the 1920s, many experimenters resisted, trusting their own fair judgement and fearing that chance might ruin a result. Experience showed the opposite: subjective assignment produces more unexpected errors, while random results can be predicted and explained by the laws of probability. "Random" is not "haphazard": in Illustration 1-7, taking the first ten rats caught is haphazard, and easily caught rats may be smaller or slower, biasing the comparison.

With three or more treatments, randomization works the same way: e.g. 30 units into three groups of 10 by drawing 10 by simple random sampling for group 1, 10 of the remaining 20 for group 2, and the rest for group 3. Dividing all units into as many groups as treatments and assigning by randomization is a **completely randomized design**.

Even with randomization, a difference in response is not automatically a treatment effect: in Illustration 1-7 rats grow at different rates, and fast growers might by chance cluster in one group. Randomization lets us judge by probability theory whether the difference is too large to be explained by chance (Chapter 6). Also, with one rat per group nothing can be concluded; using more units makes the groups more alike on average and the treatment effect easier to detect.

Third principle of experimental design (replication)

Apply each treatment repeatedly to a sufficiently large number of experimental units.

1.6 Blocking

Motivation. Complete randomization ignores known differences between units. When such differences are large, grouping similar units first and randomizing within groups gives far more precise comparisons.

Illustration 1-8: paired design

A psychologist studies the effect of room temperature (15 °C vs. 20 °C) on manual dexterity, measured by the number of pegs inserted into small holes in 30 minutes. A completely randomized design would split subjects into two groups, one per temperature; but individual dexterity varies so much that a small temperature effect may be hidden. Better: let every subject work at both temperatures, on different days (to avoid fatigue), with the order randomized (e.g. heads $\Rightarrow$ 15 °C first) to neutralize learning and boredom.

This is a **randomized paired comparison design**: it keeps comparison, randomization and replication, but each unit receives both treatments and randomization decides the order. Each unit acts as its own control, removing between-person variation, so a small number of subjects suffices.

Pairing two subjects of similar dexterity and randomizing which one works at 15 °C is also a paired design.

Definition (Blocking)

Blocking means forming, before the experiment, groups of experimental units expected to respond similarly to the same treatment; each such group is a **block**. A design that uses blocking and randomization (within blocks) together is a **randomized block design**; the randomized paired comparison design is a special case (block size 2).

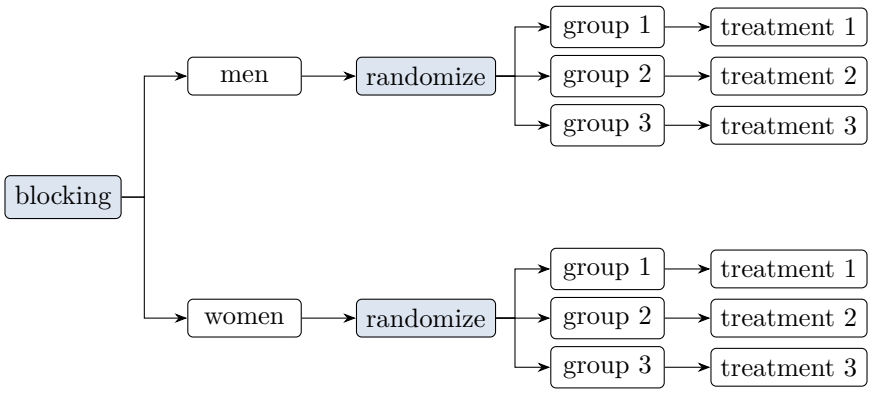
Blocks are built from factors expected to affect the response, so that their effect is controlled. Usually the block size equals the number of treatments (Illustration 1-9); sometimes blocks are larger and randomization splits each block into as many groups as treatments (Illustration 1-10).

Illustration 1-9: field blocks

To compare yields over 2 soybean varieties × 3 pesticides, fields differing in soil and fertility are each taken as a block and divided into 6 plots; within each block the 6 treatments are assigned to plots by randomization.

Illustration 1-10: blocking by sex

A cancer progresses at different speeds in men and women. A trial of three treatments uses 30 male and 30 female patients as two blocks; within each block, randomization forms three groups of 10, each receiving one treatment, and survival is compared within each block.



Randomized block design of Illustration 1-10 (Figure 1-4)

Whether by survey or experiment, the generated data are treated in analysis as a *sample from a population*. Often the population is a conceptual infinite population: in Illustration 1-9, for each variety–pesticide combination one imagines the infinite population of yields if that combination were applied to all fields, and the difference in mean yield is a difference between these six infinite populations.

Finally, a key contrast. Studying a drug's effect on blood pressure, an *experiment* lets the experimenter decide who takes the drug, so extraneous factors can be controlled; in a *sample survey* the units decide themselves and the surveyor merely observes, so control is hard. Experiments are not always possible: forcing people to take an unproven drug or to smoke is unethical. Then an **observational study** based on survey data is needed, and hidden extraneous factors must be interpreted with special care.

Chapter 2. Populations and Samples

2.1 Distribution of a population

Motivation. Before estimating anything we need a language for "what the whole population looks like". The distribution records how the characteristic values are spread, as nonnegative weights summing to one.

Table 1-1 summarized a *sample*; the actual election result summarizes the *population*:

Candidate	J. Smith	E. Johnson	M. Lee	S. Davis	Total
Vote share	0.367	0.281	0.271	0.081	1.000

The shares are nonnegative and sum to 1; drawn as bars they form a bar graph (Figure 2-1).

Definition 2-1 (Distribution of a population)

The way the characteristic values of a population are spread, expressed by nonnegative numbers summing to 1, is called the **distribution** of the population.

A main purpose of a pre-election poll is to estimate this distribution.

Illustration 2-1: Mendel's peas

Let R (round), r (wrinkled), Y (yellow), y (green) be pea genes. Crossing RRYY with rryy, the second generation shows round–yellow (C1), round–green (C2), wrinkled–yellow (C3), wrinkled–green (C4) in ratio $9:3:3:1$. Repeating the crossing indefinitely gives an infinite population with values C1–C4, whose distribution (if Mendel is right) is

Value	C1	C2	C3	C4
Proportion	9/16	3/16	3/16	1/16

Qualitative values (candidate, pea shape and colour) or discrete values (number of children of married women) have a distribution given by relative frequencies, or their limits for infinite populations.

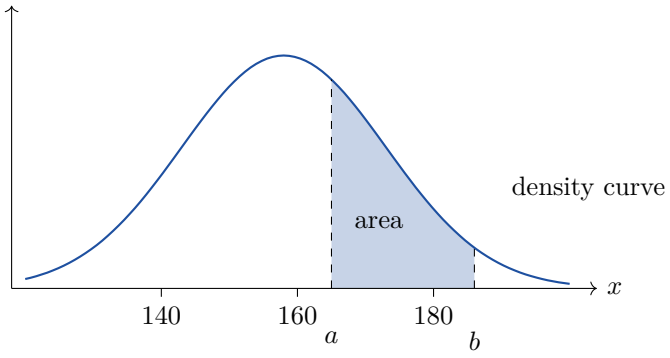
Continuous characteristics. If the characteristic is height, weight, etc., which can take any value in an interval, group the values into classes. With class width 10 cm suppose the relative frequencies are

Class (cm)	130–140	140–150	150–160	160–170	170–180	180–190	Total
Rel. freq.	0.05	0.1	0.3	0.3	0.2	0.05	1.00

(class $[a, b)$, i.e. "a or more, less than b"). Assuming values are spread uniformly in each class, draw over each class a rectangle whose *area* is the class relative frequency (height = rel. freq./width); total area is 1. With width 5 cm the picture becomes finer (Table 2-4: relative frequencies $0.01, 0.04, 0.04, 0.06, 0.2, 0.1, 0.2, 0.1, 0.05, 0.15, 0.03, 0.02$ for $130, 135, \ldots, 185$), and its shape depends on the class choice. To remove this arbitrariness one describes the distribution by one smooth curve.

Definition (Density curve)

A curve describing the distribution of an infinite population with a continuous characteristic is a **density curve**. The area under it over $[a, b]$ equals the limit of the relative frequency of values in $[a, b]$, so the total area between curve and horizontal axis is always 1. Even for a finite population, a density curve is a convenient approximation when the population is large.



Area over $[a, b]$ = limit of the relative frequency of values in $[a, b]$ (Figure 2-5)

For two or more characteristics the same idea applies: the distribution spreads total mass 1 over multivariate values—bar heights summing to 1 in the discrete case, volume 1 under a surface in the continuous case. This is a **joint distribution**.

2.2 Summary measures of a population

Motivation. A whole distribution is hard to estimate, so we summarize it by a few numbers describing location, spread and (for two characteristics) association. These only make sense for numerical characteristics.

2.2.1 Location: population mean, percentiles

For a finite population $\{c_1, \ldots, c_N\}$ (c_i the value of unit i), the **population mean** is

$$\mu = \frac{1}{N}(c_1 + c_2 + \cdots + c_N) = \frac{1}{N}\sum_{i=1}^{N} c_i.$$

If the distinct values are $c_1^*, \ldots, c_k^*$ with frequencies $f_1, \ldots, f_k$, then μ is the sum of (value)×(relative frequency):

Definition 2-2 (Population mean of a finite population)

$$\mu = \sum_{i=1}^{k} c_i^* \frac{f_i}{N}.$$

Example 2-1 (Number of children)

Number of children of the 16,549 thousand married Korean women aged 15+ (2020 census; 5* means "5 or more", counted as 5):

Children	0	1	2	3	4	5*
Rel. freq.	0.083	0.189	0.477	0.150	0.054	0.047

Find the population mean.
Solution.

$$\mu = 0(0.083) + 1(0.189) + 2(0.477) + 3(0.150) + 4(0.054) + 5(0.047) = 2.044,$$

about 2 children on average.

The mean is the **balance point**: placing at each value a mass proportional to its relative frequency, the bar graph balances at μ (Figure 2-7). The same interpretation holds for infinite populations: for a continuous characteristic, μ is the point where a plate shaped like the density curve balances.

Definition 2-3 (Population mean of an infinite population)

$$\mu = \begin{cases} \displaystyle\sum_{\text{all } x} x\,p(x) & \text{(discrete case)} \\ \displaystyle\int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous case)} \end{cases}$$

Here $p(x)$ is the limiting relative frequency of the value x (discrete case) or the density curve (continuous case); p is called the **density function**.

Example 2-2 (Mean of an infinite population)

Find the population mean of the infinite population

x	0	1	2	3	4
$p(x)$	1/10	3/10	3/10	2/10	1/10

Solution.

$$\mu = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{3}{10} + 2 \cdot \tfrac{3}{10} + 3 \cdot \tfrac{2}{10} + 4 \cdot \tfrac{1}{10} = 1.9.$$

The p**th percentile** is the point such that, with values in increasing order, $p\%$ are below it and $(100 - p)\%$ above. The 25th, 50th and 75th percentiles are the **first quartile** Q_1, the **median** Q_2 and the **third quartile** Q_3. Like μ, the median measures the centre: for a continuous infinite population it splits the area under the density curve in half. For a symmetric density $\mu = Q_2$; for a right-skewed density the long tail pulls μ to the right of Q_2 (Figure 2-9). Of these location measures, μ (together with σ below) is the key one; the median is usually not a target of inference because of limitations in estimation theory.

2.2.2 Spread: MAD, IQR, variance

For a finite population, the **mean absolute deviation** is the average distance from the median, and the **interquartile range** is the range of the middle 50%:

$$\text{MAD} = \frac{1}{N}\sum_{i=1}^{N} |c_i - Q_2|, \qquad \text{IQR} = Q_3 - Q_1.$$

Both are easy to interpret but are not usually inference targets. The standard measure is the **standard deviation** σ and its square, the **variance**:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(c_i-\mu)^2}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(c_i-\mu)^2.$$

In terms of distinct values c_i^* with frequencies f_i:

Definition 2-4 (Variance and s.d. of a finite population)

(1) Population variance: $\sigma^2 = \sum_{i=1}^{k}(c_i^*-\mu)^2\frac{f_i}{N}$.
(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Definition 2-5 (Variance and s.d. of an infinite population)

For an infinite population with density function $p(x)$:
(1) Population variance:

$$\sigma^2 = \begin{cases} \sum_{\text{all } x}(x-\mu)^2 p(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty}(x-\mu)^2 p(x)\,dx & \text{(continuous case)} \end{cases}$$

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Example 2-3 (Variance)

Find the population variance and standard deviation for Example 2-2.
Solution. With $\mu = 1.9$,

$$\begin{aligned}\sigma^2 &= (0-1.9)^2\tfrac{1}{10} + (1-1.9)^2\tfrac{3}{10} + (2-1.9)^2\tfrac{3}{10} + (3-1.9)^2\tfrac{2}{10} + (4-1.9)^2\tfrac{1}{10} \\ &= 1.29,\end{aligned}$$

and $\sigma = \sqrt{1.29} = 1.1358$.

2.2.3 Association: correlation

Table 2-5 gives two bivariate populations; each entry is the relative frequency of the pair (characteristic 1, characteristic 2).

Population A

char. 2 \ char. 1	0	1	2	Total
0	0.1	0.2	0.1	0.4
1	0.0	0.2	0.1	0.3
2	0.0	0.0	0.3	0.3
Total	0.1	0.4	0.5	1.0

Population B

char. 2 \ char. 1	0	1	2	Total
0	0.0	0.0	0.4	0.4
1	0.0	0.2	0.1	0.3
2	0.1	0.2	0.0	0.3
Total	0.1	0.4	0.5	1.0

The bottom margin gives the distribution of characteristic 1, the right margin that of characteristic 2. In both populations

$$\mu_1 = 1.4,\ \sigma_1 = 0.6633, \qquad \mu_2 = 0.9,\ \sigma_2 = 0.8307,$$

yet A's mass lies along a rising direction and B's along a falling one. Means and standard deviations do not describe a bivariate population; we need a measure of the relation between the characteristics. For a bivariate finite population $(c_{11}, c_{21}), \dots, (c_{1N}, c_{2N})$ the **correlation coefficient** (Galton, Pearson) is

$$\rho = \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1}\Big)\Big(\frac{c_{2i}-\mu_2}{\sigma_2}\Big).$$

Proposition (Range of the correlation coefficient)

$-1 \le \rho \le 1$, because

$$0 \le \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1} - \rho\,\frac{c_{2i}-\mu_2}{\sigma_2}\Big)^2 = 1-\rho^2.$$

Proof. Write $u_i = (c_{1i}-\mu_1)/\sigma_1$ and $v_i = (c_{2i}-\mu_2)/\sigma_2$. By the definitions of σ_1^2, σ_2^2 and ρ,

$$\frac{1}{N}\sum u_i^2 = 1, \qquad \frac{1}{N}\sum v_i^2 = 1, \qquad \frac{1}{N}\sum u_i v_i = \rho.$$

Expanding the square,

$$\frac{1}{N}\sum(u_i - \rho v_i)^2 = \frac{1}{N}\sum u_i^2 - 2\rho\,\frac{1}{N}\sum u_i v_i + \rho^2\,\frac{1}{N}\sum v_i^2 = 1 - 2\rho^2 + \rho^2 = 1-\rho^2.$$

A mean of squares is ≥ 0, so $\rho^2 \le 1$. ■

Grouping distinct values c_{1i}^* $(i \le k)$ of characteristic 1 and c_{2j}^* $(j \le l)$ of characteristic 2, with f_{ij} the frequency of (c_{1i}^*, c_{2j}^*):

Definition 2-6 (Correlation of a finite population)

$$\rho = \sum_{i=1}^{k}\sum_{j=1}^{l}\Big(\frac{c_{1i}^*-\mu_1}{\sigma_1}\Big)\Big(\frac{c_{2j}^*-\mu_2}{\sigma_2}\Big)\frac{f_{ij}}{N} = \frac{\sum_{i=1}^{k}\sum_{j=1}^{l}(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\,f_{ij}/N}{\sigma_1\sigma_2}.$$

Example 2-4 (Correlations of Table 2-5)

Find the population correlation coefficients of populations A and B.
Solution. For A,

$$\begin{aligned}&\sum_i\sum_j (c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\frac{f_{ij}}{N}\\ &\quad= (0-1.4)(0-0.9)(0.1) + (1-1.4)(0-0.9)(0.2) + \cdots + (2-1.4)(2-0.9)(0.3)\\ &\quad= 0.34,\end{aligned}$$

so $\rho = 0.34/(0.6633 \times 0.8307) = 0.6171$. The same computation for B gives a numerator of -0.46 and $\rho = -0.8348$.

For an infinite bivariate population let $p(x, y)$ be the limiting relative frequency of (x, y) (discrete case) or the joint density surface (continuous case): the **joint density function**. With μ_1, σ_1 and μ_2, σ_2 the means and standard deviations of the two characteristics:

Definition 2-7 (Correlation of an infinite population)

$$\rho = \begin{cases} \dfrac{\sum_{\text{all } x}\sum_{\text{all } y}(x-\mu_1)(y-\mu_2)\,p(x,y)}{\sigma_1\sigma_2} & \text{(discrete case)} \\ \dfrac{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)\,p(x,y)\,dx\,dy}{\sigma_1\sigma_2} & \text{(continuous case)} \end{cases}$$

The numerator is the **covariance** of the two characteristics; the population covariance equals $\rho\sigma_1\sigma_2$. As in Example 2-4, $\rho > 0$ when the mass lies along a rising direction and $\rho < 0$ along a falling one; the more the mass concentrates on a line of nonzero slope, the closer ρ is to ± 1.

The population mean, variance, standard deviation, covariance and correlation—numbers describing a population—are called **parameters**. Section 2.3 shows how to picture the population from a sample; Section 2.4 gives sample quantities used to estimate parameters.

2.3 Graphical display of samples

Motivation. A sample is our only window on the population; tables and graphs of the sample give a first rough picture of the population's distribution.

2.3.1 Qualitative or few-valued data

A **frequency table** lists each value with its **frequency** (count) and **relative frequency** (count / sample size). It is drawn as a **bar graph** (bar heights proportional to relative frequencies) or a **pie graph** (sector areas, i.e. central angles, proportional to relative frequencies).

Example 2-5 (Bar and pie graphs)

Draw a bar graph and a pie graph of the sample in Table 1-1.
Solution. The bar graph has heights $0.42, 0.37, 0.15, 0.06$. The central angles of the pie are

$$360^\circ \times 0.42 = 151.2^\circ, \quad 360^\circ \times 0.37 = 133.2^\circ, \quad 360^\circ \times 0.15 = 54^\circ, \quad 360^\circ \times 0.06 = 21.6^\circ.$$

2.3.2 Frequency tables, histograms, polygons

For continuous values, or discrete values with many distinct values, divide the range into classes. Class widths are arbitrary, but each class should not have too small a relative frequency; too wide classes hide the shape; and no observation should fall on a class boundary.

Constructing a frequency table

(1) Range = (largest value) − (smallest value).
(2) Choose about 5–20 classes; class width = range ÷ (number of classes), rounded *up* to the smallest unit of the data.
(3) Start the first class at (smallest value) $-\frac{1}{2}\times$(smallest unit), and build the classes with the width from (2).
(4) Count the (relative) frequency of each class.

Example 2-6 (Heights of 40 people)

Heights (cm) of 40 randomly selected people:

183	168	165	158	180	176	168	160	160	150
180	183	170	170	185	170	165	170	168	154
152	173	173	165	163	163	160	178	138	171
163	168	165	165	178	170	168	183	145	174

Make a frequency table.

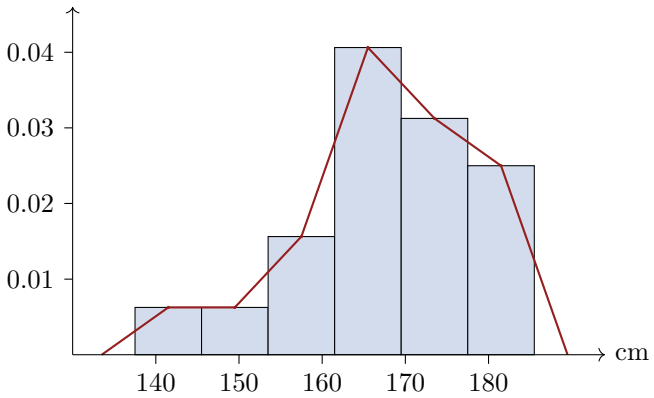
Solution. Max 185, min 138, range 47. With 6 classes, $47/6 = 7.83\ldots$, rounded up to width 8. The first class starts at $138 - \frac{1}{2} \times 1 = 137.5$.

Class (cm)	Freq.	Rel. freq.
$[137.5, 145.5)$	2	0.050
$[145.5, 153.5)$	2	0.050
$[153.5, 161.5)$	5	0.125
$[161.5, 169.5)$	13	0.325
$[169.5, 177.5)$	10	0.250
$[177.5, 185.5)$	8	0.200
Total	40	1

A **histogram** marks the class intervals on the horizontal axis and draws over each class a rectangle with

$$\text{area} = \text{relative frequency}, \qquad \text{i.e.} \qquad \text{height} = \frac{\text{relative frequency}}{\text{class width}}.$$

Total area is then 1, so the histogram estimates the density curve. Classes need not have equal widths: narrow classes where data are dense show detail, wide classes where data are sparse collect enough values (Figure 2-13 splits the 4th class into two of width 4). A **frequency polygon** joins the midpoints of the tops of the rectangles, with end points on the axis half a class width beyond the first and last classes; its total area is also 1.



Histogram (height = rel. freq./8) and frequency polygon for Example 2-6 (Figures 2-12, 2-14)

2.3.3 Stem-and-leaf display

Write the leading digits (here the first two) in a column—the **stems**—with a vertical line to their right; write the last digit of each observation as a **leaf** to the right of its stem; finally sort the leaves. For Example 2-6:

18	0 0 3 3 3 5
17	0 0 0 0 0 1 3 3 4 6 8 8
16	0 0 0 3 3 3 5 5 5 5 5 8 8 8 8 8
15	0 2 4 8
14	5
13	8

This **stem-and-leaf display** looks like a histogram rotated by 90° (clockwise). The leaf width plays the role of the class width and must be constant; above it is 10, and splitting each stem in two (14^- for 140–145, 14^+ for 145–150) gives width 5. Its advantage is that the data values are kept; its drawback is that large samples are hard to draw.

2.3.4 Bivariate samples

The bivariate analogue of the frequency table is the **contingency table**, listing the (relative) frequency of each pair of values. For instance, the favourite subject of 400 students:

	Korean	English	Math	Total
Male	0.18	0.24	0.18	0.60
Female	0.14	0.16	0.10	0.40
Total	0.32	0.40	0.28	1.00

It can be drawn as a bivariate bar graph. For continuous (or many-valued) characteristics, class both ranges to get a contingency table, and draw a **bivariate histogram**: over each rectangular cell a box whose volume is the cell's relative frequency.

Example 2-7 (Height and weight)

Adding weight to the heights of Example 2-6 (height cm, weight kg):

(183, 82)	(168, 52)	(165, 52)	(158, 55)	(180, 93)
(176, 57)	(168, 60)	(160, 48)	(160, 50)	(150, 49)
(180, 79)	(183, 73)	(170, 60)	(170, 66)	(185, 74)
(170, 82)	(165, 57)	(170, 54)	(168, 57)	(154, 55)
(152, 50)	(173, 63)	(173, 60)	(165, 49)	(163, 63)
(163, 57)	(160, 57)	(178, 77)	(138, 54)	(171, 69)
(163, 50)	(168, 57)	(165, 54)	(165, 59)	(178, 79)
(170, 54)	(168, 79)	(183, 70)	(145, 37)	(174, 65)

Make a contingency table and a bivariate histogram.
Solution. Using the 6 height classes of Example 2-6 and 6 weight classes of width 10 starting at 36.5:

Height \ Weight	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5	Total
137.5–145.5	0.025	0.025	0	0	0	0	0.050
145.5–153.5	0	0.050	0	0	0	0	0.050
153.5–161.5	0	0.100	0.025	0	0	0	0.125
161.5–169.5	0	0.125	0.175	0	0.025	0	0.325
169.5–177.5	0	0.050	0.150	0.025	0.025	0	0.250
177.5–185.5	0	0	0	0.075	0.100	0.025	0.200
Total	0.025	0.350	0.350	0.100	0.150	0.025	1.000

The bivariate histogram erects over each cell a box of volume equal to the entry (Figure 2-17).

2.3.5 Scatter plots and regressograms

A **scatter plot** plots each bivariate observation as the point (value 1, value 2). A **regressogram** classes one characteristic (as for a histogram) and, within each class, plots the mean of the other characteristic. For Example 2-7, classing weight into the 6 classes above, the mean heights are

Weight class	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5
Mean height	145.0	159.9	168.3	180.5	176.2	180.0

Plotting these as a step function over the weight classes gives the regressogram (Figure 2-19). Both graphs show how the two characteristics vary together; the regressogram is generally more effective for analysing the relationship.

2.4 Summary measures of a sample

Motivation. Parameters are unknown; we estimate them by the corresponding quantities computed from the sample. Some care (trimming, dividing by $n-1$) makes these estimates better behaved.

2.4.1 Location

Let the sample values be $x_1, \dots, x_n$.

Definition 2-8 (Sample mean)

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The sample pth percentile is defined as for the population; the sample quartiles and median are $\hat{Q}_1, \hat{Q}_2, \hat{Q}_3$. They estimate the population mean and percentiles. A drawback of $\bar{x}$: a few values far from the bulk strongly affect it, and when the population spread is large some sample values may lie far on one side, making $\bar{x}$ a poor estimate. The $100p\%$ **trimmed mean** $\bar{x}_p$ removes the smallest and the largest $100p\%$ of the ordered values and averages the remaining $100(1-2p)\%$ (this p is unrelated to the p of percentiles).

Example 2-8 (Sample mean, quartiles, trimmed mean)

For the data of Example 2-6, find the sample mean, the sample quartiles and the 10% trimmed mean.

Solution. $\bar{x} = \frac{1}{40}(183 + 168 + \cdots + 174) = \frac{6698}{40} = 167.45$. From the stem-and-leaf display, $\hat{Q}_1$ is the midpoint of the 10th and 11th values, $\hat{Q}_2$ of the 20th and 21st, $\hat{Q}_3$ of the 30th and 31st:

$$\hat{Q}_1 = 163, \qquad \hat{Q}_2 = 168, \qquad \hat{Q}_3 = 173.5.$$

The 10% trimmed mean drops 4 values at each end ($138, 145, 150, 152$ and $183, 183, 183, 185$) and averages the middle 32:

$$\bar{x}_{0.1} = \frac{1}{32}(180 + 180 + \cdots + 154) = \frac{5379}{32} = 168.0938.$$

2.4.2 Spread

The **sample interquartile range** $\widehat{\mathrm{IQR}} = \hat{Q}_3 - \hat{Q}_1$ estimates the IQR. For the variance, the analogue of σ^2 would be $\frac{1}{n}\sum(x_i-\mu)^2$; replacing the unknown μ by $\bar{x}$ gives $\frac{1}{n}\sum(x_i-\bar{x})^2$, which tends to *underestimate* σ^2 because of the following identity.

Proposition (Centring at the sample mean)

For any number μ,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2 - (\bar{x}-\mu)^2.$$

Proof. [Supplement] Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$ and square:

$$\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}(x_i-\bar{x})^2 + 2(\bar{x}-\mu)\sum_{i=1}^{n}(x_i-\bar{x}) + n(\bar{x}-\mu)^2.$$

The middle sum is $\sum x_i - n\bar{x} = 0$. Divide by n and rearrange. ∎

Hence dividing by n gives a value never larger than the "ideal" $\frac{1}{n}\sum(x_i-\mu)^2$. Dividing by $n-1$ instead slightly increases the estimate; this is known to be more accurate, and is the **sample variance**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2.$$

Definition 2-9 (Sample standard deviation)

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}.$$

Example 2-9 (Sample IQR and s.d.)

For the data of Example 2-6, find the sample interquartile range and the sample standard deviation.

Solution. From Example 2-8, $\widehat{\mathrm{IQR}} = 173.5 - 163 = 10.5$. Next

$$s^2 = \frac{1}{40-1}\{(183-167.45)^2 + (168-167.45)^2 + \cdots + (174-167.45)^2\} = 108.6128,$$

so $s = \sqrt{108.6128} = 10.4218$.

2.4.3 Bivariate samples

For a bivariate sample $(x_1, y_1), \ldots, (x_n, y_n)$, the means and standard deviations of each characteristic are estimated separately:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n}x_i, \quad s_1 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}; \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n}y_i, \quad s_2 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2}.$$

Definition 2-10 (Sample correlation coefficient)

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{s_1}\right)\left(\frac{y_i-\bar{y}}{s_2}\right) = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}.$$

(The two expressions agree since $(n-1)s_1 s_2 = \sqrt{\sum(x_i-\bar{x})^2}\sqrt{\sum(y_i-\bar{y})^2}$.) The **sample covariance**, estimating the population covariance, is

$$rs_1 s_2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}).$$

Example 2-10 (Sample correlation)

Find the sample correlation coefficient for the data of Example 2-7.

Solution. For height, $\bar{x} = 167.45$, $s_1 = 10.42$ (Examples 2-8, 2-9); similarly for weight, $\bar{y} = 61.45$, $s_2 = 11.94$. The sample covariance is

$$rs_1 s_2 = \frac{1}{40-1}\{(183-167.45)(82-61.45) + \cdots + (174-167.45)(65-61.45)\} = 90.59,$$

so $r = 90.59/(10.42 \times 11.94) = 0.73$.

Sample quantities such as the sample mean, variance, standard deviation, covariance and correlation are called **statistics**; a statistic used to estimate a parameter is an **estimator**.

Chapter 3. Probability and Probability Distributions

3.1 Definition of probability

Motivation. A survey observes part of a population and draws conclusions about the whole by induction; probability is the logical basis that makes such conclusions quantifiable. We first fix the objects to which probabilities are assigned: sample spaces and events.

Definition (Sample space and events)

The set of all possible outcomes of a statistical survey (experiment) is the **sample space** S. A subset of S—the set of outcomes satisfying some condition of interest—is an **event**; the event A *occurs* if the observed outcome is an element of A. An event consisting of a single element is an **elementary** (**simple**) **event**.
$A \cup B$, $A \cap B$, A^c are the **union**, **intersection** and **complement**. If $A \cap B = \varnothing$, A and B are **mutually disjoint** (exclusive): they cannot occur together.

Illustration 3-1: sample spaces

(a) Number shown by one die: $S = \{1, 2, 3, 4, 5, 6\}$.
(b) Number of items produced until the 10th defective: $S = \{10, 11, 12, \dots\}$.
(c) Lifetime of a light bulb drawn from a factory: $S = \{t \mid t \geq 0\}$.

When each of finitely many elementary events is equally likely (as for the six faces of a die), Laplace defined probability as follows.

Definition 3-1 (Classical definition of probability)

If $S = \{e_1, \dots, e_N\}$ has N elements whose elementary events are equally likely, the **probability** of an event A with m elements is

$$P(A) = \frac{m}{N}.$$

Example 3-1 (Table-tennis balls)

A box contains 3 white balls and 1 yellow ball. Two balls are drawn by simple random sampling. Find the probability of getting one white and one yellow ball.
Solution. Label the white balls a, b, c and the yellow one d:

$$S = \{\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}\}, \qquad A = \{\{a,d\},\{b,d\},\{c,d\}\}.$$

All 6 pairs are equally likely, so $P(A) = 3/6 = 1/2$.

The classical definition fails when outcomes are infinitely many or not equally likely. Experience shows that, as a survey is repeated, the *relative frequency* of an event tends to settle at a constant (Figure 3-1: the proportion of defectives approaches a constant as production grows), which can be called the probability of the event. Based on the basic properties of relative frequencies, Kolmogorov gave the axioms.

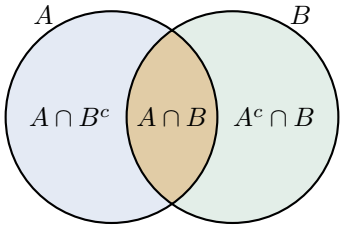
Definition 3-2 (Axiomatic definition of probability)

A rule P assigning numbers to events of S is a probability if
(1) $0 \le P(A) \le 1$ for every event A;
(2) $P(S) = 1$;
(3) for mutually disjoint events $A_1, A_2, \ldots,$

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots .$$

$P(A)$ is called the **probability** of A.

This covers empirical probability (limits of relative frequencies) as well as subjective probability (values set by personal judgement).



Partition of $A \cup B$ into three disjoint pieces (Figure 3-2)

Theorem 3-1 (Properties of probability)

(1) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
(2) $P(A^c) = 1 - P(A)$.
(3) If $A \subset B$, then $P(A) \le P(B)$.

Proof. (1) $A \cap B^c$, $A \cap B$, $A^c \cap B$ are mutually disjoint, with $A \cup B$ their union, $A = (A \cap B^c) \cup (A \cap B)$ and $B = (A \cap B) \cup (A^c \cap B)$. By axiom (3),

$$\begin{aligned} P(A \cup B) &= P(A \cap B^c) + P(A \cap B) + P(A^c \cap B), \\ P(A) &= P(A \cap B^c) + P(A \cap B), \\ P(B) &= P(A \cap B) + P(A^c \cap B). \end{aligned}$$

Subtracting, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(2) A and A^c are disjoint with union S, so $P(A) + P(A^c) = P(S) = 1$ by axioms (3) and (2).

(3) If $A \subset B$, then $B = A \cup (A^c \cap B)$ is a disjoint union, so $P(B) = P(A) + P(A^c \cap B) \ge P(A)$ by axioms (3) and (1). ■

Example 3-2 (Three bags)

Each of three bags contains 5 marbles numbered 1 to 5. One marble is drawn from each bag. Find the probability that the sum of the numbers is at least 5.
Solution. Let $A =$ "sum ≥ 5"; then $A^c =$ "sum ≤ 4" $= B \cup C$ with $B =$ "sum $= 3$", $C =$ "sum $= 4$", $B \cap C = \varnothing$. There are $5^3 = 125$ equally likely outcomes, and

$$B = \{(1,1,1)\}, \qquad C = \{(1,1,2), (1,2,1), (2,1,1)\},$$

so $P(B) = 1/125$, $P(C) = 3/125$ and

$$P(A) = 1 - \{P(B) + P(C)\} = 1 - \frac{4}{125} = \frac{121}{125}.$$

3.2 Conditional probability and independence

Motivation. Information that one event has occurred changes how likely another is. Conditional probability formalizes this update, and independence describes when no update is needed.

Given that A has occurred, outcomes in A^c are impossible, so A plays the role of the whole sample space; the probability that B occurs is the share of $P(A)$ taken by $P(A \cap B)$.

Definition 3-3 (Conditional probability)

The **conditional probability** of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad (P(A) > 0).$$

Example 3-3 (Two dice)

Two dice are thrown. Given that the first die shows more than the second, find the probability that the sum is 7.
Solution. Let $A =$ "first $>$ second", $B =$ "sum $= 7$". Of 36 outcomes, A has 15, so $P(A) = 15/36$. Also $B = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$ and $A \cap B = \{(4,3),(5,2),(6,1)\}$, so

$$P(B) = \frac{6}{36} = \frac{1}{6}, \qquad P(A \cap B) = \frac{3}{36} = \frac{1}{12}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{3}{15} = \frac{1}{5}.$$

Theorem 3-2 (Multiplication rule)

If $P(A) > 0$ and $P(B) > 0$, then

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Proof. By Definition 3-3, $P(B \mid A) = P(A \cap B)/P(A)$; multiply by $P(A)$. Exchanging the roles of A and B (and using $B \cap A = A \cap B$) gives $P(A \cap B) = P(A \mid B)P(B)$. ■

The rule is useful when conditional probabilities are easy to compute.

Example 3-4 (Prize draw)

A couple joins a draw at a supermarket: a box holds 10 winning and 50 losing balls; each person draws once and a drawn ball is not returned. Find the probability that both win.
Solution. Let $A =$ "husband wins", $B =$ "wife wins". Then

$$P(A \cap B) = P(B \mid A)P(A) = \frac{9}{59} \times \frac{10}{60} = \frac{3}{118}.$$

When $P(B)$ is hard to compute directly, split S into mutually disjoint events $A_1, \ldots, A_n$ with $A_1 \cup \cdots \cup A_n = S$ (a *partition*). Then the $B \cap A_i$ are mutually disjoint with union B.

Theorem 3-3 (Law of total probability)

If $A_i \cap A_j = \varnothing$ $(i \neq j)$, $A_1 \cup A_2 \cup \cdots \cup A_n = S$ and $P(A_i) > 0$, then

$$P(B) = P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n).$$

Proof.

Step 1. For $i \neq j$, $(B \cap A_i) \cap (B \cap A_j) \subset A_i \cap A_j = \varnothing$, and

$$(B \cap A_1) \cup \cdots \cup (B \cap A_n) = B \cap (A_1 \cup \cdots \cup A_n) = B \cap S = B.$$

Step 2. By axiom (3), $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$, and by Theorem 3-2 each term equals $P(B \mid A_i)P(A_i)$. ■

Example 3-5 (Joint training)

Three football teams train together: 50% of the players are from team 1, 30% from team 2, 20% from team 3; 20%, 15%, 30% of the respective teams are forwards. A player is chosen at random. Find the probability that he is a forward.
Solution. With A_i = "from team i" (a partition) and B = "forward",

$$P(B) = 0.2 \times 0.5 + 0.15 \times 0.3 + 0.3 \times 0.2 = 0.205.$$

If the occurrence of A does not affect the probability of B, i.e. $P(B \mid A) = P(B)$, then A and B are **(mutually) independent**. Then $P(B) = P(A \cap B)/P(A)$, i.e. $P(A \cap B) = P(A)P(B)$; and by the multiplication rule $P(A)P(B) = P(A \mid B)P(B)$, i.e. $P(A \mid B) = P(A)$.

Definition 3-4 (Independent events)

Events A and B are **independent** if one of the following holds (when the conditional probabilities are defined, they are equivalent):

$$P(A \cap B) = P(A)P(B), \qquad P(A \mid B) = P(A), \qquad P(B \mid A) = P(B).$$

Example 3-6 (Prize draw revisited)

In Example 3-4:
(a) Are "husband wins" and "wife wins" independent?
(b) Are they independent if the drawn ball is returned to the box?
Solution.
(a) By the law of total probability,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = \frac{9}{59} \cdot \frac{10}{60} + \frac{10}{59} \cdot \frac{50}{60} = \frac{1}{6}.$$

Since $P(B) \neq P(B \mid A) = 9/59$, A and B are not independent.
(b) With replacement, $P(B \mid A) = 10/60 = 1/6 = P(B)$, so A and B are independent.

Under simple random sampling with replacement successive draws are independent; without replacement they are not. Events that are not independent are **(mutually) dependent**. In general, $A_1, \ldots, A_n$ are **independent** if

$$\begin{aligned} P(A_i \cap A_j) &= P(A_i)P(A_j) && (1 \le i < j \le n), \\ P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k) && (1 \le i < j < k \le n), \\ &\vdots \\ P(A_1 \cap A_2 \cap \cdots \cap A_n) &= P(A_1)P(A_2)\cdots P(A_n). \end{aligned}$$

3.3 Random variables and probability distributions

Motivation. Outcomes may be qualitative (blood types O, AB, A, B), which makes probability calculations clumsy. Mapping outcomes to numbers turns probability on the sample space into probability on sets of numbers.

Illustration 3-2: coin toss

Tossing one coin, $S = \{\mathrm{H}, \mathrm{T}\}$ with $P(\mathrm{H}) = P(\mathrm{T}) = 1/2$. Define $X(\mathrm{H}) = 1$, $X(\mathrm{T}) = 0$. Then X is a function on S, and the events $\{X = 0\}$, $\{X = 1\}$ each have probability $1/2$: X transfers the probability on S to the set $\{0, 1\}$.

Definition (Random variable, distribution)

A real-valued function defined on a sample space is a **random variable**. If an outcome is mapped to the value a, we say "X takes the value a". The description of how probability is spread over the values of X, by nonnegative numbers summing to 1, is the **probability distribution** (or simply **distribution**) of X.

Illustration 3-3

In Illustration 3-2 the distribution of X is $P(X = 0) = P(X = 1) = 1/2$.

If the possible values $x_1, x_2, \dots$ of X can be listed (countable), X is a **discrete random variable**, and

$$p(x) = \begin{cases} P(X = x_i), & x = x_i \ (i = 1, 2, \dots) \\ 0, & \text{otherwise} \end{cases}$$

is its **probability mass function**.

Definition 3-5 (Probability mass function of a discrete r.v.)

A function $p(x)$ is the **probability mass function** of the discrete random variable X if
(1) $0 \le p(x) \le 1$ and $\sum_{\text{all } x} p(x) = 1$;
(2) $P(a < X \le b) = \sum_{a<x\le b} p(x)$.

Thus the distribution of a discrete random variable is completely determined by its probability mass function; with few values it is convenient to list it as a table.

Example 3-7 (Popular posts)

Of an influencer's 12 posts, 4 have more than 10,000 likes. Three posts are chosen by simple random sampling; X is the number among them with more than 10,000 likes. Find the distribution of X.
Solution. There are $\binom{12}{3} = 220$ equally likely choices, and $p(x) = \binom{8}{3-x}\binom{4}{x}/\binom{12}{3}$:

$$p(0) = \frac{56 \times 1}{220} = \frac{14}{55}, \qquad p(1) = \frac{28 \times 4}{220} = \frac{28}{55},$$
$$p(2) = \frac{8 \times 6}{220} = \frac{12}{55}, \qquad p(3) = \frac{1 \times 4}{220} = \frac{1}{55}.$$

If X can take every value in an interval (so its values cannot be listed), X is a **continuous random variable**.

Illustration 3-4: spinner

A pointer fixed at the centre of a dial marked from 0 to 1 spins freely; X is the mark where it stops. X can take any value in $[0, 1]$.

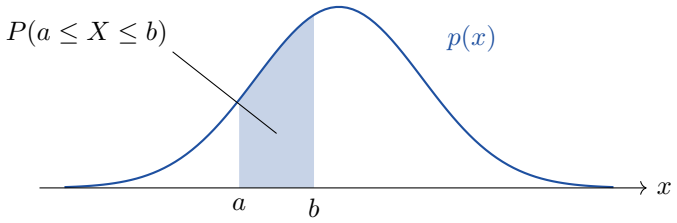
The probability that the pointer stops in $[a, b]$ is the ratio of the lengths:

$$P(a \le X \le b) = \frac{b-a}{1-0} = b - a \qquad (0 \le a < b \le 1).$$

With $p(x) = 1$ for $0 \le x \le 1$ and $p(x) = 0$ otherwise, this is $P(a \le X \le b) = \int_a^b p(x)\,dx$. So probabilities for X are determined by p, its **probability density function**.

Definition 3-6 (Probability density function of a continuous r.v.)

A function $p(x)$ is the **probability density function** of the continuous random variable X if
(1) $p(x) \ge 0$ and $\int_{-\infty}^{\infty} p(x)\,dx = 1$;
(2) $P(a \le X \le b) = \int_a^b p(x)\,dx$ $(-\infty \le a < b \le \infty)$.



$P(a \le X \le b)$ is the area under the density over $[a, b]$ (Figure 3-5)

Theorem 3-4 (Properties of continuous random variables)

For a continuous random variable X,
(1) $P(X = c) = 0$ for every c;
(2) $P(a \le X \le b) = P(a \le X < b) = P(a < X \le b) = P(a < X < b)$.

Proof. (1) $P(X = c)$ is the area at the single point c. Precisely, for every $h > 0$, $\{X = c\} \subset \{c - h \le X \le c + h\}$, so by Theorem 3-1(3)

$$0 \le P(X = c) \le \int_{c-h}^{c+h} p(x)\,dx \;\longrightarrow\; 0 \qquad (h \to 0),$$

since the integral over a shrinking interval of an integrable function tends to 0.

(2) $\{a \le X \le b\}$ is the disjoint union of $\{a < X < b\}$ with (some of) $\{X = a\}$ and $\{X = b\}$, which have probability 0 by (1); apply axiom (3). ■

Example 3-8 (Finding the constant)

A continuous random variable X has density $p(x) = bx(1 - x)$ for $0 \le x \le 1$, and 0 for $x < 0$ or $x > 1$. Find b and $P(0 \le X \le \frac{2}{3})$.
Solution. $1 = \int_0^1 bx(1-x)\,dx = b/6$, so $b = 6$. Then

$$P\Big(0 \le X \le \frac{2}{3}\Big) = \int_0^{2/3} 6x(1-x)\,dx = \Big[3x^2 - 2x^3\Big]_0^{2/3} = \frac{4}{3} - \frac{16}{27} = \frac{20}{27}.$$

When one characteristic of a population is observed, the sample space is the set of distinct characteristic values; the observed value is a random variable whose distribution equals the population distribution. So, to model a population theoretically, we use a random variable with the same distribution.

3.4 Expected value and its properties

Motivation. Chapter 2 summarized populations by mean and standard deviation. The same summaries for a probability distribution are the expected value and the variance of a random variable.

If X takes $x_1, x_2, \ldots$ with p.m.f. p, the **mean** or **expected value** $E(X)$ is $\sum_i x_i p(x_i) = x_1 P(X = x_1) + x_2 P(X = x_2) + \cdots$; for a density p it is $\int x p(x)\,dx$.

Definition 3-7 (Expected value of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$,

$$E(X) = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Example 3-9 (A mean)

X has distribution $p(1) = \frac{1}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{3}{8}$, $p(4) = \frac{1}{8}$. Find $E(X)$.
Solution. $E(X) = 1 \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{3}{8} + 4 \cdot \frac{1}{8} = 2.5$, the centre of the distribution graph (Figure 3-6).

As in mechanics, if masses $p(x_i)$ sit at $x_1, x_2, \ldots$, then $\sum x_i p(x_i)$ is the centre of mass relative to the origin; likewise $\int x p(x)\,dx$ for a mass density p. If the distribution of X models a population, $E(X)$ equals the population mean, often written μ.

Example 3-10 (A function of X)

A coin is tossed 3 times; X is the number of heads.
(a) Find the distribution of $Y = X(3 - X)$.
(b) Find $E(Y)$.
Solution.
(a) $p_X(0) = \frac{1}{8}$, $p_X(1) = \frac{3}{8}$, $p_X(2) = \frac{3}{8}$, $p_X(3) = \frac{1}{8}$. $Y = 0$ when $X \in \{0, 3\}$ and $Y = 2$ when $X \in \{1, 2\}$, so $p_Y(0) = \frac{1}{4}$, $p_Y(2) = \frac{3}{4}$.
(b) $E[X(3 - X)] = E(Y) = 0 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \frac{3}{2}$.

A function of a random variable is again a random variable. Its expectation can be computed without first finding its distribution:

$$\begin{aligned} E[X(3 - X)] &= 0(3 - 0)p_X(0) + 1(3 - 1)p_X(1) + 2(3 - 2)p_X(2) + 3(3 - 3)p_X(3) \\ &= 0 \cdot \tfrac{1}{8} + 2 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 0 \cdot \tfrac{1}{8} = \tfrac{3}{2}. \end{aligned}$$

Theorem 3-5 (Expectation of a function of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$, then for any function g

$$E[g(X)] = \begin{cases} \sum_{\text{all } x} g(x)\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} g(x)\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Proof. [Supplement]

Step 1 (discrete case). $Y = g(X)$ is discrete; for each value y, $\{Y = y\}$ is the disjoint union of $\{X = x\}$ over the x with $g(x) = y$, so $p_Y(y) = \sum_{x:\,g(x)=y} p(x)$. Hence

$$E(Y) = \sum_y y\,p_Y(y) = \sum_y \sum_{x:\,g(x)=y} g(x)\,p(x) = \sum_{\text{all } x} g(x)\,p(x),$$

since every x appears in exactly one group (the rearrangement is valid when the sum converges absolutely, which is the condition for $E(Y)$ to exist).

Step 2 (continuous case, monotone g). Let g be differentiable and strictly increasing with inverse h. Then $P(Y \le y) = P(X \le h(y)) = \int_{-\infty}^{h(y)} p(x)\,dx$, and differentiating, Y has density $p_Y(y) = p(h(y))h'(y)$. Substituting $x = h(y)$,

$$E(Y) = \int y\,p(h(y))h'(y)\,dy = \int g(x)\,p(x)\,dx.$$

For strictly decreasing g, $p_Y(y) = -p(h(y))h'(y)$ and the substitution reverses the limits, giving the same result.

Step 3 (general g). For general g split the line into intervals on which g is monotone and apply Step 2 on each piece; for completely arbitrary g the formula is the general change-of-variables rule of measure-theoretic integration, which we accept. ■

Example 3-11 (Waiting for a bus)

Buses leave a stop at 0, 15, 30 and 45 minutes past each hour. A person arrives at a random time. Find the expected waiting time.

Solution. Let X (minutes past the hour) be the arrival time; X has constant density $p(x) = \frac{1}{60}$ on $0 < x \le 60$. The waiting time is

$$g(x) = 15 - x, \quad 30 - x, \quad 45 - x, \quad 60 - x \qquad \text{on } (0,15],\ (15,30],\ (30,45],\ (45,60].$$

By Theorem 3-5,

$$\begin{aligned} E[g(X)] &= \frac{1}{60}\Big\{\int_0^{15}(15-x)\,dx + \int_{15}^{30}(30-x)\,dx + \int_{30}^{45}(45-x)\,dx + \int_{45}^{60}(60-x)\,dx\Big\} \\ &= \frac{1}{60} \times 4 \times 112.5 = 7.5, \end{aligned}$$

i.e. 7 minutes 30 seconds.

Theorem 3-6 (Properties of expected value)

(1) $E(aX + b) = aE(X) + b$ (a, b constants).
(2) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E[g_1(X)] + c_2 E[g_2(X)]$ (c_1, c_2 constants, g_1, g_2 any functions).

Proof. [Supplement] (2) By Theorem 3-5, in the discrete case

$$E[c_1 g_1(X) + c_2 g_2(X)] = \sum_x \{c_1 g_1(x) + c_2 g_2(x)\} p(x) = c_1 \sum_x g_1(x) p(x) + c_2 \sum_x g_2(x) p(x),$$

which is $c_1 E[g_1(X)] + c_2 E[g_2(X)]$; the continuous case is identical with integrals, using linearity of the integral.

(1) Take $g_1(x) = x$, $g_2(x) = 1$, $c_1 = a$, $c_2 = b$ in (2), and note $E(1) = \sum_x p(x) = 1$ (or $\int p = 1$). ■

The **standard deviation** of X (mean μ) is $\mathrm{sd}(X) = \sqrt{E[(X-\mu)^2]}$, and its square $E[(X-\mu)^2]$ is the **variance** $\mathrm{Var}(X)$.

Definition 3-8 (Variance and standard deviation of a random variable)

If X has mean μ and p.m.f. (or p.d.f.) $p(x)$,

(1) $\mathrm{Var}(X) = E[(X-\mu)^2] = \begin{cases} \sum_{\text{all } x} (x-\mu)^2 p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx & \text{(continuous)} \end{cases}$

(2) $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{E[(X-\mu)^2]}$.

When the distribution of X models a population, $\mathrm{Var}(X)$ and $\mathrm{sd}(X)$ are the population variance and standard deviation, often written σ^2 and σ.

Theorem 3-7 (Shortcut formula for the variance)

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. By Theorem 3-6(2) with $\mu = E(X)$,

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2.$$

■

Example 3-12 (Two coin tosses)

X is the number of heads in 2 tosses of a coin. Find the mean and standard deviation of X.
Solution.

x	0	1	2	Total
$p(x)$	1/4	1/2	1/4	1
$xp(x)$	0	1/2	1/2	1
$x^2 p(x)$	0	1/2	1	3/2

So $E(X) = 1$, $E(X^2) = \frac{3}{2}$, $\mathrm{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}$ and $\mathrm{sd}(X) = 1/\sqrt{2}$.

Example 3-13 (Spinner)

For the spinner of Illustration 3-4, find the mean and standard deviation of X.
Solution. $p(x) = 1$ on $[0,1]$, so

$$E(X) = \int_0^1 x\, dx = \frac{1}{2}, \qquad E(X^2) = \int_0^1 x^2\, dx = \frac{1}{3},$$

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}, \qquad \mathrm{sd}(X) = \frac{1}{\sqrt{12}}.$$

Theorem 3-8 (Property of the variance)

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X) \qquad (a, b \text{ constants}).$$

Proof. [Supplement] Let $\mu = E(X)$. By Theorem 3-6(1), $E(aX+b) = a\mu+b$, so $(aX+b) - E(aX+b) = a(X-\mu)$ and

$$\mathrm{Var}(aX + b) = E[a^2(X-\mu)^2] = a^2 E[(X-\mu)^2] = a^2 \,\mathrm{Var}(X).$$

■

Subtracting the mean and dividing by the standard deviation is **standardization**:

$$Z = \frac{X - E(X)}{\mathrm{sd}(X)}.$$

By Theorems 3-6 and 3-8 (with $a = 1/\mathrm{sd}(X)$, $b = -E(X)/\mathrm{sd}(X)$), $E(Z) = 0$ and $\mathrm{Var}(Z) = \mathrm{Var}(X)/\mathrm{sd}(X)^2 = 1$.

3.5 Joint distribution of two random variables

Motivation. Height and weight of a newborn, or English and maths scores, are observed together and are related. The individual distributions do not capture that relation; the joint distribution does.

We treat discrete random variables; the results extend to the continuous case.

Illustration 3-5: same marginals, different relations

Three different coins A, B, C are tossed. Let $X = 1$ if A shows heads (else 0); $Y = 1$ if A and B both show heads (else 0); $Z = 1$ if B and C both show heads (else 0). Y and Z have the same distribution, $P(0) = \frac{3}{4}$, $P(1) = \frac{1}{4}$. Yet the relation of X to Y differs from that of X to Z: $X = 0$ forces $Y = 0$, whereas given $X = 0$, Z is 0 or 1 with probabilities $\frac{3}{4}, \frac{1}{4}$.

Definition 3-9 (Joint distribution of two random variables)

The description of how probability is spread over all ordered pairs of values of X and Y, by nonnegative numbers summing to 1, is the **joint probability distribution** (or **joint distribution**) of X and Y.

For discrete X, Y with values $x_1, x_2, \ldots$ and $y_1, y_2, \ldots$,

$$p(x,y) = \begin{cases} P(X = x,\, Y = y), & (x,y) = (x_i, y_j)\ (i,j = 1,2,\ldots) \\ 0, & \text{otherwise} \end{cases}$$

is the **joint probability mass function**.

Definition 3-10 (Joint probability mass function)

$p(x,y)$ is the **joint probability mass function** of the discrete random variables X and Y if
(1) $0 \le p(x,y) \le 1$ and $\sum_{\text{all } x}\sum_{\text{all } y} p(x,y) = 1$;
(2) $P(a < X \le b,\ c < Y \le d) = \sum_{a<x\le b}\sum_{c<y\le d} p(x,y)$.

Example 3-14 (Joint distributions from three coins)

For the experiment of Illustration 3-5, find the joint distribution of
(a) X and Y; (b) X and Z.
Solution. Listing coins A, B, C:

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	1	0	0	0	1	0	0	0

Each outcome has probability $\frac{1}{8}$; collecting them,

(a) X and Y

$y\backslash x$	0	1	Row sum
0	2/4	1/4	3/4
1	0	1/4	1/4
Col. sum	2/4	2/4	1

(b) X and Z

$z\backslash x$	0	1	Row sum
0	3/8	3/8	6/8
1	1/8	1/8	2/8
Col. sum	4/8	4/8	1

The column and row sums reproduce the distributions of X and Y. In general the **marginal probability mass functions** are

$$p_1(x) = \sum_{\text{all } y} p(x,y), \qquad p_2(y) = \sum_{\text{all } x} p(x,y),$$

and they determine the distributions of X and Y (because $\{X = x\}$ is the disjoint union of the events $\{X = x, Y = y\}$ over y).

Example 3-15 (Two games on one die)

A game pays according to one die roll. A receives 100 won for 1 or 2, 200 won for 3 or 4, 300 won for 5 or 6; B receives 100 won for an even number and (face $\times 100$) won for an odd number. Find the distribution of Z, the total received by A and B.
Solution. Let X, Y be the amounts of A and B. E.g. $(X,Y) = (100,100)$ occurs for faces 1 and 2, so $p(100,100) = \frac{2}{6}$. The joint distribution is

$x\backslash y$	100	300	500	Row sum
100	2/6	0	0	2/6
200	1/6	1/6	0	2/6
300	1/6	0	1/6	2/6
Col. sum	4/6	1/6	1/6	1

$Z = X + Y$ takes the values 200, 300, 400, 500, 800 with probabilities $\frac{2}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$.

Theorem 3-9 (Expectation of a function of two random variables)

Let X, Y be discrete with joint p.m.f. $p(x, y)$. For any functions g, g_1, g_2:
(1) $E[g(X,Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x,y)\, p(x,y)$;
(2) $E[c_1 g_1(X,Y) + c_2 g_2(X,Y)] = c_1 E[g_1(X,Y)] + c_2 E[g_2(X,Y)]$ (c_1, c_2 constants).

Proof. [Supplement] (1) $W = g(X,Y)$ is discrete, and $\{W = w\}$ is the disjoint union of the events $\{X = x, Y = y\}$ over pairs with $g(x,y) = w$, so $p_W(w) = \sum_{(x,y):\, g(x,y)=w} p(x,y)$. Hence

$$E(W) = \sum_w w\, p_W(w) = \sum_w \sum_{(x,y):\, g(x,y)=w} g(x,y)\, p(x,y) = \sum_x \sum_y g(x,y)\, p(x,y),$$

each pair (x, y) appearing exactly once (absolute convergence justifies the regrouping).

(2) By (1) applied to $c_1 g_1 + c_2 g_2$, then splitting the double sum:

$$\sum_x \sum_y \{c_1 g_1 + c_2 g_2\}(x,y)\, p(x,y) = c_1 \sum_x \sum_y g_1(x,y) p(x,y) + c_2 \sum_x \sum_y g_2(x,y) p(x,y).$$

■

Extending independence of events (Section 3.2): X and Y are independent if, for all x, y, the events $\{X = x\}$ and $\{Y = y\}$ are independent.

Definition 3-11 (Independence of two random variables)

Let $p(x,y)$ be the joint p.m.f. (joint p.d.f.) of discrete (continuous) X, Y and p_1, p_2 the marginals. X and Y are **independent** if for all x, y

$$p(x,y) = p_1(x)\, p_2(y).$$

Otherwise they are **dependent**.

Example 3-16 (Independence check)

A coin is tossed three times; X is the number of heads in the first two tosses and Y the number of heads on the third toss. Are X and Y independent?
Solution. The joint distribution is

$x \backslash y$	0	1	Row sum
0	1/8	1/8	1/4
1	2/8	2/8	1/2
2	1/8	1/8	1/4
Col. sum	1/2	1/2	1

Every cell equals the product of its margins, e.g. $p_1(0)p_2(0) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} = p(0,0)$ and $p_1(1)p_2(1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = p(1,1)$; the other four cells check the same way. Hence X and Y are independent.

3.6 Covariance and correlation

Motivation. When X and Y tend to be large together, the product of their standardized deviations is typically positive. Averaging this product gives a single number measuring the direction and strength of linear association.

Let X, Y have means μ_1, μ_2 and standard deviations σ_1, σ_2. The **correlation coefficient** $\mathrm{Corr}(X,Y)$ is the expected value of the product of standardized variables $\left(\frac{X-\mu_1}{\sigma_1}\right)\left(\frac{Y-\mu_2}{\sigma_2}\right)$. It measures the strength and direction of a *linear* relation, and always $-1 \le \mathrm{Corr}(X,Y) \le 1$. In the extreme case $Y = aX + b$ ($a \neq 0$), the correlation is 1 if $a > 0$ and -1 if $a < 0$. The expected value of $(X-\mu_1)(Y-\mu_2)$, i.e. the correlation times $\sigma_1\sigma_2$, is the **covariance** $\mathrm{Cov}(X,Y)$.

Definition 3-12 (Covariance and correlation coefficient)

(1) $\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$.

(2) $\mathrm{Corr}(X,Y) = E\left[\left(\frac{X-\mu_1}{\sigma_1}\right)\left(\frac{Y-\mu_2}{\sigma_2}\right)\right] = \frac{\mathrm{Cov}(X,Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)}$.

Proposition (Range of the correlation)

$-1 \le \mathrm{Corr}(X,Y) \le 1$; and if $Y = aX + b$ with $a \neq 0$, then $\mathrm{Corr}(X,Y) = 1$ for $a > 0$ and -1 for $a < 0$.

Proof. [Supplement]

Step 1. Let $U = (X-\mu_1)/\sigma_1$, $V = (Y-\mu_2)/\sigma_2$ and $\rho = \mathrm{Corr}(X,Y) = E(UV)$. Then $E(U^2) = E(V^2) = 1$, and by Theorem 3-9(2)

$$0 \le E[(U-\rho V)^2] = E(U^2) - 2\rho E(UV) + \rho^2 E(V^2) = 1 - \rho^2,$$

so $\rho^2 \le 1$.

Step 2. If $Y = aX + b$, then $\mu_2 = a\mu_1 + b$ and $\sigma_2 = |a|\sigma_1$ (Theorem 3-8), so $V = \frac{a}{|a|}U$ and $\rho = \frac{a}{|a|}E(U^2) = \pm 1$ according to the sign of a. ■

Theorem 3-10 (Shortcut formula for the covariance)

$$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y).$$

Proof. By Theorem 3-9(2),

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X-\mu_1)(Y-\mu_2)] = E(XY - \mu_2 X - \mu_1 Y + \mu_1\mu_2) \\ &= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1\mu_2 = E(XY) - \mu_1\mu_2.\end{aligned}$$

■

Example 3-17 (Covariance and correlation)

X and Y have the joint distribution below. Find $\mathrm{Cov}(X,Y)$ and $\mathrm{Corr}(X,Y)$.

$x \backslash y$	0	1	2	3	Row sum
0	0.05	0.05	0.10	0.00	0.20
1	0.05	0.10	0.25	0.10	0.50
2	0.00	0.15	0.10	0.05	0.30
Col. sum	0.10	0.30	0.45	0.15	1.00

Solution. From the margins,

$$\mu_1 = 1.1, \qquad \sigma_1^2 = 1.7 - (1.1)^2 = 0.49, \qquad \sigma_1 = 0.7,$$

$$\mu_2 = 1.65, \qquad \sigma_2^2 = 3.45 - (1.65)^2 = 0.7275, \qquad \sigma_2 = \sqrt{0.7275} = 0.8529.$$

Summing $xy\,p(x,y)$ over the cells with $x, y, p \neq 0$,

$$E(XY) = 0.1 + 0.5 + 0.3 + 0.3 + 0.4 + 0.3 = 1.9.$$

Hence $\mathrm{Cov}(X,Y) = 1.90 - 1.65 \times 1.1 = 0.085$ and

$$\mathrm{Corr}(X,Y) = \frac{0.085}{0.70 \times 0.8529} = 0.1424.$$

Theorem 3-11 (Properties of covariance and correlation)

For constants a, b, c, d:
(1) $\mathrm{Cov}(aX+b,\, cY+d) = ac\ \mathrm{Cov}(X,Y)$;
(2) $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{Corr}(X,Y)$ if $ac > 0$, and $-\mathrm{Corr}(X,Y)$ if $ac < 0$.

Proof. [Supplement] (1) $E(aX+b) = a\mu_1 + b$ and $E(cY+d) = c\mu_2 + d$, so the centred variables are $a(X-\mu_1)$ and $c(Y-\mu_2)$, and

$$\mathrm{Cov}(aX+b, cY+d) = E[ac(X-\mu_1)(Y-\mu_2)] = ac\ \mathrm{Cov}(X,Y).$$

(2) By Theorem 3-8, $\mathrm{sd}(aX+b) = |a|\,\mathrm{sd}(X)$ and $\mathrm{sd}(cY+d) = |c|\,\mathrm{sd}(Y)$. With (1),

$$\mathrm{Corr}(aX+b, cY+d) = \frac{ac\ \mathrm{Cov}(X,Y)}{|a||c|\,\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \frac{ac}{|ac|}\,\mathrm{Corr}(X,Y),$$

and $ac/|ac| = \pm 1$ according to the sign of ac. ■

Theorem 3-12 (Variance of a sum of two random variables)

(1) $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
(2) $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X,Y)$.

Proof. [Supplement] $E(X \pm Y) = \mu_1 \pm \mu_2$ by Theorem 3-9(2), so $(X \pm Y) - E(X \pm Y) = (X-\mu_1) \pm (Y-\mu_2)$. Squaring and taking expectations (Theorem 3-9(2)),

$$\mathrm{Var}(X \pm Y) = E[(X-\mu_1)^2] + E[(Y-\mu_2)^2] \pm 2E[(X-\mu_1)(Y-\mu_2)],$$

which is $\mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$. ■

Theorem 3-13 (Independent random variables)

If X and Y are independent, then
(1) $E(XY) = E(X)E(Y)$;
(2) $\mathrm{Cov}(X,Y) = 0$ and $\mathrm{Corr}(X,Y) = 0$;
(3) $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Proof. (1) By Theorem 3-9(1) and $p(x,y) = p_1(x)p_2(y)$,

$$E(XY) = \sum_x \sum_y xy\,p(x,y) = \sum_x \sum_y xy\,p_1(x)p_2(y) = \sum_x x\,p_1(x) \sum_y y\,p_2(y) = E(X)E(Y).$$

(2) By Theorem 3-10, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0$, hence also $\mathrm{Corr}(X,Y) = 0$.

(3) Insert $\mathrm{Cov}(X,Y) = 0$ into Theorem 3-12. ■

Chapter 4. Sampling Distributions

4.1 The Bernoulli and normal distributions as population distributions

Motivation. Statistical inference judges an estimate by the probability distribution of the estimator, and that distribution is driven by the population distribution. The two most important population models are the Bernoulli distribution (a two-valued characteristic) and the normal distribution (a continuous characteristic).

4.1.1 Bernoulli trials

Many everyday experiments have exactly two outcomes: pass/fail, win/lose, cured/not cured. Such an experiment is called a **Bernoulli trial**; its outcomes are labelled success s and failure f.

Definition 4-1 (Bernoulli trial)

(a) A Bernoulli trial has exactly two outcomes, success s and failure f, so its sample space is the two-point set $S = \{s, f\}$.
(b) The success probability is $p = P(\{s\})$ and the failure probability is $q = P(\{f\}) = 1 - p$, where $0 \le p \le 1$.

Definition 4-2 (Bernoulli random variable)

On the sample space $S = \{s, f\}$ of a Bernoulli trial, the random variable X with $X(s) = 1$, $X(f) = 0$ is called a **Bernoulli random variable**. Its distribution, $P(X = 1) = p$, $P(X = 0) = 1 - p$, is the **Bernoulli distribution** with success probability p.

The Bernoulli distribution models a population whose characteristic is dichotomous (e.g. patients "cured"/"not cured" coded 1/0).

Example 4-1 (Mean and variance of the Bernoulli distribution)

Find the mean and variance of the Bernoulli distribution.
Solution. $E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$ and $E(X^2) = 1^2 \cdot p + 0^2 \cdot (1 - p) = p$. Hence

$$\mathrm{Var}(X) = E(X^2) - \{E(X)\}^2 = p - p^2 = p(1 - p).$$

Bernoulli trials are **independent** if the result of one trial does not affect the others, and **repeated** if the success probability is the same in every trial.

Illustration 4-1: sampling with replacement

A deck of 10 cards contains 3 winning cards; two cards are drawn by simple random sampling *with* replacement. Let X_1, X_2 be the Bernoulli variables of the first and second draws (1 = winning card). Then

$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1) = P(X_2 = 1 \mid X_1 = 0) = \tfrac{3}{10},$$

and similarly $P(X_2 = 0) = \tfrac{7}{10}$ for either value of X_1. Also $P(X_1 = 1) = P(X_2 = 1) = \tfrac{3}{10}$, so the two trials are independent and repeated: X_1, X_2 are independent with the same distribution. More generally, n draws with replacement give independent, identically distributed

$X_1, \ldots, X_n$.

Illustration 4-2: sampling without replacement

Draw the two cards of Illustration 4-1 *without* replacement. Then $P(X_1 = 1) = \frac{3}{10}$, $P(X_2 = 1 \mid X_1 = 1) = \frac{2}{9}$, $P(X_2 = 1 \mid X_1 = 0) = \frac{3}{9}$, and by the law of total probability

$$P(X_2 = 1) = \tfrac{2}{9} \cdot \tfrac{3}{10} + \tfrac{3}{9} \cdot \tfrac{7}{10} = \tfrac{3}{10}.$$

So each draw wins with probability $\frac{3}{10}$, but $P(X_2 = 1) \neq P(X_2 = 1 \mid X_1 = 1)$: the trials are not independent.

Illustration 4-3: general finite deck

With N cards of which M are winning, sampling two without replacement gives

$$P(X_1 = 1) = P(X_2 = 1) = \frac{M}{N}, \qquad P(X_2 = 1 \mid X_1 = 1) = \frac{M-1}{N-1},$$

so again the draws are identically distributed but dependent. If N (and M) is large relative to the sample, $P(X_2 = 1) \approx P(X_2 = 1 \mid X_1 = 1)$ and the draws are nearly independent.

Sampling with vs. without replacement

From a finite population, sampling without replacement is practically indistinguishable from sampling with replacement when the population size N is large compared with the sample size.

4.1.2 The normal distribution

The normal distribution is the leading model for an infinite population whose characteristic is continuous. Its density curve

(a) is symmetric about a single point, the population mean μ;

(b) is highest at the centre μ;

(c) decreases as one moves away from μ;

(d) has inflection points at $\mu \pm \sigma$.

Definition 4-3 (Normal density with mean μ and standard deviation σ)

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

Changing μ shifts the curve without changing its shape; a smaller σ makes the curve narrower and taller near μ, a larger σ flatter. If one characteristic is observed at random from a normal population, the observation X has the population distribution, so $X \sim N(\mu, \sigma^2)$.

Theorem 4-1 (Properties of the normal distribution 1)

If $X \sim N(\mu, \sigma^2)$, then for any constants $a \neq 0$ and b,

$$aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

Proof. [Supplement] We use moment generating functions (mgf), $M_X(t) = E(e^{tX})$; an mgf finite near 0 determines the distribution.

Step 1. For $X \sim N(\mu, \sigma^2)$, substituting $x = \mu + \sigma z$ and completing the square $tz\sigma - \frac{z^2}{2} = -\frac{1}{2}(z - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$,

$$M_X(t) = e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma t z - z^2/2}\, dz = e^{\mu t + \sigma^2 t^2/2} \int_{-\infty}^{\infty} \frac{e^{-(z-\sigma t)^2/2}}{\sqrt{2\pi}}\, dz = e^{\mu t + \sigma^2 t^2/2}.$$

Step 2. Hence $M_{aX+b}(t) = e^{bt} M_X(at) = e^{(a\mu+b)t + a^2\sigma^2 t^2/2}$, which is the mgf of $N(a\mu + b, a^2\sigma^2)$. By uniqueness of mgfs the claim follows. ■

(If $a = 0$ then $aX + b = b$ is constant, a "normal distribution with variance 0".)

In particular, taking $a = 1/\sigma$, $b = -\mu/\sigma$,

$$X \sim N(\mu, \sigma^2) \implies Z = \frac{X - \mu}{\sigma} \sim N(0, 1). \tag{4.1}$$

$N(0, 1)$ is the **standard normal distribution**; a standard normal variable is usually written Z. Appendix C tabulates $P(Z \leq z)$.

Definition (Probabilities from the standard normal table)

(a) $P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$.
(b) $P(Z \geq z) = 1 - P(Z < z)$.

Example 4-2 (Two tails)

Find $P(Z < -1.9 \text{ or } Z > 2.1)$.
Solution. The events are disjoint, so the probability is $P(Z < -1.9) + P(Z > 2.1) = 0.0287 + (1 - 0.9821) = 0.0287 + 0.0179 = 0.0466$.

Example 4-3 (A quantile)

Find z with $P(Z > z) = 0.025$.
Solution. $P(Z \leq z) = 1 - 0.025 = 0.975$, so the table gives $z = 1.96$.

Definition (Upper quantile z_α)

For $0 < \alpha < 1$, z_α is the number with $P(Z > z_\alpha) = \alpha$, $Z \sim N(0, 1)$; it is the $100(1 - \alpha)$th percentile (quantile) of $N(0, 1)$. Frequently used values:

$$z_{0.005} = 2.58, \qquad z_{0.025} = 1.96, \qquad z_{0.05} = 1.645.$$

If $X \sim N(\mu, \sigma^2)$, standardizing with (4.1) gives

$$P(X \leq b) = P\Big(Z \leq \frac{b - \mu}{\sigma}\Big),$$

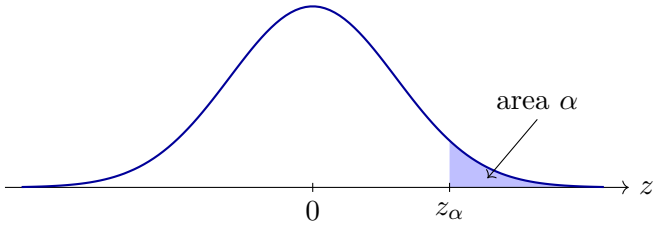


Figure 1: The upper quantile z_α of $N(0,1)$.

$$P(a \le X \le b) = P\Big(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\Big).$$

Example 4-4 (Proportion of A grades)

Grades in a statistics class are approximately $N(60, 10^2)$. Students scoring 75 or more get an A. Approximate the proportion of A grades.
Solution. $P(X \ge 75) = P(\frac{X-60}{10} \ge \frac{75-60}{10}) \approx P(Z \ge 1.5) = 0.0668$.

Theorem 4-2 (Properties of the normal distribution 2)

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a_1, a_2,

$$a_1X_1 + a_2X_2 \sim N(a_1\mu_1 + a_2\mu_2,\ a_1^2\sigma_1^2 + a_2^2\sigma_2^2).$$

Proof. [Supplement] By independence the mgf of a sum is the product of the mgfs. Using Step 1 of the proof of Theorem 4-1,

$$\begin{aligned} M_{a_1X_1+a_2X_2}(t) &= M_{X_1}(a_1t)\, M_{X_2}(a_2t) = e^{a_1\mu_1 t + a_1^2\sigma_1^2t^2/2}\, e^{a_2\mu_2 t + a_2^2\sigma_2^2t^2/2} \\ &= \exp\Big\{(a_1\mu_1 + a_2\mu_2)t + \tfrac{1}{2}(a_1^2\sigma_1^2 + a_2^2\sigma_2^2)t^2\Big\}, \end{aligned}$$

the mgf of the stated normal distribution. By induction the same holds for any finite number of independent normal variables: $\sum a_iX_i \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$. ∎

Example 4-5 (Calories burned)

Calories burned running each morning are approximately $N(200, 15^2)$ and calories burned cycling approximately $N(80, 5^2)$, independently. Find the probability that the total burned in a day is at least 300 kcal.
Solution. The total S is approximately normal with mean $200 + 80 = 280$ and variance $15^2 + 5^2 = 250$. So

$$P(S \ge 300) = P\Big(\frac{S-280}{\sqrt{250}} \ge \frac{300-280}{\sqrt{250}}\Big) \approx P(Z \ge 1.27) = 1 - 0.898 = 0.102.$$

4.2 Sampling distributions

Motivation. A single estimate, say a sample proportion of 76.4%, says nothing by itself about its error. The accuracy of an estimation method is judged by looking at *all* possible samples, i.e. by the probability distribution of the statistic.

4.2.1 Sampling distribution of a statistic

A survey estimates the population proportion p by the sample proportion (number in favour)/(sample size). Accuracy is expressed as the probability that the estimation error is at most a given amount, over random selection of the sample.

Illustration: all samples of size 3

Population $\{\text{yes}, \text{no}, \text{yes}, \text{no}, \text{yes}\}$ (true $p = 0.6$), sample of size 3 without replacement.

Sample	Probability	Sample proportion	Error
{yes, yes, yes}	1/10	1	0.4
{yes, yes, no}	6/10	2/3	0.067
{yes, no, no}	3/10	1/3	−0.267

So $P(|\text{error}| \le 0.1) = 0.6$ and $P(|\text{error}| \le 0.3) = 0.9$.

Definition (Statistic and sampling distribution)

A **statistic** is a random variable computed from the sample. The probability distribution of a statistic is its **sampling distribution**; it quantifies how reliable an estimate is.

Example 4-6 (Sample mean from a small population)

From the finite population $\{2, 2, 4, 5\}$ draw a sample of size 2 without replacement. Find the distribution of the sample mean.
Solution. The six equally likely samples (probability $\frac{1}{6}$ each) and their means are

Sample	$\{2,2\}$	$\{2,4\}$	$\{2,5\}$	$\{2,4\}$	$\{2,5\}$	$\{4,5\}$
Mean	2	3	3.5	3	3.5	4.5

Hence $P(\bar{X} = 2) = \frac{1}{6}$, $P(\bar{X} = 3) = \frac{1}{3}$, $P(\bar{X} = 3.5) = \frac{1}{3}$, $P(\bar{X} = 4.5) = \frac{1}{6}$.

The sampling distribution depends on both the population distribution and the sampling method (with or without replacement).

4.2.2 Random samples

A sample drawn by simple random sampling without replacement from a finite population is a **random sample**, written $X_1, \ldots, X_n$ (X_i is the value of the ith unit; categorical answers are coded 1/0). Sampling with replacement gives $X_1, \ldots, X_n$ that

(a) each have the population distribution, and

(b) are mutually independent.

When the population is large, a random sample without replacement nearly has properties (a) and (b); this motivates the definition for infinite populations.

Definition 4-4 (Random sample)

(a) Finite population: a sample drawn by simple random sampling without replacement.
(b) Infinite population: random variables $X_1, \ldots, X_n$ such that (i) each X_i has the population

distribution and (ii) $X_1, \ldots, X_n$ are independent.

Exact sampling distributions for finite populations are complicated, so a large finite population is treated as infinite. From Chapter 5 on, all inference assumes a random sample from an infinite population.

4.3 The binomial distribution

Motivation. When each unit is classified into two categories, the number of "successes" in a random sample is the basic statistic; its distribution is the binomial distribution.

Consider an infinite population of 1s and 0s with proportion p of 1s ($0 < p < 1$). A random sample $X_1, \ldots, X_n$ consists of independent Bernoulli(p) variables, i.e. n independent repeated Bernoulli trials, and the number of successes is

$$X = X_1 + \cdots + X_n.$$

Its distribution is the **binomial distribution** $B(n, p)$; $B(1, p)$ is the Bernoulli distribution.

Definition 4-5 (Binomial distribution)

With n trials and success probability p, the probability mass function is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

The **binomial coefficient** $\binom{n}{x}$ is the coefficient of $a^{n-x}b^x$ in $(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^{n-x}b^x$ and can be read off Pascal's triangle. For $p = 0.5$ the binomial distribution is symmetric; for $p < 0.5$ it is **skewed to the right** (long right tail), for $p > 0.5$ **skewed to the left**.

Example 4-7 (Guessing on a multiple-choice test)

A test has 20 five-choice questions and every answer is guessed at random. Find the probability of
(a) no correct answer;
(b) 4 to 6 correct answers;
(c) at least 2 correct answers.
Solution. The number correct is $X \sim B(20, 0.2)$.
(a) $P(X = 0) = \binom{20}{0} 0.2^0\, 0.8^{20} = 0.012$.
(b) $P(4 \le X \le 6) = \sum_{x=4}^{6} \binom{20}{x} 0.2^x\, 0.8^{20-x} = 0.502$.
(c) $P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.8^{20} - 20(0.2)(0.8^{19}) = 0.931$.

Example 4-8 (On-time deliveries)

A courier delivers on time 95% of the time. In a random sample of 20 deliveries, find the probability that the sample on-time rate is at most 90%.
Solution. The number of late deliveries X is approximately $B(20, 0.05)$. An on-time rate $\le 90\%$ means a late rate $X/20 \ge 0.1$, so

$$P\Big(\frac{X}{20} \ge 0.1\Big) = P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 0.264.$$

Theorem 4-3 (Mean and variance of $B(n,p)$)

If $X \sim B(n,p)$, then
(a) $E(X) = np$;
(b) $\mathrm{Var}(X) = np(1-p)$.

Proof. Write $X = X_1 + \cdots + X_n$ with X_i independent Bernoulli(p), so $E(X_i) = p$ and $\mathrm{Var}(X_i) = p(1-p)$ (Example 4-1).

(a) By linearity of expectation, $E(X) = \sum_i E(X_i) = np$.

(b) For independent variables the covariances vanish, so $\mathrm{Var}(X) = \sum_i \mathrm{Var}(X_i) = np(1-p)$. ■

4.4 The distribution of the sample mean

Motivation. The sample mean is the basic estimator of a population mean. Its mean and variance follow from simple rules; for normal populations it is exactly normal, and by the central limit theorem it is approximately normal for any population when n is large.

For a 0/1 population the population mean is the proportion and the sample mean is the sample proportion (§4.3). Now let $X_1, \ldots, X_n$ be a random sample from an infinite population with mean μ and variance σ^2, and

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n).$$

Theorem 4-4 (Distribution of the sample mean 1)

For a random sample of size n from a population with mean μ and variance σ^2,
(a) $E(\bar{X}) = \mu$;
(b) $\mathrm{Var}(\bar{X}) = \sigma^2/n$;
(c) $\mathrm{sd}(\bar{X}) = \sigma/\sqrt{n}$.

Proof.

(a) Since $E(X_i) = \mu$, $E(\bar{X}) = \frac{1}{n}\{E(X_1) + \cdots + E(X_n)\} = \frac{1}{n} \cdot n\mu = \mu$.

(b) Since the X_i are independent with $\mathrm{Var}(X_i) = \sigma^2$, $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\{\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)\} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

(c) Take square roots in (b). ■

Finite population

For a sample of size n drawn without replacement from a finite population of size N with mean μ and variance σ^2,

$$E(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n}.$$

Thus $\bar{X}$ is centred at μ and concentrates around μ as n grows.

Theorem 4-5 (Distribution of the sample mean 2)

If the population is $N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.

Proof. By the extension of Theorem 4-2 to n variables, $\bar{X} = \sum_{i=1}^{n} \frac{1}{n} X_i$ is normal with mean $\sum \frac{1}{n}\mu = \mu$ and variance $\sum \frac{1}{n^2}\sigma^2 = \sigma^2/n$. ■

Theorem 4-6 (Distribution of the sample mean 3: central limit theorem)

For a random sample of size n from any population with mean μ and variance σ^2, when n is large $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0,1).$$

Proof. [Supplement] We prove the precise statement $P(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z) \to P(Z \le z)$ for every z, under the extra assumption that the mgf $M(t) = E(e^{t(X_1-\mu)/\sigma})$ is finite for $|t| < h$. We use the *continuity theorem*: if $M_{W_n}(t) \to M_W(t)$ for all $|t| < h$, then the cdf of W_n converges to that of W at every continuity point.

Step 1. Let $Y_i = (X_i - \mu)/\sigma$, so the Y_i are i.i.d. with $E(Y_i) = 0$, $E(Y_i^2) = 1$, and

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i, \qquad M_{W_n}(t) = \Big\{ M\Big(\frac{t}{\sqrt{n}}\Big) \Big\}^n.$$

Step 2. By Taylor's theorem, $M(s) = 1 + M'(0)s + \frac{1}{2}M''(0)s^2 + r(s)$ with $r(s)/s^2 \to 0$ as $s \to 0$. Here $M'(0) = E(Y_1) = 0$ and $M''(0) = E(Y_1^2) = 1$, so

$$M\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + r\Big(\frac{t}{\sqrt{n}}\Big), \qquad n\, r\Big(\frac{t}{\sqrt{n}}\Big) = t^2 \cdot \frac{r(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Step 3. Hence $M_{W_n}(t) = (1 + \frac{a_n}{n})^n$ with $a_n = \frac{t^2}{2} + n\, r(t/\sqrt{n}) \to \frac{t^2}{2}$, and since $(1 + a_n/n)^n \to e^a$ whenever $a_n \to a$, $M_{W_n}(t) \to e^{t^2/2}$, the mgf of $N(0,1)$.

Step 4. By the continuity theorem, and since the cdf of $N(0,1)$ is continuous, $P(W_n \le z) \to P(Z \le z)$ for every z. (Without the mgf assumption the same argument works with characteristic functions $E(e^{itY})$.) ■

The striking point is that the conclusion holds whatever the shape of the population; this is why the normal distribution is central to inference.

Illustration 4-4: the CLT in action

The population is uniform on the digits $0, 1, \ldots, 9$ (each with probability 0.1), far from normal, with $\mu = 4.5$ and $\sigma^2 = 8.25$. One hundred random samples of size 5 were drawn and $\bar{X}$ computed for each. The histogram of the 100 sample means is roughly bell-shaped and centred near 4.5 (relative frequencies 0.29 and 0.27 in the classes $[4,5)$ and $[5,6)$, tapering to 0.03 and 0.07 in the outer classes), even though $n = 5$. Larger n brings it still closer to normal.

How large n must be depends on how far the population is from normal. For a normal population $\bar{X}$ is exactly normal; otherwise $n > 30$ usually gives a satisfactory normal approximation.

Example 4-9 (Calories of a new drink)

A new drink has mean 168 kcal and standard deviation 6 kcal. For a random sample of 100 drinks, find $P(167 \le \bar{X} \le 169)$.
Solution. With $n = 100$ the CLT applies; $\bar{X}$ has mean 168 and standard deviation $6/\sqrt{100} =$

0.6. So

$$P(167 \le \bar{X} \le 169) \approx P\Big(\frac{167-168}{0.6} \le Z \le \frac{169-168}{0.6}\Big)$$
$$= P(-1.67 \le Z \le 1.67) = 0.9525 - 0.0475 = 0.9050.$$

4.4.1 Normal approximation to the binomial

If $X_1, \dots, X_n$ is a random sample from $B(1,p)$, then $X = X_1 + \cdots + X_n \sim B(n,p)$, the population mean is p, the variance is $p(1-p)$ and the sample mean is the sample proportion $\hat{p} = X/n$. The CLT therefore gives the following.

Definition (Normal approximation to the binomial)

If $X \sim B(n,p)$ and n is large (rule of thumb: $np \ge 5$ and $n(1-p) \ge 5$), then X is approximately $N(np, np(1-p))$:

$$\frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \dot{\sim} N(0,1).$$

For example $B(15, 0.6)$ is already well approximated by $N(9, 3.6)$ since $np = 9 > 5$.

Example 4-10 (Normal approximation)

Let $X \sim B(100, \frac{1}{5})$. Approximate
(a) $P(X \le 21)$;
(b) $P(15 \le X \le 19)$.
Solution. $np = 20$ and $\sqrt{np(1-p)} = \sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}} = 4$.
(a) $P(X \le 21) \approx P(Z \le \frac{21-20}{4}) = P(Z \le 0.25) = 0.5987$.
(b) $P(15 \le X \le 19) \approx P(-1.25 \le Z \le -0.25) = 0.4013 - 0.1056 = 0.2957$.

Continuity correction

A binomial probability is a sum of bars of width 1 centred at the integers. For $X \sim B(15, 0.6)$, $P(7 \le X \le 10)$ is better approximated by the normal area from 6.5 to 10.5:

$$P(7 \le X \le 10) = P(6.5 \le X \le 10.5) \approx P\Big(\frac{6.5-9}{\sqrt{3.6}} \le Z \le \frac{10.5-9}{\sqrt{3.6}}\Big)$$
$$= P(-1.318 \le Z \le 0.791) = 0.692,$$

very close to the exact value 0.688. This adjustment is the **continuity correction**.

4.5 The chi-square, t and F distributions

Motivation. Inference about a normal variance, and about a normal mean when σ is unknown, relies on statistics built from the sample variance. Their sampling distributions are the chi-square, t and F distributions.

For a random sample $X_1, \dots, X_n$ from $N(\mu, \sigma^2)$ the **sample variance** is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

4.5.1 The chi-square distribution

Definition 4-6 (Chi-square distribution)

If $Z_1, \ldots, Z_k$ is a random sample from $N(0,1)$, the distribution of $Z_1^2 + \cdots + Z_k^2$ is the **chi-square distribution with k degrees of freedom**:

$$Z_1^2 + \cdots + Z_k^2 \sim \chi^2(k).$$

Because each Z_i^2 is mostly near 0, the chi-square density is concentrated on the left with a long right tail. The upper quantile $\chi^2_\alpha(k)$ is defined by $P(V \geq \chi^2_\alpha(k)) = \alpha$ for $V \sim \chi^2(k)$; e.g. $\chi^2_{0.995}(4) = 0.21$, $\chi^2_{0.975}(4) = 0.48$, $\chi^2_{0.05}(4) = 9.49$.

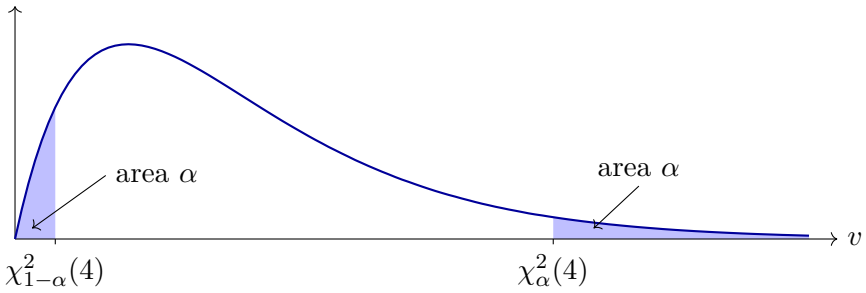


Figure 2: The $\chi^2(4)$ density $ve^{-v/2}/4$ with $\alpha = 0.05$: $\chi^2_{0.95}(4) = 0.71$, $\chi^2_{0.05}(4) = 9.49$.

Theorem 4-7 (Additivity of the chi-square distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, then $V_1 + V_2 \sim \chi^2(k_1 + k_2)$.

Proof. [Supplement]

Step 1. For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, substituting $u = z\sqrt{1-2t}$,

$$E(e^{tZ^2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(1-2t)z^2/2}\, dz = (1-2t)^{-1/2}.$$

Step 2. Hence, by independence, $V \sim \chi^2(k)$ has mgf $M_V(t) = (1-2t)^{-k/2}$, $t < \frac{1}{2}$; conversely any variable with this mgf has the $\chi^2(k)$ distribution (uniqueness of mgfs).

Step 3. For independent V_1, V_2, $M_{V_1+V_2}(t) = (1-2t)^{-k_1/2}(1-2t)^{-k_2/2} = (1-2t)^{-(k_1+k_2)/2}$, so $V_1 + V_2 \sim \chi^2(k_1 + k_2)$. ■

Theorem 4-8 (Distribution of the sample variance from a normal population)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with sample variance S^2,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Moreover $\bar{X}$ and S^2 are independent.

Proof. The book's argument: since $(X_i - \mu)/\sigma \sim N(0,1)$ independently, $\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(n)$. Adding and subtracting $\bar{X}$ (the cross term vanishes because $\sum(X_i - \bar{X}) = 0$),

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2,$$

and the last term is $\chi^2(1)$. "Subtracting" degrees of freedom gives $\chi^2(n-1)$. We make this rigorous.

[Supplement]

Step 1. Let $Z_i = (X_i - \mu)/\sigma$, independent $N(0,1)$, so $\mathbf{Z} = (Z_1, \dots, Z_n)^\top$ has joint density $(2\pi)^{-n/2} e^{-\|\mathbf{z}\|^2/2}$. Let A be an $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$ (complete this unit vector to an orthonormal basis), and $\mathbf{W} = A\mathbf{Z}$.

Step 2. The map $\mathbf{z} \mapsto A\mathbf{z}$ has Jacobian $|\det A| = 1$ and preserves $\|\mathbf{z}\|$, so $\mathbf{W}$ has the same density: $W_1, \dots, W_n$ are independent $N(0,1)$.

Step 3. $W_1 = \sqrt{n}\,\bar{Z}$ and $\sum_{i=1}^n W_i^2 = \sum_{i=1}^n Z_i^2$, hence

$$\sum_{i=2}^{n} W_i^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = \sum_{i=1}^{n} (Z_i - \bar{Z})^2 = \frac{(n-1)S^2}{\sigma^2}.$$

Step 4. By Definition 4-6 this is $\chi^2(n-1)$. It is a function of $W_2, \dots, W_n$, while $\bar{X} = \mu + \sigma W_1/\sqrt{n}$ is a function of W_1; so $\bar{X}$ and S^2 are independent. ■

Next take independent random samples $X_1, \dots, X_{n_1}$ from $N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2}$ from $N(\mu_2, \sigma^2)$ (common variance), with

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \qquad S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2.$$

Definition (Pooled sample variance)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

the weighted average of S_1^2, S_2^2 with weights $\frac{n_j - 1}{(n_1 - 1) + (n_2 - 1)}$; it estimates the common variance σ^2.

Theorem 4-9 (Sample variances from two normal populations with equal variance)

Under the above assumptions,

$$\frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2).$$

Proof. By Theorem 4-8, $(n_1 - 1)S_1^2/\sigma^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma^2 \sim \chi^2(n_2 - 1)$, and they are independent since they come from independent samples. Their sum is $(n_1 + n_2 - 2)S_p^2/\sigma^2$, which is $\chi^2(n_1 + n_2 - 2)$ by Theorem 4-7. ■

4.5.2 The t distribution

When σ is unknown we replace it by the sample standard deviation S and obtain the **studentized** sample mean $\frac{\bar{X} - \mu}{S/\sqrt{n}}$.

Definition 4-7 (t distribution)

If $Z \sim N(0,1)$ and $V \sim \chi^2(k)$ are independent, the distribution of

$$T = \frac{Z}{\sqrt{V/k}}$$

is the t **distribution with k degrees of freedom**, $T \sim t(k)$.

Like $N(0,1)$, $t(k)$ is symmetric about 0, but it has heavier tails; it is used especially for small samples. The upper quantile $t_\alpha(k)$ satisfies $P(T \geq t_\alpha(k)) = \alpha$; e.g. $t_{0.1}(4) = 1.533$, $t_{0.05}(4) = 2.132$, $t_{0.01}(4) = 3.747$.

Theorem 4-10 (Distribution of the studentized sample mean)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof. Dividing numerator and denominator by $\sigma/\sqrt{n}$,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\dfrac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}}.$$

The numerator is $N(0,1)$ (Theorem 4-5), the quantity under the root is $\chi^2(n-1)$ divided by $n-1$ (Theorem 4-8), and they are independent because $\bar{X}$ and S^2 are (Theorem 4-8; the book omits this proof). By Definition 4-7 the ratio is $t(n-1)$. ■

For two normal populations with equal variance, a short computation gives

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\dfrac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}}{\sqrt{\dfrac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} \Big/ (n_1 + n_2 - 2)}}.$$

Theorem 4-11 (t distribution for two normal populations with equal variance)

Under the assumptions of Theorem 4-9,

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

Proof.

Step 1. Numerator. $\bar{X} \sim N(\mu_1, \sigma^2/n_1)$ and $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$ are independent, so by Theorem 4-2 $\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \sigma^2(\frac{1}{n_1} + \frac{1}{n_2}))$, and the standardized numerator in the display is $N(0,1)$.

Step 2. Denominator. By Theorem 4-9, $(n_1 + n_2 - 2)S_p^2/\sigma^2 \sim \chi^2(n_1 + n_2 - 2)$.

Step 3. Independence. By Theorem 4-8, $\bar{X}$ is independent of S_1^2 and $\bar{Y}$ of S_2^2; as the two samples are independent, $(\bar{X}, \bar{Y})$ is independent of (S_1^2, S_2^2), hence of S_p^2.

Step 4. Definition 4-7 gives $t(n_1 + n_2 - 2)$. ■

4.5.3 The F distribution

Definition 4-8 (F distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, the distribution of

$$F = \frac{V_1/k_1}{V_2/k_2}$$

is the F **distribution with** (k_1, k_2) **degrees of freedom**, $F \sim F(k_1, k_2)$.

The F density is right-skewed on $(0, \infty)$. Its upper quantile $F_\alpha(k_1, k_2)$ satisfies $P(F \geq F_\alpha(k_1, k_2)) = \alpha$; e.g. $F_{0.10}(3, 4) = 4.19$, $F_{0.05}(3, 4) = 6.59$.

Theorem 4-12 (Property of the F distribution)

If $F \sim F(k_1, k_2)$, then $1/F \sim F(k_2, k_1)$. Consequently

$$F_{1-\alpha}(k_2, k_1) = \frac{1}{F_\alpha(k_1, k_2)}.$$

Proof.

Step 1. Write $F = \frac{V_1/k_1}{V_2/k_2}$ as in Definition 4-8. Then $1/F = \frac{V_2/k_2}{V_1/k_1}$ is a ratio of the same form with the roles of (V_1, k_1) and (V_2, k_2) exchanged, so $1/F \sim F(k_2, k_1)$.

Step 2. Let $F' = 1/F \sim F(k_2, k_1)$. Since $F > 0$, $\alpha = P(F \geq F_\alpha(k_1, k_2)) = P(F' \leq 1/F_\alpha(k_1, k_2))$, so $P(F' \geq 1/F_\alpha(k_1, k_2)) = 1-\alpha$ (F' is continuous). By definition of the upper quantile, $F_{1-\alpha}(k_2, k_1) = 1/F_\alpha(k_1, k_2)$. ■

For example, $F_{0.95}(4, 3) = 1/F_{0.05}(3, 4) = 1/6.59 = 0.15$.

For independent samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, $(n_1 - 1)S_1^2/\sigma_1^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma_2^2 \sim \chi^2(n_2 - 1)$ independently.

Theorem 4-13 (Ratio of sample variances from two normal populations)

Under these assumptions,

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1).$$

Proof. Put $V_j = (n_j - 1)S_j^2/\sigma_j^2 \sim \chi^2(n_j - 1)$ (Theorem 4-8); V_1, V_2 are independent. Then $\frac{V_1/(n_1-1)}{V_2/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$, which is $F(n_1 - 1, n_2 - 1)$ by Definition 4-8. ■

Theorem 4-14 (Relation between the F and t distributions)

If $T \sim t(k)$, then $T^2 \sim F(1, k)$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim N(0, 1)$, $V \sim \chi^2(k)$ independent. Then $T^2 = \frac{Z^2/1}{V/k}$, where $Z^2 \sim \chi^2(1)$ (Definition 4-6) and Z^2 is independent of V. By Definition 4-8, $T^2 \sim F(1, k)$. ■

4.6 Normal quantile plots

Motivation. Most inference in later chapters assumes a normal population. A normal quantile (Q–Q) plot is a quick graphical check of that assumption.

A q-quantile of a distribution divides its total area in the ratio $q : (1-q)$. A **normal quantile plot** (Q–Q plot) plots the quantiles of the standard normal distribution against the corresponding quantiles of the data.

Definition (Normal quantile plot)

Order the data $x_{(1)} < x_{(2)} < \cdots < x_{(n)}$. The value $x_{(i)}$ is roughly the $\frac{i}{n+1}$-quantile of the data, whose standard normal counterpart is $z_{1-i/(n+1)}$. Plot the points

$$\big(z_{1-i/(n+1)},\, x_{(i)}\big), \qquad i = 1, \ldots, n.$$

If the points lie close to a straight line, the normality assumption is reasonable. Software often uses $\frac{i-3/8}{n+1/4}$ instead of $\frac{i}{n+1}$.

For instance, with $n = 99$ observations from $N(2, 3^2)$, the population quantiles are $3z_{0.99} + 2$, $3z_{0.98} + 2, \ldots, 3z_{0.01} + 2$, a linear function of the standard normal quantiles, so the points $(z_{0.99}, x_{(1)}), \ldots, (z_{0.01}, x_{(99)})$ lie near a line with slope 3 and intercept 2.

Reading a Q–Q plot

Systematic departures from the line indicate a non-normal population:
(a) light tails: the extreme points bend toward the centre (an S-shape flatter than the line at both ends);
(b) heavy tails: the extreme points fall below the line on the left and above it on the right;
(c) long right tail: the points curve upward on the right;
(d) long left tail: the points curve downward on the left.

Chapter 5. Statistical Inference

5.1 Estimation

Motivation. Statistical inference draws conclusions about a population from a sample and attaches a probability-based measure of reliability to them. Its two basic forms are estimation and significance testing; this section explains estimation through the population mean.

The usual targets of inference are population characteristics such as the mean, standard deviation, proportion or correlation. Computing an estimate of such a characteristic from a sample *together with its margin of error* is called **estimation**.

5.1.1 Estimating a population mean

Illustration 5-1: tire lifetimes

A tire maker develops a new process expected to raise the mean tire life, while the standard deviation stays at the old value $\sigma = 5000$ km. A trial of $n = 100$ tires gives $\bar{x} = 38000$ km. The natural estimate of the mean life μ is $\hat{\mu} = \bar{x} = 38000$ km.

How reliable is this method? By the CLT, $\bar{X}$ is approximately $N(\mu, \sigma^2/100)$ with standard deviation $\sigma/\sqrt{n} = 5000/10 = 500$ km. Hence

(a) with probability about 0.95, $\bar{X}$ lies within $1.96\,\sigma/\sqrt{n} = 980$ of μ;

(b) the event $|\bar{X} - \mu| < 980$ is the same as $\mu \in (\bar{X} - 980, \bar{X} + 980)$;

(c) if the experiment were repeated indefinitely, about 95% of the intervals $(\bar{x} - 980, \bar{x} + 980)$ would contain μ.

For the observed sample this interval is $(37020, 38980)$. Either μ lies in it, or the sample is one of the unusual 5% for which $\bar{x}$ is more than 980 from μ. We cannot tell which, but the *method* is right 95 times in 100, so we state $37020 < \mu < 38980$ with **confidence level** 95%; the interval is a 95% **confidence interval**. In short, $\hat{\mu} = \bar{x} = 38000$ with $|\mu - \hat{\mu}| \le 1.96\,\sigma/\sqrt{n} = 980$.

Definition (Estimate, estimator, standard error)

The number $\hat{\mu} = \bar{x}$ is an **estimate** of μ, and $1.96\,\sigma/\sqrt{n}$ is its 95% **margin of error**. The statistic $\hat{\mu} = \bar{X}$ used for estimation is an **estimator** (the estimator relates to the estimate as a random variable to its observed value). The standard deviation of an estimator is its **standard error**:

$$\mathrm{SE}(\hat{\mu}) = \sqrt{\mathrm{Var}(\hat{\mu})} = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}.$$

For large n,

$$P\Big(-z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\Big) \approx P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1 - \alpha.$$

Definition (Estimating μ when σ is known)

(a) Estimator and standard error: $\hat{\mu} = \bar{X}$, $\mathrm{SE}(\hat{\mu}) = \sigma/\sqrt{n}$.
(b) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\mathrm{SE}(\hat{\mu}) = z_{\alpha/2}\,\sigma/\sqrt{n}$.
(c) $100(1-\alpha)\%$ confidence interval: $\Big(\bar{x} - z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\Big)$.

These statements are exact for a normal population and approximate (large n) otherwise. Common confidence levels are 90%, 95%, 99%, with $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.005} = 2.576$.

Illustration 5-2

From a normal population with $\sigma = 10$, a sample of $n = 25$ gives $\bar{x} = 40$. Then $\hat{\mu} = 40$, the 90% margin of error is $1.645 \cdot 10/\sqrt{25} = 3.29$, and the 90% confidence interval is (36.71, 43.29).

Meaning of the confidence level. In the setting of Illustration 5-2, twenty samples of size 25 give twenty intervals $\bar{x} \pm 3.29$ with different centres; in the book's simulation 18 of the 20 contain μ. A 90% confidence level means that, over repeated sampling, 90% of such intervals contain μ.

Example 5-1 (Cholesterol of low-income residents)

A survey of $n = 49$ low-income Central American residents found mean blood cholesterol $\bar{x} = 157.02$ mg/L. Assume cholesterol is approximately normal with $\sigma = 30$ mg/L. Find a 95% confidence interval for μ.
Solution. $\bar{x} \pm 1.96\,\sigma/\sqrt{n} = 157.02 \pm 1.96 \cdot 30/7 = 157.02 \pm 8.40$, i.e. (148.62, 165.42) mg/L.

At 99% the interval is $\bar{x} \pm 2.576\,\sigma/\sqrt{n} = (145.98, 168.06)$: higher confidence gives a longer interval, and excessively high confidence makes the interval uselessly wide. The length of a $100(1-\alpha)\%$ interval is

$$\left(\bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) - \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

twice the margin of error, so it can be controlled through n.

Definition (Sample size for estimating μ (σ known))

To make the $100(1-\alpha)\%$ margin of error at most d (interval length at most $2d$), take

$$n \geq \left(z_{\alpha/2}\frac{\sigma}{d}\right)^2.$$

(Indeed $z_{\alpha/2}\sigma/\sqrt{n} \leq d \iff \sqrt{n} \geq z_{\alpha/2}\sigma/d$.)

Example 5-2 (Required sample size)

In Example 5-1, how many residents must be surveyed for the 95% margin of error to be at most 5 mg/L?
Solution. With $\sigma = 30$, $n \geq (1.96 \cdot 30/5)^2 = 138.2976$, so at least 139 residents are needed.

5.1.2 Parameters, point and interval estimation

In practice σ is rarely known; the general case is treated in Chapter 6. A numerical summary of a population, such as μ or σ, is a **parameter**, generically θ. An estimator of θ based on $X_1, \ldots, X_n$ is written $\hat{\theta}(X_1, \ldots, X_n)$, or simply $\hat{\theta}$ (also for the estimate).

Definition (Confidence interval for a parameter)

If statistics $L = L(X_1, \ldots, X_n)$ and $U = U(X_1, \ldots, X_n)$ satisfy

$$P(L < \theta < U) = 1 - \alpha,$$

then (L, U) is a $100(1-\alpha)\%$ confidence interval for θ. The confidence level has the same repeated-sampling meaning as before.

Illustration 5-3: estimating σ and σ^2

The usual estimator of σ is the sample standard deviation, and that of σ^2 the sample variance:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2} = S, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = S^2.$$

Giving only an estimate of the parameter is **point estimation**; giving a confidence interval is **interval estimation**. Together they constitute **estimation of parameters**.

Example 5-3 (Thickness of plastic sheets)

The variability of thickness is a key quality characteristic. Ten sheets measured (mm):

226	228	226	225	232	228	227	229	225	230

Estimate the population standard deviation σ.
Solution. $\bar{x} = \frac{1}{10}(226 + 228 + \cdots + 230) = 227.6$ mm and

$$s^2 = \frac{1}{10-1}\{(226-227.6)^2 + (228-227.6)^2 + \cdots + (230-227.6)^2\} = 5.1556,$$

so $\hat{\sigma} = s = \sqrt{5.1556} = 2.27$ mm.

Other estimators exist (e.g. the trimmed mean for μ, more stable when there are outliers), but this book uses $\bar{X}$ and S.

5.2 Significance tests and hypothesis tests

Motivation. Sometimes the question is not "how large is μ?" but "do the data give clear evidence against a stated claim?". Significance testing measures the strength of such evidence by a probability, the P-value.

5.2.1 Null and alternative hypotheses

Illustration 5-4: testing Mendel's law

A scientist crossed genetically modified peas and observed in the second generation 860 round-yellow, 340 round-green, 240 wrinkled-yellow and 160 wrinkled-green peas. Mendel's theory predicts the ratio $9:3:3:1$. Do the data strongly suggest a new theory?

Science advances by finding strong evidence against established theories. Here the claim under scrutiny is the **hypothesis** "the four types occur in ratio $9:3:3:1$".

Definition (Null and alternative hypotheses; significance test)

A hypothesis set up in order to look for evidence *against* it is the **null hypothesis** H_0; the hypothesis proposed as the alternative to H_0 is the **alternative hypothesis** H_1. A **significance test** evaluates the strength of the evidence against H_0. Its logic resembles proof by contradiction: if H_0 were true, the observed data would be very unlikely.

Illustration 5-5: sprinkler temperature

Sprinklers are designed to activate at 55°C. The process is abnormal if the mean activation temperature μ satisfies $\mu < 55$ or $\mu > 55$. Since we seek clear evidence of abnormality,

$$H_0 : \mu = 55, \qquad H_1 : \mu \neq 55.$$

Illustration 5-6: paint drying time

A paint dries in 75 minutes on average ($\sigma = 9.4$). A lab claims a new additive shortens the drying time, and the company will adopt it only on clear evidence. With μ the mean drying time with the additive,

$$H_0 : \mu = 75, \qquad H_1 : \mu < 75.$$

An alternative containing values on both sides of the reference value (as $\mu \neq 55$) is **two-sided**; one containing one side only (as $\mu < 75$) is **one-sided**. The claim we want to establish from the data goes into H_1.

Choice of the null value

With $H_1 : \mu < 75$ the logical null is its negation $H_0 : \mu \geq 75$. Using instead the boundary case $H_0 : \mu = 75$, the value hardest to distinguish from H_1, does not change the testing procedure, so the null is stated at the boundary value.

Evidence for H_1 is evaluated through an estimator of the parameter: the farther the estimate lies from the value specified by H_0 in the direction of H_1, the stronger the evidence.

Illustration 5-7

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, the evidence is sought in $\bar{X}$; since H_1 is two-sided, $\bar{X}$ far from 55 in *either* direction supports H_1.

Illustration 5-8

For $H_0 : \mu = 75$, $H_1 : \mu < 75$ (left-sided), values of $\bar{X}$ far *below* 75 support H_1.

A statistic such as $\bar{X}$ used in a test is a **test statistic**.

5.2.2 The P-value

Illustration 5-9: P-value for the sprinklers

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, $n = 9$ sprinklers gave $\bar{x} = 55.63$°C; assume normality with $\sigma = 0.9$. The observed deviation is $\bar{x} - 55 = 0.63$. The probability, under H_0, of evidence at

least as strong as observed is

$$P\Big(\Big|\frac{\bar{X}-55}{\sigma/\sqrt{n}}\Big| \geq 2.1 \;\Big|\; \mu = 55\Big) = P(|Z| \geq 2.1) = 0.0357,$$

since $0.63/(0.9/3) = 2.1$. Repeating the survey, stronger evidence against H_0 would arise fewer than 4 times in 100 if H_0 were true; the data strongly suggest $\mu \neq 55$.

Definition 5-1 (Significance probability (P-value))

The ***P*-value** is the probability, computed under H_0, that the test statistic takes a value at least as extreme as the observed value in the direction supporting H_1. The smaller the P-value, the stronger the evidence against H_0.

Illustration 5-10: P-value for the paint

For $H_0 : \mu = 75$, $H_1 : \mu < 75$, $n = 100$ samples with the additive gave $\bar{x} = 73.5$ ($\sigma = 9.4$). By the CLT, $\bar{X} \approx N(\mu, 9.4^2/100)$, so

$$P = P(\bar{X} \leq 73.5 \mid \mu = 75) = P\Big(\frac{\bar{X}-75}{9.4/\sqrt{100}} \leq -1.5957 \;\Big|\; \mu = 75\Big) \approx P(Z \leq -1.5957) = 0.0553.$$

The data fairly strongly suggest a shorter drying time; the strength of evidence is about 5.5%.

5.2.3 Significance level

"Strong" is relative, so a reference value α, the **significance level**, is often fixed in advance and the P-value compared with it. Usually $\alpha = 0.05$ or 0.01. With $\alpha = 0.05$, a P-value ≤ 0.05 is reported as "statistically significant ($P \leq 0.05$)".

Definition 5-2 (Statistical significance)

If the P-value is at most the significance level α, the result is **statistically significant at level** α; the evidence against H_0 is stronger than the prescribed level.

Definition (Procedure of a significance test)

(a) State H_0 and H_1.
(b) [Optional] Fix the significance level α.
(c) Compute the observed value of the test statistic (based on an estimator of the relevant parameter).
(d) Compute the P-value from the sampling distribution of the test statistic under H_0; it measures the evidence against H_0.
(e) [Optional] Compare the P-value with α: if $P \leq \alpha$ the result is significant.

Steps (c) and (d) are mechanical (tables or software), but H_0, H_1 and the interpretation are the investigator's responsibility.

5.2.4 Tests for μ with σ known

For $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ the standardized statistic and its observed value are

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}},$$

and $P = P(\bar{X} \le \bar{x} \mid \mu = \mu_0) \approx P(Z \le z)$ (in Illustration 5-10, $z = -1.5957$ and $P = 0.0553$). For $H_1 : \mu > \mu_0$, $P \approx P(Z \ge z)$; for $H_1 : \mu \ne \mu_0$, deviations of $\bar{X}$ from μ_0 in both directions count, so

$$P(\bar{X} - \mu_0 \le -|\bar{x} - \mu_0| \text{ or } \bar{X} - \mu_0 \ge |\bar{x} - \mu_0| \mid \mu = \mu_0) \approx P(|Z| \ge |z|).$$

Definition (Significance test for μ (σ known))

For $H_0 : \mu = \mu_0$ use $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$. With observed value z, the P-value is
(a) $P = P(Z \ge z)$ if $H_1 : \mu > \mu_0$;
(b) $P = P(Z \le z)$ if $H_1 : \mu < \mu_0$;
(c) $P = P(|Z| \ge |z|)$ if $H_1 : \mu \ne \mu_0$.

The P-value is exact for a normal population and approximate for large n otherwise.

Example 5-4 (New tire process)

Current tires last on average 37,000 km with standard deviation 5,000 km. A trial of 100 tires from a new process gives $\bar{x} = 38000$ km. Assuming $\sigma = 5000$ is unchanged, do the data show that the new process is successful?
Solution. We want to establish $\mu > 37000$: $H_0 : \mu = 37000$, $H_1 : \mu > 37000$. The observed statistic is

$$z = \frac{38000 - 37000}{5000/\sqrt{100}} = 2.$$

Z is approximately standard normal ($n = 100$), and H_1 is right-sided, so $P = P(Z \ge 2) = 1 - 0.9772 = 0.0228$. The data strongly suggest that the new process increases tire life.

Example 5-5 (Concentration of an active ingredient)

A drug should contain 0.85 g/L of an active ingredient; repeated measurements on one specimen are $N(\mu, 0.0123^2)$. Three measurements of a specimen from a new method average $\bar{x} = 0.8404$ g/L. Is there evidence that $\mu \ne 0.85$?
Solution. $H_0 : \mu = 0.85$, $H_1 : \mu \ne 0.85$, and

$$z = \frac{0.8404 - 0.85}{0.0123/\sqrt{3}} = -1.35.$$

The population is normal, so $Z \sim N(0,1)$ exactly, and $P = P(|Z| \ge 1.35) = 2P(Z \le -1.35) = 2 \times 0.0885 = 0.177$. A P-value above 17% gives no clear evidence that the true concentration differs from 0.85 g/L.

5.2.5 Rejection regions and hypothesis testing

If the required strength of evidence α is fixed beforehand, the P-value need not be computed.

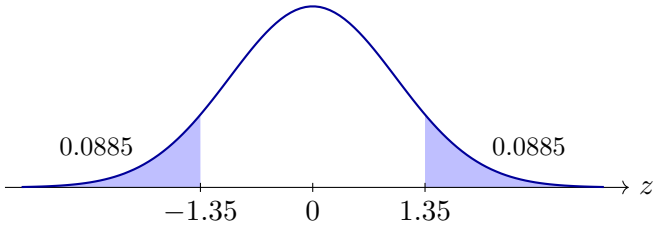


Figure 3: Two-sided P-value in Example 5-5: $P = P(|Z| \geq 1.35) = 0.177$.

Illustration 5-11

In Example 5-4 ($H_0 : \mu = 37000$, $H_1 : \mu > 37000$) take $\alpha = 0.05$. We must decide whether $P(Z \geq z) \leq 0.05$. Two equivalent ways:
(a) compute the P-value from $z = 2$ as in Example 5-4;
(b) compare z with the upper 5% point $z_{0.05} = 1.645$. Since $z = 2 > 1.645$, i.e. $z \geq 1.645$ holds, the result is significant at 5%. Had $z < 1.645$, it would not be.

Definition (Rejection region; hypothesis test)

For a fixed level α, the set of values of the test statistic for which significance at level α is declared (e.g. $z \geq 1.645$) is the **rejection region** of H_0. The conclusion is stated as "H_0 is rejected at level α" or "H_0 cannot be rejected at level α". Deciding in this way is called **hypothesis testing**:
(a) find the rejection region corresponding to α;
(b) compute the observed value of the test statistic;
(c) check whether it lies in the rejection region.

Definition (Hypothesis test for μ (σ known, level α))

For $H_0 : \mu = \mu_0$ with $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ and observed value z, the rejection region is
(a) $z \geq z_\alpha$ if $H_1 : \mu > \mu_0$;
(b) $z \leq -z_\alpha$ if $H_1 : \mu < \mu_0$;
(c) $|z| \geq z_{\alpha/2}$ if $H_1 : \mu \neq \mu_0$.

Frequently $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

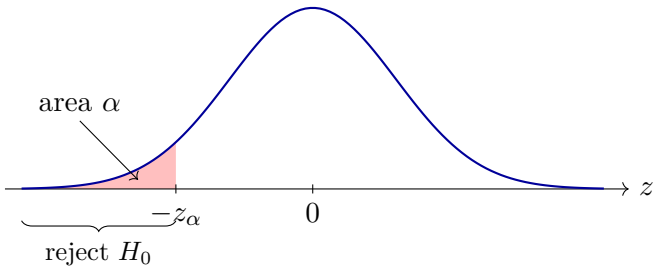


Figure 4: Rejection region $z \leq -z_\alpha$ for $H_1 : \mu < \mu_0$.

Example 5-6 (A faster pain reliever)

An existing painkiller takes on average 30 minutes to act ($\sigma = 5$). For a new painkiller, 50 randomly chosen patients gave $\bar{x} = 28.5$ minutes; assume normality with $\sigma = 5$. Test at level

5% whether the new drug acts faster.
Solution. $H_0 : \mu = 30$, $H_1 : \mu < 30$. The rejection region at 5% is

$$z = \frac{\bar{x} - 30}{5/\sqrt{50}} \le -z_{0.05} = -1.645.$$

Here $z = \frac{28.5-30}{5/\sqrt{50}} = -2.12$ lies in the rejection region, so H_0 is rejected at 5%: the evidence that the new drug is faster exceeds the prescribed level.

Example 5-7 (Sprinklers at level 1%)

In Illustrations 5-5 and 5-9 ($n = 9$, $\bar{x} = 55.63$, $\sigma = 0.9$), test $H_0 : \mu = 55$ against $H_1 : \mu \neq 55$ at level 1%.
Solution. $z = \frac{55.63-55}{0.9/\sqrt{9}} = 2.1$, and the rejection region is $|z| \ge z_{0.005} = 2.576$. Since $2.1 < 2.576$, H_0 cannot be rejected at 1%: the evidence of a process abnormality is not stronger than the 1% level.

The conclusion depends on α. In Example 5-6, $P = P(Z \le -2.12) = 0.0170$: rejected at $\alpha = 0.05$ but not at $\alpha = 0.01$. In Example 5-7, $P = P(|Z| \ge 2.1) = 0.0357$: not rejected at 0.01 but rejected at 0.05. The P-value reports the evidence without fixing α; when a decision is required, α must be chosen with the practical context in mind.

5.2.6 Two kinds of error

A level-α test is a rule for choosing between H_0 and H_1, and it can err in two ways.

Decision \ Truth	H_0 true	H_1 true
Do not reject H_0 (choose H_0)	correct	Type II error
Reject H_0 (choose H_1)	Type I error	correct

Illustration 5-12: Type I error probability

In Example 5-6 the level-α rejection region is $\frac{\bar{X}-30}{5/\sqrt{50}} \le -z_\alpha$, so

$$P(\text{Type I error}) = P\Big(\frac{\bar{X} - 30}{5/\sqrt{50}} \le -z_\alpha \;\Big|\; \mu = 30\Big) = P(Z \le -z_\alpha) = \alpha.$$

In general a test with the level-α rejection region has Type I error probability at most α (with $H_0 : \mu \ge \mu_0$, the probability is largest at the boundary $\mu = \mu_0$). Fixing α thus sets the tolerated Type I error rate.

Illustration 5-13: Type II error probability

In Example 5-6, a true mean of 28 minutes would be a notable improvement. If $\mu = 28$, a Type II error occurs when H_0 is not rejected:

$$\begin{aligned} P\Big(\frac{\bar{X} - 30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) &= P\Big(\frac{\bar{X} - 28}{5/\sqrt{50}} + \frac{28 - 30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) \\ &= P(Z > 2\sqrt{2} - 1.645) = P(Z > 1.183) \approx 0.1190. \end{aligned}$$

Just as α bounds the Type I error, one may require the Type II error probability at a specified alternative to be at most β; this is achieved by choosing n.

Example 5-8 (Sample size for given α and β)

In Example 5-6, find n such that the level-0.05 test has Type II error probability at most $\beta = 0.1$ when $\mu = 28$.
Solution. As in Illustration 5-13,

$$P\Big(\frac{\bar{X}-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big) = P\Big(\frac{\bar{X}-28}{5/\sqrt{n}} + \frac{28-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big)$$
$$= P\Big(Z > \frac{30-28}{5/\sqrt{n}} - 1.645\Big) \le 0.1.$$

Since $z_{0.1} = 1.282$, this requires $\frac{30-28}{5/\sqrt{n}} - 1.645 \ge 1.282$, i.e.

$$n \ge \Big\{\frac{1.645 + 1.282}{(30-28)/5}\Big\}^2 = 53.546.$$

The required sample size is $n = 54$.

Definition (Sampling inspection; power)

Choosing n and the rejection region so that the Type I error probability is at most α and the Type II error probability at $\mu = \mu_1$ is at most β is called **sampling inspection** (widely used in statistical quality control). The probability of *not* making a Type II error when $\mu = \mu_1$, i.e. of rejecting H_0, is the **power** at $\mu = \mu_1$.

Illustration 5-14: power

For the 5% test of Illustration 5-13, the power at $\mu = 28$ is

$$P\Big(\frac{\bar{X}-30}{5/\sqrt{50}} \le -1.645 \;\Big|\; \mu = 28\Big) = 1 - 0.1190 = 0.8810.$$

Conditions for valid inference

For the methods of this chapter to be valid:
(a) the sample must be drawn at random;
(b) the assumptions on the population distribution must be checked.

Chapter 6. Inference about Distributions

6.1 Inference about a population mean

Motivation. Chapter 5 assumed the population standard deviation σ known, which is rarely the case. Replacing σ by the sample standard deviation S leads to the t distribution and to the t interval and t test for a normal mean.

With σ known we used $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. When σ is unknown it is replaced by its estimator S; this is called **studentizing**. For a random sample from $N(\mu,\sigma^2)$, Theorem 4-10 gives

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Hence

$$P\Big(-t_{\alpha/2}(n-1) \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{\alpha/2}(n-1)\Big) = 1-\alpha, \quad \text{i.e.}$$
$$P\Big(\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}\Big) = 1-\alpha.$$

Definition (Confidence interval for μ (normal population, σ unknown))

A $100(1-\alpha)\%$ confidence interval for μ is

$$\Big(\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}\Big).$$

For large n the normality assumption can be relaxed: by the CLT and since $t(n-1)$ approaches $N(0,1)$, the quantile $t_{\alpha/2}(n-1)$ may be replaced by $z_{\alpha/2}$.

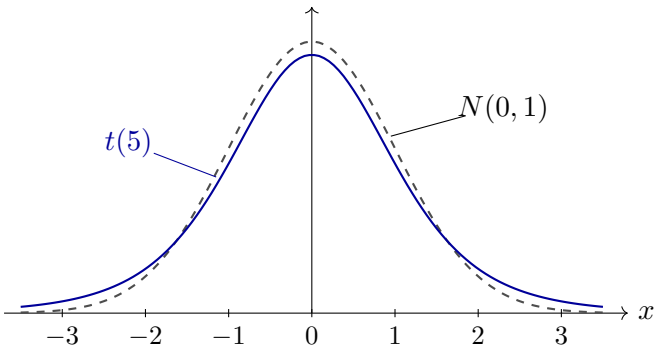


Figure 5: The $t(5)$ density (solid) has heavier tails than $N(0,1)$ (dashed).

Example 6-1 (Trees per 100 m^2)

A forest manager counted trees in 50 randomly chosen 100 m^2 plots of a 9 km^2 forest:

43	28	23	18	46	39	40	43	36	25
26	28	20	22	17	26	17	12	16	35
37	12	32	47	20	25	13	20	46	23
37	29	23	37	34	27	33	29	24	19
37	23	37	28	27	35	30	26	33	35

Find a 99% confidence interval for the mean number μ of trees per $100\,\mathrm{m}^2$.
Solution. With $n = 50$,

$$\bar{x} = \frac{43 + 28 + \cdots + 35}{50} = 28.76, \qquad s = \sqrt{\frac{(43 - 28.76)^2 + \cdots + (35 - 28.76)^2}{49}} = 9.146.$$

The normal quantile plot is close to a line, so a normal population is reasonable. The table lacks 49 degrees of freedom; using the nearest value $t_{0.005}(50) = 2.678$,

$$\bar{x} \pm t_{0.005}(49)\frac{s}{\sqrt{n}} \approx 28.76 \pm 2.678 \times \frac{9.146}{\sqrt{50}} = 28.76 \pm 3.464,$$

so the 99% confidence interval is (25.296, 32.223).

Testing with the studentized mean is the t **test**.

Definition (t test for μ (normal population, σ unknown))

For $H_0 : \mu = \mu_0$ use $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, which is $t(n-1)$ under H_0. With observed value t:

Alternative	P-value	Rejection region at level α
(a) $H_1 : \mu > \mu_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
(b) $H_1 : \mu < \mu_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
(c) $H_1 : \mu \neq \mu_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

(The rejection regions have Type I error exactly α because $T \sim t(n-1)$ when $\mu = \mu_0$; e.g. in (c), $P(|T| \geq t_{\alpha/2}(n-1)) = 2 \cdot \frac{\alpha}{2}$ by symmetry.)

Example 6-2 (Light bulb lifetimes)

Current bulbs last 1950 hours on average. Nine prototypes of a new bulb lasted (hours)

2000	1975	1900	2000	1950	1850	1950	2100	1975

(a) State suitable hypotheses.
(b) Assuming normal lifetimes, test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 1950$, $H_1 : \mu > 1950$.
(b) $\bar{x} = 1966.7$, $s = 69.6$, so $t = \dfrac{1966.7 - 1950}{69.6/\sqrt{9}} = 0.718$. The rejection region is $t \geq t_{0.05}(8) = 1.860$; since $0.718 < 1.860$, H_0 is not rejected at 5%.
(c) $P = P(T \geq 0.718)$, $T \sim t(8)$. From the table, 0.718 lies between the upper 25% point (0.706) and 20% point (0.889), so $0.20 < P < 0.25$; software gives $P = 0.247$. The evidence that the new bulb lasts longer is quite weak.

The normal quantile plot of these data looks roughly linear, but with so few observations such a check has limited power; one may instead rely on subject knowledge, or use the nonparametric methods of Chapter 10.

Example 6-3 (Cholesterol of immigrant women)

The national mean cholesterol for women is 190 mg/dL. For 100 randomly selected women who immigrated in the past year, $\bar{x} = 178.2$ and $s = 45.3$.
(a) State suitable hypotheses.
(b) Test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 190$, $H_1 : \mu \neq 190$.
(b) $t = \dfrac{178.2 - 190}{45.3/\sqrt{100}} = -2.60$. With $n = 100$ the t test applies without assuming normality, and $z_{0.025} = 1.96$ can replace the $t(99)$ quantile: the rejection region is $|t| \geq 1.96$. Since $|t| = 2.60 \geq 1.96$, H_0 is rejected at 5%.
(c) $P = P(|T| \geq 2.60) \approx 2{\times}0.0047 = 0.0094$ from the normal table; software gives $P = 0.0107$. The mean cholesterol of immigrant women differs significantly from the national mean, so follow-up measures (lifestyle changes, medication) may be recommended.

6.2 Comparing means by paired comparison

Motivation. Many questions compare two populations or treatments. When the units are heterogeneous, applying both treatments to each unit (or to matched pairs) removes the unit-to-unit variation, and the comparison reduces to a one-sample problem for the differences.

Comparing two means by observing pairs and analysing the within-pair differences is called **paired comparison**.

Illustration 6-1: two sleeping pills

Patients' health varies greatly, so each of 6 randomly chosen patients took both pills A and B once, in an order decided by a coin toss, and sleeping hours were recorded:

Patient	1	2	3	4	5	6
Pill A	4.8	4.0	5.8	4.9	5.3	7.4
Pill B	4.0	4.2	5.2	4.9	5.6	7.1

What can be inferred about $\mu_A - \mu_B$?

Illustration 6-2: wearable walking robot

To see whether a gait-assist robot increases exercise effect, 15 rehabilitation patients each walked 10 minutes with and without the robot; calories burned (kcal):

Patient	1	2	3	4	5	6	7	8
Without	23.9	23.9	20.0	19.2	22.2	11.8	21.7	27.7
With	25.2	28.2	24.9	20.2	19.7	16.9	22.6	20.8
Patient	9	10	11	12	13	14	15	
Without	15.6	22.5	23.2	19.3	15.2	10.0	11.2	
With	23.8	18.3	23.7	22.5	16.2	12.1	22.5	

In both illustrations the objects receiving the treatments (patients) are **experimental units**; the factors applied (pills, robot) are **treatments**; the change they produce is the **treatment**

effect. Pairing homogeneous units sharpens the comparison of two treatments. With data $(X_1, Y_1), \ldots, (X_n, Y_n)$, $\mu_1 = E(X_i)$, $\mu_2 = E(Y_i)$, form

$$D_i = X_i - Y_i, \qquad E(D_i) = \mu_1 - \mu_2, \qquad i = 1, \ldots, n,$$

and apply §6.1 to the sample $D_1, \ldots, D_n$ with mean $\bar{D}$ and standard deviation S_D.

Definition (Paired comparison of two means)

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\Big(\bar{d} - t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}},\ \bar{d} + t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}\Big).$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ the test statistic is $T = \dfrac{\bar{D} - \delta_0}{S_D/\sqrt{n}}$; with observed value t:

Alternative	P-value	Rejection region at level α
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

These are exact when $D_1, \ldots, D_n$ is a random sample from a normal distribution (Theorem 4-10 applied to the D_i), and approximate for large n. To meet the assumptions, the experimental units should be selected at random and the order of treatments randomized (e.g. by a coin toss): **randomization**.

Example 6-4 (Sleeping pills: confidence interval)

Using Illustration 6-1, find a 95% confidence interval for $\mu_A - \mu_B$.
Solution. The differences A−B are 0.8, −0.2, 0.6, 0.0, −0.3, 0.3, so $\bar{d} = 0.20$ and $s_D = 0.443$. With $t_{0.025}(5) = 2.571$,

$$\bar{d} \pm t_{0.025}(5)\frac{s_D}{\sqrt{n}} = 0.20 \pm 2.571 \times \frac{0.443}{\sqrt{6}} = 0.20 \pm 0.46,$$

so the 95% confidence interval is $(-0.26, 0.66)$.

A normal quantile plot can check normality here, but more important is randomization in planning and observation.

Example 6-5 (Walking robot: test)

Using Illustration 6-2:
(a) test at level 1% whether wearing the robot increases calorie consumption;
(b) find the P-value of (a).
Solution.
(a) With μ_1, μ_2 the mean calories without and with the robot, $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$. The $n = 15$ differences (without − with) are

$$-1.3,\ -4.3,\ -4.9,\ -1.0,\ 2.5,\ -5.1,\ -0.9,\ 6.9,\ -8.2,\ 4.2,\ -0.5,\ -3.2,\ -1.0,\ -2.1,\ -11.3,$$

with $\bar{d} = -2.01$ and $s_D = 4.58$, so $t = \dfrac{-2.01 - 0}{4.58/\sqrt{15}} = -1.70$. Since $t_{0.01}(14) = 2.624$, the rejection region is $t \leq -2.624$; as $-1.70 > -2.624$, H_0 is not rejected at 1%.
(b) $P = P(T \leq -1.70)$, $T \sim t(14)$: the table gives $0.05 < P < 0.1$, and software $P = 0.056$.

At the 1% level there is no clear evidence that the robot increases calorie consumption.

A normal quantile plot of these differences supports the normality assumption.

6.3 Comparing means with two independent samples

Motivation. When the same unit cannot receive both treatments, the units are split into two groups (or two populations are sampled separately). Inference about $\mu_1 - \mu_2$ then rests on the difference of two independent sample means.

Definition (Data structure for two-sample comparison)

(a) The observations in each group are a random sample from that group's population.
(b) Observations in different groups are independent.

Random samples from two populations satisfy this automatically; when one pool of units receives two treatments, the units must be split into the groups at random (randomization).

Label the populations $1, 2$ with means μ_1, μ_2 and standard deviations σ_1, σ_2, and let $X_{11}, \ldots, X_{1n_1}$ and $X_{21}, \ldots, X_{2n_2}$ be the samples, with

$$\bar{X}_1 = \frac{X_{11} + \cdots + X_{1n_1}}{n_1}, \quad S_1 = \sqrt{\frac{1}{n_1 - 1}\sum_{j=1}^{n_1}(X_{1j} - \bar{X}_1)^2},$$

and $\bar{X}_2, S_2$ likewise. We estimate $\mu_1 - \mu_2$ by $\widehat{\mu_1 - \mu_2} = \bar{X}_1 - \bar{X}_2$. By independence,

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \text{so} \qquad \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

is the standardized difference, which is exactly $N(0,1)$ for normal populations (Theorem 4-2). With σ_1^2, σ_2^2 unknown we use the studentized difference

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}.$$

6.3.1 Large samples

When n_1, n_2 are both large, $S_1^2 \approx \sigma_1^2$ and $S_2^2 \approx \sigma_2^2$, so T is approximately $N(0,1)$ by the CLT.

Definition (Two-sample comparison of means (large n_1, n_2))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$, $T \dot{\sim} N(0,1)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq z_\alpha$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -z_\alpha$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq z_{\alpha/2}$

Example 6-6 (Tennis rackets and tennis elbow)

A new racket is designed to reduce the stress on the elbow. Of 220 intermediate players, 100 were randomly assigned the new racket and 120 the old one; the force on the elbow at impact in a forehand stroke (kgf) was:

	n_i	$\bar{x}_i$	s_i
New racket	100	25.2	5.1
Old racket	120	27.9	5.6

(a) Test at level 5% whether the mean forces differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. The samples are large, and

$$t = \frac{(25.2 - 27.9) - 0}{\sqrt{5.1^2/100 + 5.6^2/120}} = -3.74.$$

The rejection region is $|t| \geq z_{0.025} = 1.96$; since $3.74 \geq 1.96$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 3.74)$ with $T \mathrel{\dot\sim} N(0,1)$: $P < 0.0006$ from the table, and $P = 0.00018$ by software.
(c) $(25.2 - 27.9) \pm 1.96\sqrt{5.1^2/100 + 5.6^2/120} = -2.7 \pm 1.42$, i.e. $(-4.12, -1.28)$.

6.3.2 Normal populations: Welch's approximation

Even when n_1, n_2 are not large, T is approximately a t distribution with suitably chosen degrees of freedom. For normal populations with $n_1, n_2 \geq 5$,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \mathrel{\dot\sim} t(df), \qquad df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2}.$$

Definition (Two-sample comparison of means (normal populations))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(df)\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

with df computed from s_1^2, s_2^2 by the formula above.
(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \mathrel{\dot\sim} t(df)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(df)$

The normality assumption matters little once n_1, n_2 are moderately large, so these methods may be used unless the Q–Q plots strongly contradict normality. Since df is generally not an integer, use the nearest tabulated value, interpolate, or use software.

Example 6-7 (Grocery spending offline vs. online)

An economic psychologist claims that grocery spending is lower offline than online. Randomly chosen 18 and 20 people shopped two weeks' groceries for a family of four offline and online, respectively (thousand won):

Offline	169	206	257	294	252	283	240	207	230	183
	298	269	256	277	300	126	318	184		
Online	289	334	278	268	336	438	388	388	394	394
	425	386	356	342	305	365	355	312	209	458

(a) Test at level 5% whether offline spending is lower.
(b) Find the P-value of (a).
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$ (1 = offline, 2 = online). Summary: $n_1 = 18$, $\bar{x}_1 = 241.6$, $s_1 = 52.82$; $n_2 = 20$, $\bar{x}_2 = 351.0$, $s_2 = 61.92$. The Q–Q plots of both groups are close to lines. Then

$$t = \frac{(241.6 - 351.0) - 0}{\sqrt{52.82^2/18 + 61.92^2/20}} = -5.875, \qquad df = \frac{\left(\frac{52.82^2}{18} + \frac{61.92^2}{20}\right)^2}{\frac{1}{17}\left(\frac{52.82^2}{18}\right)^2 + \frac{1}{19}\left(\frac{61.92^2}{20}\right)^2} = 35.91.$$

The rejection region is $t \leq -t_{0.05}(35.91)$, where $t_{0.05}(35.91) \approx 1.620$ by interpolation in the table (1.688 by software). In either case $t = -5.875$ lies in the rejection region, so H_0 is rejected at 5%.
(b) $P = P(T \leq -5.875)$, $T \sim t(35.91)$; the table gives $P < 0.0005$. There is clear evidence that offline grocery spending is lower.

6.3.3 Normal populations with a common variance

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, both S_1^2 and S_2^2 estimate σ^2, and combining them is more efficient. Weighting by degrees of freedom gives the **pooled sample variance**

$$S_p^2 = \frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} S_1^2 + \frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} S_2^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

with $n_1 + n_2 - 2$ degrees of freedom (Theorem 4-9).

	Population 1	Population 2	Pooled
Variance	σ^2	σ^2	
Sample variance	S_1^2	S_2^2	S_p^2
Degrees of freedom	$n_1 - 1$	$n_2 - 1$	$n_1 + n_2 - 2$

Under a common variance the standardized difference is $\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+1/n_2}}$; replacing σ by S_p,

$$T=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}\sim t(n_1+n_2-2)$$

for normal populations, by Theorem 4-11.

Definition (Two-sample comparison (normal populations, common variance))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1-\mu_2$ is

$$(\bar{x}_1-\bar{x}_2)\pm t_{\alpha/2}(n_1+n_2-2)\,s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}.$$

(b) For $H_0:\mu_1-\mu_2=\delta_0$ use $T=\dfrac{(\bar{X}_1-\bar{X}_2)-\delta_0}{S_p\sqrt{1/n_1+1/n_2}}$; with observed value t:

Alternative	P-value	Rejection region
$H_1:\mu_1-\mu_2>\delta_0$	$P(T\geq t)$	$t\geq t_\alpha(n_1+n_2-2)$
$H_1:\mu_1-\mu_2<\delta_0$	$P(T\leq t)$	$t\leq -t_\alpha(n_1+n_2-2)$
$H_1:\mu_1-\mu_2\neq\delta_0$	$P(\lvert T\rvert\geq\lvert t\rvert)$	$\lvert t\rvert\geq t_{\alpha/2}(n_1+n_2-2)$

This is the **pooled two-sample t test**; the previous one, with adjusted degrees of freedom and no common-variance assumption, is the **unpooled two-sample t test**.

Pooled or unpooled?

The pooled test requires equal variances, and its validity deteriorates badly if they differ, especially when n_1 and n_2 are very different. Equal variances are hard to verify from box plots or histograms, so the unpooled test should also be examined; use the pooled test only when confident of a common variance.

Example 6-8 (Semaglutide and weight loss)

Twelve similar volunteers were split at random into two groups of 6: one received weekly semaglutide injections, the other a placebo, for 12 weeks with diet and exercise controlled. Weight loss (kg):

Semaglutide	5.12	4.87	5.64	5.01	5.49	5.25
Placebo	3.62	3.98	4.14	3.25	4.42	3.60

(a) Test at level 5% whether the mean weight losses differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0:\mu_1=\mu_2$, $H_1:\mu_1\neq\mu_2$. Summary: $n_1=n_2=6$, $\bar{x}_1=5.23$, $s_1=0.29$; $\bar{x}_2=3.84$, $s_2=0.42$. The Q–Q plots support normality and the standard deviations are similar, so we assume a common variance:

$$s_p^2=\frac{(6-1)\times 0.29^2+(6-1)\times 0.42^2}{6+6-2}=0.13025,\qquad s_p=\sqrt{0.13025}=0.361,$$

$$t=\frac{(5.23-3.84)-0}{0.361\sqrt{\frac{1}{6}+\frac{1}{6}}}=6.67.$$

The rejection region is $|t| \geq t_{0.025}(10) = 2.228$; since $6.67 > 2.228$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 6.67)$, $T \sim t(10)$: $P < 0.001$. There is strong evidence that the mean weight losses differ.
(c) $(5.23 - 3.84) \pm 2.228 \times 0.361 \times \sqrt{\frac{1}{6} + \frac{1}{6}} = 1.39 \pm 0.46$, i.e. (0.93, 1.85).

The two-sample t tests and intervals are not very sensitive to the normality assumption and are approximately valid for large samples. What really matters is that the experiment was randomized.

6.4 Inference about population variances

Motivation. Variability itself is often the target: a process producing parts of the right average size but with large variation still produces many defects. For normal populations, inference on σ^2 uses the chi-square distribution and comparison of two variances uses the F distribution.

6.4.1 One variance

For a random sample of size n from $N(\mu, \sigma^2)$ with μ, σ unknown, Theorem 4-8 gives

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1).$$

With the quantiles $\chi^2_{1-\alpha/2}(n-1)$ and $\chi^2_{\alpha/2}(n-1)$,

$$P\Big(\chi^2_{1-\alpha/2}(n-1) \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2}(n-1)\Big) = 1 - \alpha,$$

and solving the inequalities for σ^2,

$$P\Big((n-1)\frac{S^2}{\chi^2_{\alpha/2}(n-1)} \leq \sigma^2 \leq (n-1)\frac{S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) = 1 - \alpha.$$

Definition (Confidence interval for σ^2 (normal population))

A $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\Big((n-1)\frac{s^2}{\chi^2_{\alpha/2}(n-1)},\ (n-1)\frac{s^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

The test of a variance based on the chi-square distribution is the **chi-square test**.

Definition (Chi-square test for σ^2 (normal population))

For $H_0 : \sigma^2 = \sigma_0^2$ use $\chi^2 = (n-1)\frac{S^2}{\sigma_0^2}$, which is $\chi^2(n-1)$ under H_0. With observed value χ_0^2:

Alternative	P-value	Rejection region at level α
$H_1 : \sigma^2 > \sigma_0^2$	$P(\chi^2 \geq \chi_0^2)$	$\chi_0^2 \geq \chi^2_\alpha(n-1)$
$H_1 : \sigma^2 < \sigma_0^2$	$P(\chi^2 \leq \chi_0^2)$	$\chi_0^2 \leq \chi^2_{1-\alpha}(n-1)$
$H_1 : \sigma^2 \neq \sigma_0^2$	$2P(\chi^2 \geq \chi_0^2)$ or $2P(\chi^2 \leq \chi_0^2)$	$\chi_0^2 \leq \chi^2_{1-\alpha/2}(n-1)$ or $\chi_0^2 \geq \chi^2_{\alpha/2}(n-1)$

For the two-sided alternative, compute both $2P(\chi^2 \geq \chi_0^2)$ and $2P(\chi^2 \leq \chi_0^2)$ and use the one smaller than 1.

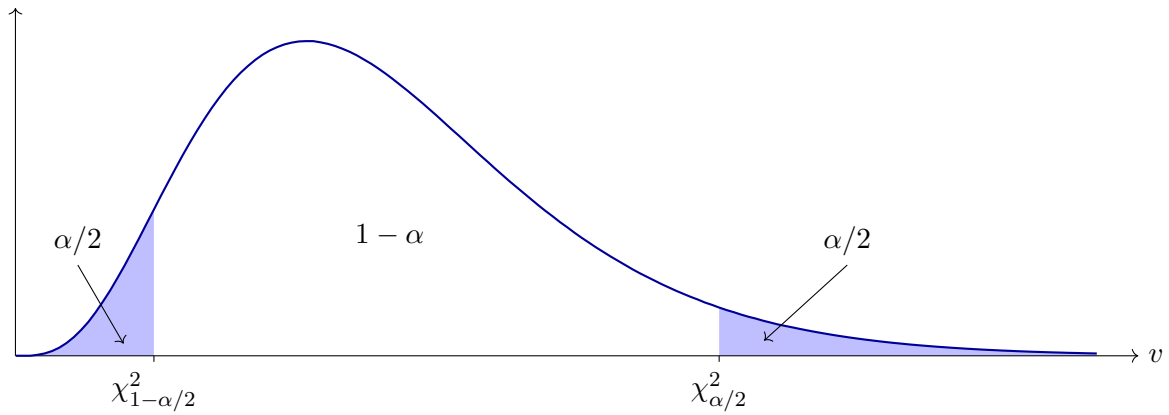


Figure 6: Quantiles of $\chi^2(9)$ used for a 90% interval for σ^2 ($\alpha = 0.1$): $\chi^2_{0.95}(9) = 3.33$, $\chi^2_{0.05}(9) = 16.92$.

Example 6-9 (Coffee grinder)

A café uses 10 g of beans per Americano, but the grinder dose varies; a standard deviation above 1.5 g is considered abnormal. Ten doses (g):

11.15	9.62	9.83	10.85	12.66	9.51	11.23	9.38	9.11	8.56

Past records indicate a normal distribution.
(a) Test at level 5% whether the grinder is abnormal.
(b) Find the P-value of (a).
(c) Find a 90% confidence interval for the standard deviation of the dose.
Solution.
(a) $H_0 : \sigma^2 = 1.5^2$, $H_1 : \sigma^2 > 1.5^2$. Here $\bar{x} = 10.19$, $s = 1.24$, and

$$\chi_0^2 = (n-1)\frac{s^2}{\sigma_0^2} = 9 \times \frac{1.24^2}{1.5^2} = 6.15.$$

The rejection region is $\chi_0^2 \geq \chi^2_{0.05}(9) = 16.92$; since $6.15 < 16.92$, H_0 is not rejected at 5%.
(b) $P = P(\chi^2 \geq 6.15)$, $\chi^2 \sim \chi^2(9)$; software gives $P = 0.72$. There is no clear evidence of abnormal variation.
(c) With $\chi^2_{0.05}(9) = 16.92$ and $\chi^2_{0.95}(9) = 3.33$, the 90% interval for σ^2 is

$$\left(9 \times \frac{1.24^2}{16.92},\ 9 \times \frac{1.24^2}{3.33}\right) = (0.818,\ 4.156),$$

so the interval for σ is $(\sqrt{0.818}, \sqrt{4.156}) = (0.90,\ 2.04)$.

Sensitivity to normality

Unlike the t test and t interval for a mean, chi-square inference on a variance depends heavily on the normality assumption; check it (e.g. by a Q–Q plot, which is roughly linear for the data of Example 6-9) before using these methods.

6.4.2 Ratio of two variances

For independent random samples $X_{11},\dots,X_{1n_1}$ and $X_{21},\dots,X_{2n_2}$ from normal populations, with sample variances S_1^2, S_2^2, Theorem 4-13 gives

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1).$$

Therefore

$$P\Big(F_{1-\alpha/2}(n_1-1,n_2-1) \le \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \le F_{\alpha/2}(n_1-1,n_2-1)\Big) = 1-\alpha,$$

and solving for σ_1^2/σ_2^2,

$$P\Big(\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}\Big) = 1-\alpha.$$

By Theorem 4-12, $F_{1-\alpha/2}(n_1-1,n_2-1) = 1/F_{\alpha/2}(n_2-1,n_1-1)$.

Definition (Confidence interval for σ_1^2/σ_2^2 (normal populations))

A $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{s_1^2}{s_2^2}\times\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)},\ \frac{s_1^2}{s_2^2}\times F_{\alpha/2}(n_2-1,n_1-1)\Big).$$

The test of σ_1^2/σ_2^2 based on the F distribution is the F **test**.

Definition (F test for the ratio of variances (normal populations))

For $H_0:\sigma_1^2=\sigma_2^2$ use $F=S_1^2/S_2^2$, which is $F(n_1-1,n_2-1)$ under H_0. With observed value f:

Alternative	P-value	Rejection region at level α
$H_1:\sigma_1^2>\sigma_2^2$	$P(F\ge f)$	$f \ge F_{\alpha}(n_1-1,n_2-1)$
$H_1:\sigma_1^2<\sigma_2^2$	$P(F\le f)$	$f \le 1/F_{\alpha}(n_2-1,n_1-1)$
$H_1:\sigma_1^2\ne\sigma_2^2$	$2P(F\ge f)$ or $2P(F\le f)$	$f \le 1/F_{\alpha/2}(n_2-1,n_1-1)$ or $f \ge F_{\alpha/2}(n_1-1,n_2-1)$

Example 6-10 (Polymer injection pressure)

Cracks in concrete are repaired by injecting a polymer. Injection pressures (10^6 psi) for two polymers:

Epoxy	1.75	2.12	2.05	1.97
MMA prepolymer	1.77	1.59	1.70	1.69

(a) Test at level 5% whether the variabilities of the injection pressures differ.
(b) Find a 95% confidence interval for the ratio of standard deviations.
Solution.
(a) $H_0:\sigma_1^2=\sigma_2^2$, $H_1:\sigma_1^2\ne\sigma_2^2$. Summary: $n_1=n_2=4$, $\bar{x}_1=1.97$, $s_1=0.160$; $\bar{x}_2=1.69$, $s_2=0.074$. The Q–Q plots do not contradict normality. The observed statistic is $f=0.160^2/0.074^2=4.675$. Since $F_{0.025}(3,3)=15.44$, the 5% rejection region is $f\le 1/15.44$ or $f\ge 15.44$. As $1/15.44<4.675<15.44$, H_0 is not rejected at 5%.

(b) The 95% interval for σ_1^2/σ_2^2 is

$$\Big(\frac{0.160^2}{0.074^2} \times \frac{1}{15.44},\ \frac{0.160^2}{0.074^2} \times 15.44\Big) = (0.301,\ 72.182),$$

so that for σ_1/σ_2 is $(\sqrt{0.301}, \sqrt{72.182}) = (0.549,\ 8.496)$.

Chapter 7. Analysis of Discrete Data

7.1 Estimation and testing of a population proportion

Motivation. Many quantities of interest — defect rates, unemployment rates, approval ratings — are proportions: the fraction of a population having some attribute. The count of such units in a random sample is binomial, and for large samples the normal approximation gives confidence intervals and tests.

Let p be the proportion of an (infinite) population having an attribute, and let X be the number of units with that attribute in a random sample of size n. Then $X \sim B(n,p)$, and the natural estimator of p is the **sample proportion**

$$\hat{p} = \frac{X}{n}, \qquad \mathrm{Var}(\hat{p}) = \frac{\mathrm{Var}(X)}{n^2} = \frac{p(1-p)}{n}.$$

The standard error $\mathrm{SE}(\hat{p}) = \sqrt{p(1-p)/n}$ depends on the unknown p, so it is estimated by plugging in $\hat{p}$:

$$\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

By the normal approximation to the binomial, $X \approx N(np, np(1-p))$ for large n, and

$$\frac{X - np}{\sqrt{n\hat{p}(1-\hat{p})}} = \frac{\hat{p}-p}{\sqrt{\hat{p}(1-\hat{p})/n}} \;\approx\; N(0,1), \qquad \text{so} \qquad P\Big(-z_{\alpha/2} \le \frac{\hat{p}-p}{\sqrt{\hat{p}(1-\hat{p})/n}} \le z_{\alpha/2}\Big) \approx 1-\alpha.$$

Definition (Estimating p (large sample))

(i) Estimator and standard error: $\hat{p} = X/n$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$.
(ii) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\widehat{\mathrm{SE}}(\hat{p})$.
(iii) $100(1-\alpha)\%$ confidence interval:

$$\Big(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\,,\;\; \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\Big).$$

Rule of thumb

The normal approximation is usually considered adequate when $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$.

Example 7-1 (Homecoming rate in Seoul)

To estimate the proportion p of Seoul households that will travel home for Chuseok, 500 households were sampled by simple random sampling; 79 said they would go. Find a 95% CI for p.

Solution. The population is huge, so treat the sample as a random sample from an infinite population. $\hat{p} = 79/500 = 0.158$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{0.158(0.842)/500} = 0.016$. Since $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$, the 95% CI is

$$(0.158 - 1.96 \times 0.016,\; 0.158 + 1.96 \times 0.016) = (0.126,\; 0.190).$$

7.1.1 Choosing the sample size

To keep the $100(1-\alpha)\%$ margin of error at most d we need $z_{\alpha/2}\sqrt{p(1-p)/n} \le d$. Since p is unknown and $p(1-p) \le 1/4$ for all p, it suffices that $z_{\alpha/2}\sqrt{1/4}/\sqrt{n} \le d$.

Definition (Sample size for estimating p)

To make the margin of error at most d (equivalently, the CI length at most $2d$):
(i) with a prior estimate p^* of p: $n \geq p^*(1-p^*)(z_{\alpha/2}/d)^2$;
(ii) with no prior information: $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$.

Why (ii) suffices. $p(1-p) = \frac{1}{4} - (p-\frac{1}{2})^2 \leq \frac{1}{4}$, so if $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$ then $z_{\alpha/2}\sqrt{p(1-p)/n} \leq z_{\alpha/2}\sqrt{1/(4n)} \leq d$ whatever the true p. ■

Example 7-2 (Poll for a district election)

Two candidates A and B run for district mayor. With no prior information, how large a sample is needed so that the 95% margin of error for A's support rate is within 3%?
Solution. $n \geq \frac{1}{4}(1.96/0.03)^2 = 1067.1$, so sample $n = 1068$ people.

7.1.2 Tests for p

Consider $H_0 : p = p_0$ vs. $H_1 : p > p_0$. Large observed counts x are evidence against H_0, so the P-value is $P = P(X \geq x)$ with $X \sim B(n, p_0)$, computed exactly from the binomial; reject at level α iff $P \leq \alpha$.

Definition (Small-sample test for p)

Test statistic: the count X, observed value x; $X \sim B(n, p_0)$ under $H_0 : p = p_0$.

H_1	P-value	reject at level α
(a) $p > p_0$	$P(X \geq x)$	$P \leq \alpha$
(b) $p < p_0$	$P(X \leq x)$	$P \leq \alpha$
(c) $p \neq p_0$	$P(X \leq y(x)) + P(X \geq x)$ if $x \geq np_0$	$P \leq \alpha$
	$P(X \leq x) + P(X \geq y(x))$ if $x < np_0$	

In (c), $y(x)$ is the integer on the opposite side of np_0 from x that is the largest ($x \geq np_0$) or smallest ($x < np_0$) nonnegative integer k with $P(X = k) \leq P(X = x)$.

Since the exact P-value is practical only for small n, this is called the **small-sample test** for a proportion.

Example 7-3 (A cheaper cutting tool)

With the current tool the defect rate is 5%. A cheaper tool may raise it, so 20 items were produced with it; 3 were defective. Test at level 5%.
Solution. $H_0 : p = 0.05$ vs. $H_1 : p > 0.05$, $n = 20$, $x = 3$. $P = P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.925 = 0.075$ for $X \sim B(20, 0.05)$. As $0.075 > 0.05$, H_0 is not rejected: this trial does not show that the cheaper tool raises the defect rate.

For large n, under H_0,

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \approx N(0,1),$$

usable when $np_0 \geq 5$ and $n(1-p_0) \geq 5$. Note that the null value p_0 (not $\hat{p}$) is used in the standard error.

Definition (Large-sample test for p ($np_0 \geq 5$, $n(1-p_0) \geq 5$))

$Z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$ with observed value z.

H_1	P-value	rejection region
(a) $p > p_0$	$P(Z \geq z)$	$z \geq z_\alpha$
(b) $p < p_0$	$P(Z \leq z)$	$z \leq -z_\alpha$
(c) $p \neq p_0$	$P(\lvert Z\rvert \geq \lvert z\rvert)$	$\lvert z\rvert \geq z_{\alpha/2}$

Example 7-4 (Has the homecoming rate dropped?)

In the past the Chuseok homecoming rate was 20%. Using the data of Example 7-1, test at level 1% whether the rate has decreased.
Solution. $H_0 : p = 0.2$ vs. $H_1 : p < 0.2$; $n = 500$, $X = 79$, $np_0 = 100 \geq 5$, $n(1-p_0) = 400 \geq 5$.

$$z = \frac{79/500 - 0.2}{\sqrt{0.2 \times 0.8/500}} = -2.348 \;\leq\; -z_{0.01} = -2.327,$$

so H_0 is rejected at level 1%: the homecoming rate is significantly lower than before.

7.2 Comparing two population proportions

Motivation. Comparing cure rates of two treatments or defect rates of two processes requires inference on $p_1 - p_2$ from two independent samples; the same normal approximation, applied to the difference, does the job.

Independent random samples of sizes n_1, n_2 are drawn from two populations; $X_1 \sim B(n_1, p_1)$ and $X_2 \sim B(n_2, p_2)$ are independent. The estimator of $p_1 - p_2$ is

$$\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}, \qquad \mathrm{Var}(\hat{p}_1 - \hat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2},$$

by independence. For large n_1, n_2, $\hat{p}_i \approx N(p_i, p_i(1-p_i)/n_i)$ independently, so $\hat{p}_1 - \hat{p}_2$ is approximately normal and

$$\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}} \;\approx\; N(0,1).$$

Definition (Estimating $p_1 - p_2$ (large n_1, n_2))

(i) $\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2$, $\widehat{\mathrm{SE}} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$.
(ii) $100(1-\alpha)\%$ CI for $p_1 - p_2$: $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\,\widehat{\mathrm{SE}}(\widehat{p_1 - p_2})$.

Example 7-5 (1954 Salk polio vaccine trial)

In a double-blind trial, a control group received placebo and a treatment group the vaccine. Let p_1, p_2 be the polio rates in the control and treatment groups.

	sample size	children with polio
Control	201,229	115
Treatment	200,745	33

Estimate $p_1 - p_2$ and give a 95% CI.

Solution. $\hat p_1 = 115/201229 = 5.71\times10^{-4}$, $\hat p_2 = 33/200745 = 1.64\times10^{-4}$, so $\hat p_1-\hat p_2 = 4.07\times10^{-4}$ with $\widehat{\mathrm{SE}} = 0.60\times10^{-4}$. The 95% CI is $4.07\times10^{-4}\pm1.96\times0.60\times10^{-4}$, i.e.

$$2.89\times10^{-4} < p_1 - p_2 < 5.25\times10^{-4}.$$

7.2.1 Testing $H_0: p_1 = p_2$

Under H_0 write $p = p_1 = p_2$; then $\mathrm{Var}(\hat p_1-\hat p_2) = p(1-p)(1/n_1+1/n_2)$. Pooling the two samples, X_1+X_2 of the n_1+n_2 units have the attribute, so p is estimated by the **pooled sample proportion**

$$\hat p = \frac{X_1+X_2}{n_1+n_2}, \qquad \widehat{\mathrm{Var}}(\hat p_1-\hat p_2) = \hat p(1-\hat p)\Big(\frac{1}{n_1}+\frac{1}{n_2}\Big).$$

Definition (Large-sample test of $H_0: p_1 = p_2$)

$$Z = \frac{\hat p_1-\hat p_2}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}}, \qquad \hat p = \frac{X_1+X_2}{n_1+n_2}.$$

With observed value z and $Z\approx N(0,1)$ under H_0:
(a) $H_1: p_1 > p_2$: $P = P(Z\ge z)$, reject if $z\ge z_\alpha$;
(b) $H_1: p_1 < p_2$: $P = P(Z\le z)$, reject if $z\le -z_\alpha$;
(c) $H_1: p_1 \ne p_2$: $P = P(|Z|\ge|z|)$, reject if $|z|\ge z_{\alpha/2}$.

Example 7-6 (Is the vaccine effective?)

With the data of Example 7-5, test whether the vaccine is effective.
Solution. $H_0: p_1 = p_2$ vs. $H_1: p_1 > p_2$ (p_1 placebo, p_2 vaccine). $\hat p = (115+33)/(201229+200745) = 3.68\times10^{-4}$,

$$z = \frac{5.71\times10^{-4}-1.64\times10^{-4}}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}} = 6.727, \qquad P = P(Z\ge6.727) = 8.66\times10^{-12}.$$

H_0 is rejected decisively: the 1954 trial strongly indicates that the vaccine prevents polio.

Example 7-7 (Parole success of murderers)

Random samples of 42 impulsive and 40 premeditated murderers released on parole were followed; success means no reoffence. Test at level 5% whether the success rates differ.

	success	failure	size
Impulsive	13	29	$n_1 = 42$
Premeditated	22	18	$n_2 = 40$
Total	35	47	$n = 82$

Solution. $H_0: p_1 = p_2$ vs. $H_1: p_1 \ne p_2$. $\hat p_1 = 13/42 = 0.310$, $\hat p_2 = 22/40 = 0.550$, $\hat p = 35/82 = 0.427$, and

$$z = \frac{\hat p_1-\hat p_2}{\sqrt{\hat p(1-\hat p)(n_1^{-1}+n_2^{-1})}} = -2.20, \qquad |z|\ge z_{0.025} = 1.96,$$

so H_0 is rejected: the parole success rates differ significantly.

7.3 Comparing several populations with categorical data

Motivation. When there are more than two populations, or more than two categories, the z-test no longer applies. Comparing observed counts with the counts expected under H_0 leads to the chi-square statistic, which handles any number of populations and categories.

The idea: reject H_0 (all populations have the same category proportions) when the **observed frequencies** differ much from the **estimated expected frequencies** under H_0, where

$$\text{estimated expected frequency} = (\text{sample size}) \times (\text{estimated probability under } H_0).$$

Illustration 7-1: expected counts for Example 7-7

Under $H_0 : p_1 = p_2$ the common success rate is estimated by $\hat{p} = 0.427$, so

$$\hat{E}_{11} = n_1\hat{p} = 42 \times 0.427 = 17.93, \qquad \hat{E}_{12} = n_1(1-\hat{p}) = 24.07,$$
$$\hat{E}_{21} = n_2\hat{p} = 40 \times 0.427 = 17.08, \qquad \hat{E}_{22} = n_2(1-\hat{p}) = 22.92.$$

Observed (expected): impulsive 13 (17.93), 29 (24.07); premeditated 22 (17.08), 18 (22.92). The discrepancy statistic is

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}} = \frac{(13-17.93)^2}{17.93} + \cdots + \frac{(18-22.92)^2}{22.92} = 4.84 = (-2.20)^2 = z^2.$$

The identity $\sum\sum(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij} = z^2$ holds in general for 2×2 tables; since $Z \approx N(0,1)$ under H_0, the statistic is approximately $\chi^2(1)$. It is called the **chi-square statistic**,

$$\chi^2 = \sum_i\sum_j \frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}}.$$

In Example 7-7 the 5% critical region is $\chi^2 \geq \chi^2_{0.05}(1) = 3.84$, equivalently $z^2 \geq z^2_{0.025} = 1.96^2 = 3.84$.

Fact ($\chi^2 = z^2$ for a 2×2 table)

For two samples with pooled proportion $\hat{p}$, the chi-square statistic of the 2×2 table equals the square of the two-sample z-statistic.

Proof. [Supplement] Write $q = 1 - \hat{p}$ and $D = X_1 - n_1\hat{p}$. Because $X_1 + X_2 = (n_1 + n_2)\hat{p}$, we have $X_2 - n_2\hat{p} = -D$, and for the failures $(n_i - X_i) - n_iq = -(X_i - n_i\hat{p})$. Hence every $|O_{ij} - \hat{E}_{ij}| = |D|$, and

$$\chi^2 = D^2\Big(\frac{1}{n_1\hat{p}} + \frac{1}{n_1q} + \frac{1}{n_2\hat{p}} + \frac{1}{n_2q}\Big) = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big).$$

On the other hand $\hat{p}_1 - \hat{p}_2 = D/n_1 + D/n_2 = D(n_1^{-1} + n_2^{-1})$, so

$$z^2 = \frac{D^2(n_1^{-1}+n_2^{-1})^2}{\hat{p}q(n_1^{-1}+n_2^{-1})} = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big) = \chi^2.$$

■

7.3.1 Several binomial populations

Observing r binomial populations gives an $r \times 2$ **contingency table** with counts O_{i1} (success), O_{i2} (failure), row totals n_i, column totals $O_{\cdot 1}, O_{\cdot 2}$ and grand total n. We test

$$H_0 : p_1 = p_2 = \cdots = p_r \qquad \text{vs.} \qquad H_1 : \text{not all } p_i \text{ are equal.}$$

Under H_0 the common rate is estimated by the pooled proportion $\hat{p} = (O_{11} + \cdots + O_{r1})/(n_1 + \cdots + n_r) = O_{\cdot 1}/n$, and

$$\hat{E}_{i1} = n_i \hat{p} = n_i \frac{O_{\cdot 1}}{n}, \qquad \hat{E}_{i2} = n_i(1-\hat{p}) = n_i \frac{O_{\cdot 2}}{n} \qquad (i = 1, \ldots, r).$$

For large samples, $\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2(r-1)$ under H_0 (this is a known result; it rests on the multivariate normal approximation).

Definition (Comparing r binomial populations ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_1 = \cdots = p_r$ vs. H_1: at least one differs, use

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = n_i \frac{O_{\cdot j}}{n}.$$

With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$ for $\chi^2 \sim \chi^2(r-1)$; reject at level α iff $\chi_0^2 \geq \chi_\alpha^2(r-1)$.

Example 7-8 (Siting an LNG storage base)

Residents of three candidate regions were sampled (400, 350, 350 people) and asked whether they favour an LNG storage base. Do the approval rates differ?

	for	against	size
Region 1	198	202	400
Region 2	140	210	350
Region 3	133	217	350
Total	471	629	1100

Solution. $H_0 : p_1 = p_2 = p_3$. Expected counts: $\hat{E}_{11} = 400 \cdot 471/1100 = 171.27$, $\hat{E}_{12} = 228.73$, $\hat{E}_{21} = \hat{E}_{31} = 149.86$, $\hat{E}_{22} = \hat{E}_{32} = 200.14$. Then

$$\chi_0^2 = \frac{(198 - 171.27)^2}{171.27} + \cdots + \frac{(217 - 200.14)^2}{200.14} = 11.75,$$

and $P = P(\chi^2(2) \geq 11.75) = 0.003$ (from the table, $0.0025 < P < 0.005$). The approval rates differ significantly across the regions.

7.3.2 Several multinomial populations

If each population is classified into c **categories**, we have **multinomial populations**; observing r of them gives an $r \times c$ table of **categorical data**. Let $p_{i1}, \ldots, p_{ic}$ ($\sum_j p_{ij} = 1$) be the category proportions in population i. The test of

$$H_0 : p_{1j} = p_{2j} = \cdots = p_{rj} \ (j = 1, \ldots, c) \qquad \text{vs.} \qquad H_1 : \text{not } H_0$$

is the **test of homogeneity**. Under H_0 the common proportion p_j of category j is estimated by $\hat{p}_j = O_{\cdot j}/n$, so $\hat{E}_{ij} = n_i \hat{p}_j = n_i O_{\cdot j}/n$, and for large samples

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \approx \chi^2((r-1)(c-1)).$$

Counting degrees of freedom

Each of the r rows has $c-1$ free proportions, giving $r(c-1)$ parameters in general; under H_0 only the $c-1$ common p_j remain. The difference $r(c-1)-(c-1)=(r-1)(c-1)$ is the degrees of freedom. For $c=2$ this is $r-1$, as above.

Definition (Test of homogeneity ($\hat{E}_{ij} \geq 5$))

$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij}$ with $\hat{E}_{ij} = n_i O_{\cdot j}/n$. With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$, $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-9 (Odour from an industrial complex)

In three regions next to an industrial complex, 97, 95 and 99 people were asked how often they notice a bad odour: 1 = daily, 2 = at least weekly, 3 = at least monthly, 4 = less than monthly, 5 = never. Test at level 1% whether the frequency differs by region.

	1	2	3	4	5	size
Region 1	20	28	23	14	12	97
Region 2	14	34	21	14	12	95
Region 3	4	12	10	20	53	99
Total	38	74	54	48	77	291

Solution. $\hat{p}_j = 38/291, 74/291, 54/291, 48/291, 77/291$; $\hat{E}_{1j} = 97\hat{p}_j$, $\hat{E}_{2j} = 95\hat{p}_j$, $\hat{E}_{3j} = 99\hat{p}_j$:

	1	2	3	4	5
Region 1	12.67	24.67	18.00	16.00	25.67
Region 2	12.41	24.16	17.63	15.67	25.14
Region 3	12.93	25.18	18.37	16.33	26.20

All expected counts exceed 5. $\chi_0^2 = (20-12.67)^2/12.67 + \cdots + (53-26.20)^2/26.20 = 71.36 \geq \chi_{0.01}^2(8) = 20.09$, so H_0 is rejected: odour perception differs by region.

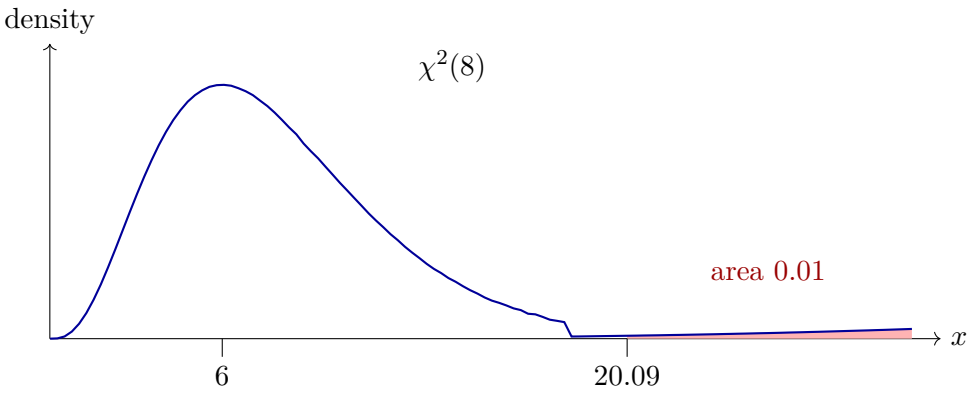


Figure 7: The $\chi^2(8)$ density with the 1% rejection region of Example 7-9; the observed $\chi_0^2 = 71.36$ lies far to the right.

7.4 Test of independence with categorical data

Motivation. Sometimes a single sample is cross-classified by two attributes, and the question is whether the attributes are related. The same chi-square statistic answers it, with expected counts computed from the product of the margins.

Each unit of one population is classified by attribute A (r categories $A_1, \dots, A_r$) and attribute B (c categories $B_1, \dots, B_c$). Let p_{ij} be the cell probabilities, $p_{i\cdot} = \sum_j p_{ij}$ and $p_{\cdot j} = \sum_i p_{ij}$ the marginal probabilities; a random sample of size n gives counts O_{ij} with margins $O_{i\cdot}, O_{\cdot j}$. Independence of A and B means

$$H_0 : p_{ij} = p_{i\cdot}\, p_{\cdot j} \ \text{ for all } i, j, \qquad H_1 : p_{ij} \neq p_{i\cdot}\, p_{\cdot j} \ \text{ for at least one } (i, j).$$

Under H_0 we estimate each margin separately and multiply:

$$\hat{p}_{ij} = \hat{p}_{i\cdot}\, \hat{p}_{\cdot j} = \frac{O_{i\cdot}}{n} \cdot \frac{O_{\cdot j}}{n}, \qquad \hat{E}_{ij} = n\hat{p}_{ij} = \frac{O_{i\cdot}\, O_{\cdot j}}{n}.$$

For large n, $\chi^2 = \sum_i \sum_j (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2((r-1)(c-1))$ under H_0: in general there are $rc - 1$ free cell probabilities, under H_0 only $(r-1) + (c-1)$, and $rc - 1 - (r-1) - (c-1) = (r-1)(c-1)$.

Definition (Test of independence ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$ $(i = 1, \dots, r,\ j = 1, \dots, c)$ vs. H_1 : not H_0,

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = \frac{O_{i\cdot} O_{\cdot j}}{n}.$$

$P = P(\chi^2 \geq \chi_0^2)$ with $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-10 (Car size and commuting distance)

250 solo car commuters in a metropolitan area were classified by car size and commuting distance. Test at level 5% whether they are related.

	<15 km	15–30 km	≥ 30 km	total
Light	6	27	19	52
Small	8	36	17	61
Mid-size	21	45	33	99
Large	14	18	6	38
Total	49	126	75	250

Solution. $\hat{E}_{ij} = O_{i\cdot} O_{\cdot j}/250$: row 1: 10.19, 26.21, 15.60; row 2: 11.96, 30.74, 18.30; row 3: 19.40, 49.90, 29.70; row 4: 7.45, 19.15, 11.40 — all ≥ 5. Then

$$\chi_0^2 = \frac{(6 - 10.19)^2}{10.19} + \cdots + \frac{(6 - 11.40)^2}{11.40} = 14.16 \ \geq \ \chi_{0.05}^2(6) = 12.59,$$

so H_0 is rejected: car size and commuting distance are related.

Homogeneity vs. independence

The test of independence and the test of homogeneity (§7.3) use the same statistic and the same procedure, but the sampling differs: for homogeneity the row totals n_i are fixed in advance (a separate sample from each population); for independence only n is fixed and one sample is cross-classified. Choose the sampling plan that fits the question and then the matching inference.

Chapter 8. Correlation and Regression Analysis

8.1 Introduction

Motivation. Data often come in pairs — IQ and grades, smoking and lifespan, process temperature and product strength — and we want to know whether, and how, the two variables are related. Scatter plots give a first look; correlation and regression analysis make the inference precise.

Correlation analysis studies whether two variables are (linearly) related and how strongly, usually through inference on the population correlation coefficient based on the sample correlation coefficient. **Regression analysis** models one variable as a function of another so that its value can be predicted. When the function is a straight line we speak of **simple linear regression**; when several explanatory variables jointly affect the response, of **multiple regression**.

The terms go back to Francis Galton, who studied the inheritance of sweet-pea seed size and of heights of parents and children; "regression" comes from his phrase *regression toward the mean.*

8.2 Correlation analysis

Motivation. The population correlation ρ measures the strength of a linear relation; its sample analogue r estimates it, and under bivariate normality a t-test decides whether $\rho \neq 0$.

For two characteristics X, Y of the units of a population, the **population correlation coefficient** $\rho = \mathrm{Corr}(X, Y)$ satisfies $-1 \le \rho \le 1$; values near $+1$ (-1) indicate a strong linear relation with positive (negative) slope, values near 0 almost no linear relation. From a random sample $(x_1, y_1), \dots, (x_n, y_n)$, ρ is estimated by the **sample correlation coefficient**

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

Also $-1 \le r \le 1$, and $r \to \pm 1$ as the points cluster around a line. A value $r \approx 0$ means a *weak linear* relation, not the absence of any relation: a strongly curved (e.g. parabolic) pattern can have $r = 0$.

Illustration 8-1: cycling distance and calories

Twenty randomly chosen adult men cycled for 10 minutes; distance x (km) and calories burned y (kcal) were recorded.

x	3.4	3.0	4.3	4.5	3.8	4.3	3.2	3.4	3.1	3.1
y	96	84	112	123	111	104	95	105	91	79
x	3.2	3.5	3.2	3.2	3.1	3.4	3.2	3.1	3.9	4.6
y	98	103	96	82	97	87	83	86	100	126

The scatter plot shows an increasing linear trend, and

$$r = \frac{110.88}{\sqrt{5.10}\,\sqrt{3258.55}} = 0.860.$$

Inference on ρ assumes the population is **bivariate normal**, i.e. (X, Y) has density

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\Big(\tfrac{x-\mu_X}{\sigma_X}\Big)^2 - 2\rho\,\tfrac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \Big(\tfrac{y-\mu_Y}{\sigma_Y}\Big)^2\Big]\Big\},$$

whose marginals are $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$.

Definition (Test for correlation (bivariate normal population))

For $H_0 : \rho = 0$ use $T = \sqrt{n-2}\,\dfrac{r}{\sqrt{1-r^2}}$, which is $t(n-2)$ under H_0. With observed value t:

H_1	P-value	rejection region
(a) $\rho > 0$	$P(T \geq t)$	$t \geq t_\alpha(n-2)$
(b) $\rho < 0$	$P(T \leq t)$	$t \leq -t_\alpha(n-2)$
(c) $\rho \neq 0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-2)$

Why $T \sim t(n-2)$

[Supplement] Regress y on x (§8.3). One checks algebraically that $\sqrt{n-2}\,r/\sqrt{1-r^2} = \hat\beta/(\hat\sigma/\sqrt{S_{(xx)}})$, using $r^2 = \mathrm{SSR}/\mathrm{SST}$ and $1-r^2 = \mathrm{SSE}/\mathrm{SST}$. Under bivariate normality with $\rho = 0$, conditionally on the x_i the y_i follow the simple linear regression model with $\beta = 0$, so by §8.4 this ratio is $t(n-2)$ given the x's; since this conditional law does not depend on the x's, it is also the unconditional law.

Example 8-1 (Is distance correlated with calories?)

In Illustration 8-1 assume bivariate normality and test at level 5% whether the two variables are correlated.

Solution. $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$; $n = 20$, $r = 0.860$:

$$t = \sqrt{20-2}\,\frac{0.860}{\sqrt{1-0.860^2}} = 7.150, \qquad |t| \geq t_{0.025}(18) = 2.101.$$

H_0 is rejected; the P-value is 0.000001, strong evidence of correlation.

8.3 The simple linear regression model and its fit

Motivation. To predict a response (e.g. product strength) from an explanatory variable (e.g. injection temperature) we postulate a straight-line model for the mean response and fit it by least squares.

The variable that influences is the **explanatory variable**; the one influenced is the **response variable**. Regression first examines the scatter plot to choose a tentative model.

Illustration 8-2: temperature and hot-drink sales

A coffee shop recorded the daily maximum temperature x (°C) and hot-drink sales y (10,000 won) on 12 random days.

x	27.1	9.1	19.2	29.1	22.0	34.9
y	46.8	63.7	46.0	46.1	50.3	34.7
x	16.5	7.6	13.1	2.6	23.5	5.6
y	57.8	63.2	52.7	62.9	45.3	69.9

The points scatter around an unknown line $y = \alpha + \beta x$ with negative slope (Figure 8).

We regard y as an observed value of $Y = \alpha + \beta x + e$, where $E(e) = 0$, $\mathrm{Var}(e) = \sigma^2$; equivalently

$E(Y \mid X = x) = \alpha + \beta x$ and $\mathrm{Var}(Y \mid X = x) = \sigma^2$. This is the **simple linear regression model**; the responses $y_1, \dots, y_n$ at $x_1, \dots, x_n$ are observed independently.

Definition (Assumptions of the simple linear regression model)

$Y_i = \alpha + \beta x_i + e_i$ $(i = 1, \dots, n)$, where
(i) linearity: $E(e_i) = 0$, i.e. $E(Y_i \mid X = x_i) = \alpha + \beta x_i$ $(-\infty < \alpha, \beta < \infty)$;
(ii) equal variance: $\mathrm{Var}(e_1) = \cdots = \mathrm{Var}(e_n) = \sigma^2 > 0$;
(iii) independence: $e_1, \dots, e_n$ are independent.

The line $y = \alpha + \beta x$ is the **population regression line**, α and β are the **population regression coefficients**, and σ^2 is the **error variance**. Fitting the model means estimating these from data, by the **method of least squares**.

Definition 8-1 (Method of least squares)

In the model $Y_i = \alpha + \beta x_i + e_i$, the estimators of α, β are the values minimizing

$$\sum_{i=1}^{n} \{Y_i - (\alpha + \beta x_i)\}^2.$$

Setting the partial derivatives in α and β to zero gives the **normal equations**

$$\begin{cases} -2\sum(y_i - \hat\alpha - \hat\beta x_i) = 0 \\ -2\sum x_i(y_i - \hat\alpha - \hat\beta x_i) = 0 \end{cases} \quad \text{i.e.} \quad \begin{cases} n\hat\alpha + (\sum x_i)\hat\beta = \sum y_i \\ (\sum x_i)\hat\alpha + (\sum x_i^2)\hat\beta = \sum x_i y_i. \end{cases}$$

The solutions are the **least squares estimators** (LSE), and $\hat y = \hat\alpha + \hat\beta x$ is the **least squares regression line**. Note that $\hat y_i$ estimates the *mean* response $E(Y_i \mid X = x_i)$, not an individual response. With the shorthand

$$S_{(xy)} = \sum(x_i - \bar x)(y_i - \bar y) = \sum x_i y_i - \tfrac{1}{n}(\sum x_i)(\sum y_i), \qquad S_{(xx)} = \sum(x_i - \bar x)^2 = \sum x_i^2 - \tfrac{1}{n}(\sum x_i)^2,$$

and $S_{(yy)}$ defined likewise, the solution is as follows.

Proposition (Least squares estimators and line)

(i) $\hat\alpha = \bar y - \hat\beta \bar x, \quad \hat\beta = S_{(xy)}/S_{(xx)}$.
(ii) The least squares line is $\hat y = \hat\alpha + \hat\beta x = \bar y + \hat\beta(x - \bar x)$.
(iii) The minimum value of the sum of squares is $\mathrm{SSE} = S_{(yy)} - \hat\beta^2 S_{(xx)} = S_{(yy)} - S_{(xy)}^2/S_{(xx)}$.

Proof. [Supplement] Assume $S_{(xx)} > 0$ (not all x_i equal). Put $Q(a, b) = \sum(y_i - a - bx_i)^2$.

Step 1. Write $y_i - a - bx_i = \{(y_i - \bar y) - b(x_i - \bar x)\} + (\bar y - a - b\bar x)$. The cross term vanishes because $\sum(y_i - \bar y) = \sum(x_i - \bar x) = 0$, so

$$Q(a, b) = \sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 + n(\bar y - a - b\bar x)^2.$$

Step 2. Expanding the first sum and completing the square in b,

$$\sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 = S_{(yy)} - 2bS_{(xy)} + b^2 S_{(xx)} = S_{(xx)}\Big(b - \frac{S_{(xy)}}{S_{(xx)}}\Big)^2 + S_{(yy)} - \frac{S_{(xy)}^2}{S_{(xx)}}.$$

Step 3. Both squares in Q are ≥ 0, and both vanish exactly when $b = S_{(xy)}/S_{(xx)}$ and $a = \bar y - b\bar x$. Hence this is the unique minimizer, giving (i) and (ii), and the minimum is $S_{(yy)} - S_{(xy)}^2/S_{(xx)}$; since $\hat\beta^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$, (iii) follows. The minimizer of course also solves the normal equations. ■

Example 8-2 (Fitting the hot-drink data)

For Illustration 8-2, with temperature as X and sales as Y, find the LSE of α, β and draw the fitted line.

Solution. $\bar{x} = 210.3/12 = 17.525$, $\bar{y} = 639.4/12 = 53.283$,

$$S_{(xy)} = 10134.21 - \tfrac{1}{12} \times 210.3 \times 639.4 = -1071.275,$$
$$S_{(xx)} = 4826.67 - \tfrac{1}{12} \times 210.3^2 = 1141.162.$$

So $\hat{\beta} = S_{(xy)}/S_{(xx)} = -0.939$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 69.739$, and $\hat{y} = 69.739 - 0.939x$ (Figure 8).

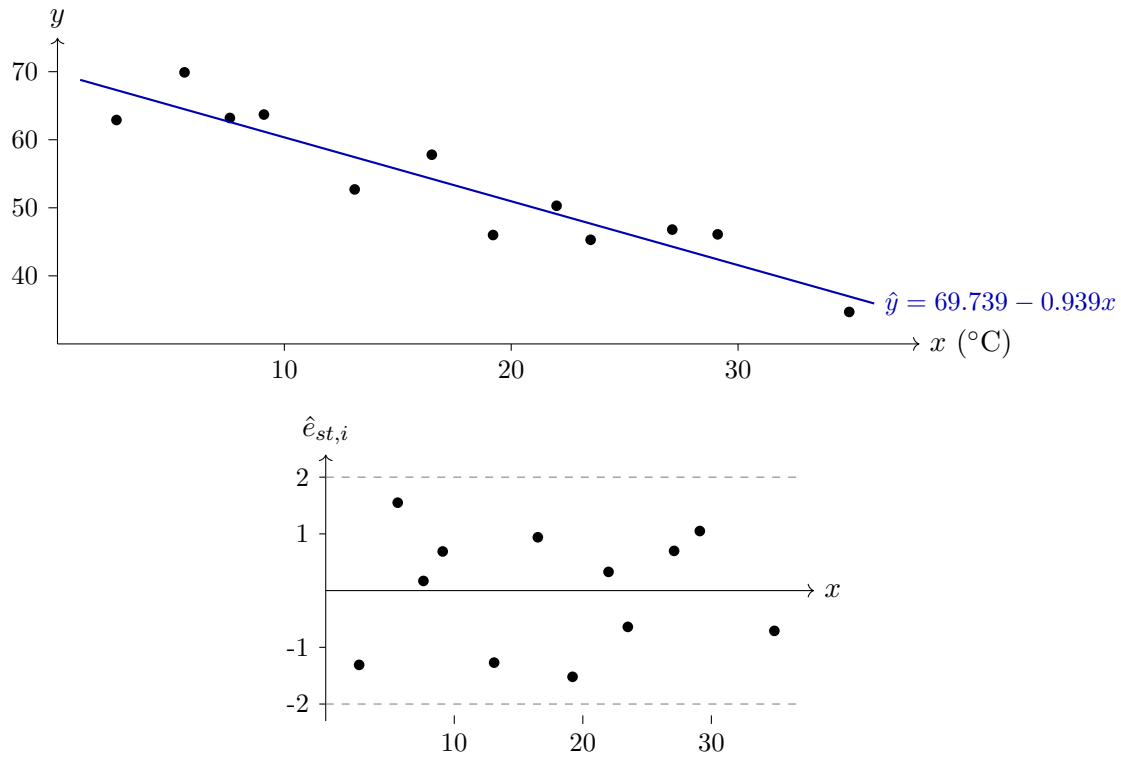


Figure 8: Top: the hot-drink data with the least squares line (Example 8-2). Bottom: studentized residuals against temperature (§8.5); they are symmetric about 0, show no pattern, and lie within ±2.

8.3.1 Residuals and the estimate of σ^2

The **residuals** are $\hat{e}_i = y_i - \hat{y}_i$; small residuals mean a good fit. They behave like observed errors e_i, so their size reflects σ^2. The **residual** (or **error**) **sum of squares** is SSE $= \sum(y_i - \hat{y}_i)^2$, and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

The divisor $n - 2$ is the degrees of freedom of SSE, and $\hat{\sigma}^2$ is called the **mean squared error**, MSE.

Definition (Residual sum of squares and the estimate of σ^2)

(i) $\text{SSE} = \sum(y_i - \hat{y}_i)^2 = \sum\{y_i - \bar{y} - \hat{\beta}(x_i - \bar{x})\}^2 = S_{(yy)} - \hat{\beta}^2 S_{(xx)} = S_{(yy)} - S^2_{(xy)}/S_{(xx)}$.

(ii) $\hat{\sigma}^2 = \text{MSE} = \text{SSE}/(n-2)$.

Proposition (MSE is unbiased)

Under the model assumptions, $E(\text{SSE}) = (n-2)\sigma^2$, so $E(\text{MSE}) = \sigma^2$.

Proof. [Supplement] Treat the x_i as fixed.

Step 1. $\sum(Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2$ with $E(Y_i^2) = \sigma^2 + \mu_i^2$, $\mu_i = \alpha + \beta x_i$, and $E(\bar{Y}^2) = \sigma^2/n + \bar{\mu}^2$. Hence $E\,S_{(YY)} = (n-1)\sigma^2 + \sum(\mu_i - \bar{\mu})^2 = (n-1)\sigma^2 + \beta^2 S_{(xx)}$.

Step 2. By the next proposition, $E(\hat{\beta}) = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, so $E(\hat{\beta}^2)S_{(xx)} = \sigma^2 + \beta^2 S_{(xx)}$.

Step 3. Subtracting, $E(\text{SSE}) = E\,S_{(YY)} - E(\hat{\beta}^2)S_{(xx)} = (n-2)\sigma^2$. ■

Example 8-3 (Estimating σ^2)

For the hot-drink data, estimate the error variance.
Solution. $S_{(xy)} = -1071.275$, $S_{(xx)} = 1141.162$, $S_{(yy)} = 1160.837$, so

$$\text{SSE} = 1160.837 - \frac{(-1071.275)^2}{1141.162} = 155.169, \qquad \hat{\sigma}^2 = \text{MSE} = \frac{155.169}{10} = 15.517.$$

8.3.2 Decomposition of the total sum of squares

How much of the variation of y does the fitted line explain? The deviation of y_i from $\bar{y}$ splits as

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}), \tag{8.1}$$

where $y_i - \hat{y}_i$ is the residual (due to error) and $\hat{y}_i - \bar{y}$ is the part explained by the regression line. $y_i - \bar{y}$ is the **total deviation** and (8.1) its **decomposition**. Squaring and summing gives a Pythagoras-type identity. Define the **total sum of squares** $\text{SST} = \sum(y_i - \bar{y})^2$ (d.f. $n-1$) and the **regression sum of squares** $\text{SSR} = \sum(\hat{y}_i - \bar{y})^2$ (d.f. 1).

Proposition (Decomposition of the total sum of squares)

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{\text{SST},\ n-1} = \underbrace{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{SSE},\ n-2} + \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{\text{SSR},\ 1}.$$

Moreover $\text{SSR} = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$ and $\text{SST} = S_{(yy)}$.

Proof. [Supplement]

Step 1. The normal equations say $\sum \hat{e}_i = 0$ and $\sum x_i \hat{e}_i = 0$, where $\hat{e}_i = y_i - \hat{y}_i$.

Step 2. Since $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$, the cross term is

$$\sum(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = \hat{\beta}\sum \hat{e}_i(x_i - \bar{x}) = \hat{\beta}\Big(\sum x_i\hat{e}_i - \bar{x}\sum \hat{e}_i\Big) = 0.$$

Step 3. Squaring (8.1) and summing therefore gives $\text{SST} = \text{SSE} + \text{SSR}$, and $\text{SSR} = \hat{\beta}^2\sum(x_i - \bar{x})^2 = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$. The degrees of freedom $n-1$ and $n-2$ reflect the one and two estimated parameters ($\bar{y}$; $\hat{\alpha}, \hat{\beta}$), and $1 = (n-1) - (n-2)$. ■

The larger the share of SSR in SST, the better the model explains the data. This share is the **coefficient of determination**

$$r^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{S_{(xy)}^2}{S_{(xx)}S_{(yy)}} = \Big(\frac{S_{(xy)}}{\sqrt{S_{(xx)}}\sqrt{S_{(yy)}}}\Big)^2,$$

which is exactly the square of the sample correlation coefficient. Values near 1 mean the points cluster tightly around the line.

Example 8-4 (Coefficient of determination)

Compute and interpret r^2 for the hot-drink data.
Solution. $r^2 = (-1071.275)^2/(1141.162 \times 1160.837) = 0.866$: the line explains 86.6% of the variation in sales, and together with the scatter plot the model describes the data well.

Statistics alone can mislead (Anscombe)

The two data sets below have nearly identical statistics: the same fitted line $\hat{y} = 3 + 0.5x$, SST = 41.29, SSR = 27.51, SSE = 13.78, $\hat{\sigma}^2 = 1.53$, $r^2 = 0.67$. Yet A is roughly linear with scatter, while B lies on a smooth curve. Always look at the scatter plot and the residuals (§8.5).

A	x	10	8	13	9	11	14	6	4	12	7	5
	y	8.04	6.95	7.58	8.81	8.33	9.96	7.24	4.26	10.84	4.82	5.68
B	x	10	8	13	9	11	14	6	4	12	7	5
	y	9.14	8.14	8.74	8.77	9.26	8.10	6.13	3.10	9.13	7.26	4.74

8.4 Inference in simple linear regression

Motivation. Adding the assumption that the errors are normal turns the least squares fit into a basis for confidence intervals and tests about the slope, the intercept and the mean response.

From now on assume, in addition,

$$Y_i = \alpha + \beta x_i + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \qquad (i = 1, \ldots, n).$$

8.4.1 Significance of the regression line

The line is useful iff $\beta \neq 0$, so we test $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$. A significant line makes SSR large relative to SSE. Under H_0,

$$F = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \sim F(1, n-2),$$

and with observed value f the P-value is $P(F \geq f)$; the level-α critical region is $f \geq F_\alpha(1, n-2)$. The test is summarized in an **analysis of variance table**; a **mean square** is a sum of squares divided by its degrees of freedom.

Definition (ANOVA table for the significance of the regression line)

Source	SS	d.f.	Mean square	F	P-value
Regression	SSR	1	$\text{MSR} = \text{SSR}/1$	$f = \text{MSR}/\text{MSE}$	$P(F \geq f)$
Residual	SSE	$n-2$	$\text{MSE} = \text{SSE}/(n-2)$		
Total	SST	$n-1$			

Example 8-5 (Is the regression significant?)

Test the significance of the regression line for the hot-drink data.
Solution. SST $= S_{(yy)} = 1160.837$, SSR $= (-1071.275)^2/1141.162 = 1005.668$, SSE $= 155.169$, so MSR $= 1005.668$, MSE $= 15.517$ and $f = 64.811$. Since $P(F(1,10) \geq 64.811) = 0.00001 < 0.001$, the line is highly significant.

Source	SS	d.f.	MS	F	P
Regression	1005.668	1	1005.668	64.811	0.00001
Residual	155.169	10	15.517		
Total	1160.837	11			

8.4.2 Inference on the slope β

Proposition (Distribution of $\hat{\beta}$)

$E(\hat{\beta}) = \beta$, $\text{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, $\hat{\beta}$ is normal, and

$$\frac{\hat{\beta} - \beta}{\hat{\sigma}/\sqrt{S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. Let $c_i = (x_i - \bar{x})/S_{(xx)}$. Since $\sum(x_i - \bar{x})\bar{Y} = 0$, $\hat{\beta} = \sum c_i Y_i$, a linear combination of independent normal variables, hence normal. Note $\sum c_i = 0$, $\sum c_i x_i = \sum c_i(x_i - \bar{x}) = 1$, $\sum c_i^2 = 1/S_{(xx)}$.

Step 2. $E(\hat{\beta}) = \sum c_i(\alpha + \beta x_i) = \alpha \sum c_i + \beta \sum c_i x_i = \beta$.

Step 3. By independence, $\text{Var}(\hat{\beta}) = \sum c_i^2 \sigma^2 = \sigma^2/S_{(xx)}$. Thus $Z = (\hat{\beta} - \beta)/(\sigma/\sqrt{S_{(xx)}}) \sim N(0,1)$.

Step 4. It is a standard fact of normal linear models (proved via Cochran's theorem) that $\text{SSE}/\sigma^2 \sim \chi^2(n-2)$ and that SSE is independent of $(\hat{\alpha}, \hat{\beta})$. Then $(\hat{\beta}-\beta)/(\hat{\sigma}/\sqrt{S_{(xx)}}) = Z/\sqrt{(\text{SSE}/\sigma^2)/(n-2)}$ is $t(n-2)$ by definition of the t distribution. (Independence is at least easy to see at the level of covariances: $\text{Cov}(\hat{e}_j, \hat{\beta}) = \sigma^2(c_j - \frac{1}{n}\sum_i c_i - (x_j - \bar{x})\sum_i c_i^2) = 0$, and jointly normal uncorrelated variables are independent.) ∎

Definition (Inference on β)

(i) $100(1-\alpha)\%$ CI: $\hat{\beta} \pm t_{\alpha/2}(n-2)\,\hat{\sigma}/\sqrt{S_{(xx)}}$.

(ii) For $H_0 : \beta = \beta_0$: $T = (\hat{\beta} - \beta_0)/(\hat{\sigma}/\sqrt{S_{(xx)}}) \sim t(n-2)$. With observed t:

(a) $H_1 : \beta > \beta_0$: $P = P(T \geq t)$, reject if $t \geq t_\alpha(n-2)$;

(b) $H_1 : \beta < \beta_0$: $P = P(T \le t)$, reject if $t \le -t_\alpha(n-2)$;
(c) $H_1 : \beta \ne \beta_0$: $P = P(|T| \ge |t|)$, reject if $|t| \ge t_{\alpha/2}(n-2)$.

Fact ($F = T^2$)

For $H_0 : \beta = 0$, the ANOVA statistic equals the square of the t-statistic, so the two-sided t-test and the F-test are the same test.

Proof. Using $\mathrm{SSR} = \hat\beta^2 S_{(xx)}$ and $\mathrm{SSE}/(n-2) = \mathrm{MSE} = \hat\sigma^2$,

$$F = \frac{\mathrm{SSR}/1}{\mathrm{SSE}/(n-2)} = \frac{\hat\beta^2 S_{(xx)}}{\mathrm{MSE}} = \frac{\hat\beta^2}{\mathrm{MSE}/S_{(xx)}} = \Big(\frac{\hat\beta - 0}{\hat\sigma/\sqrt{S_{(xx)}}}\Big)^2 = T^2.$$

Since $t_{\alpha/2}(n-2)^2 = F_\alpha(1, n-2)$, the rejection regions coincide. ■

Example 8-6 (Inference on the slope)

For the hot-drink data ($n = 12$, $\hat\beta = -0.939$, $S_{(xx)} = 1141.162$, $\hat\sigma^2 = 15.517$):
(a) test the significance of the line by a t-test at level 1% and relate it to the F-test;
(b) test at level 5% whether a 10°C rise lowers mean sales by more than 90,000 won;
(c) give a 95% CI for β and interpret it.
Solution.
(a) $H_0 : \beta = 0$ vs. $H_1 : \beta \ne 0$: $t = -0.939/\sqrt{15.517/1141.162} = -8.053$, and $|t| \ge t_{0.005}(10) = 3.169$, so the line is significant at 1%. Also $F = T^2$ (here $8.053^2 \approx 64.8$), so the F-test is the same test.
(b) Sales are in units of 10,000 won, so the claim is a drop of more than 0.9 per 1°C: $H_0 : \beta = -0.9$ vs. $H_1 : \beta < -0.9$. $t = (-0.939 + 0.9)/\sqrt{15.517/1141.162} = -0.334 > -t_{0.05}(10) = -1.812$, so H_0 is not rejected; there is no clear evidence of a drop exceeding 90,000 won per 10°C.
(c) $t_{0.025}(10) = 2.228$: $-0.939 \pm 2.228\sqrt{15.517/1141.162} = -0.939 \pm 0.260$, i.e. $(-1.199, -0.679)$. With 95% confidence, each 1°C rise lowers mean hot-drink sales by between 6,790 and 11,990 won.

8.4.3 Inference on the mean response $\alpha + \beta x$

The mean response $E(Y \mid X = x) = \alpha + \beta x$ is estimated by $\hat\alpha + \hat\beta x$.

Proposition (Distribution of $\hat\alpha + \hat\beta x$)

$E(\hat\alpha + \hat\beta x) = \alpha + \beta x$, $\mathrm{Var}(\hat\alpha + \hat\beta x) = \Big(\frac{1}{n} + \frac{(x - \bar x)^2}{S_{(xx)}}\Big)\sigma^2$, it is normal, and

$$\frac{(\hat\alpha + \hat\beta x) - (\alpha + \beta x)}{\hat\sigma\sqrt{n^{-1} + (x - \bar x)^2/S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. $\hat\alpha + \hat\beta x = \bar Y + \hat\beta(x - \bar x) = \sum(\frac{1}{n} + (x - \bar x)c_i)Y_i$ is a linear combination of independent normals, hence normal, with mean $(\alpha + \beta\bar x) + \beta(x - \bar x) = \alpha + \beta x$ (using $E\hat\beta = \beta$).

Step 2. $\mathrm{Cov}(\bar{Y}, \hat{\beta}) = \sum_i \frac{1}{n} c_i \sigma^2 = 0$ since $\sum c_i = 0$. Therefore

$$\mathrm{Var}(\hat{\alpha} + \hat{\beta} x) = \mathrm{Var}(\bar{Y}) + (x - \bar{x})^2 \,\mathrm{Var}(\hat{\beta}) = \frac{\sigma^2}{n} + \frac{(x - \bar{x})^2 \sigma^2}{S_{(xx)}}.$$

Step 3. Standardizing and replacing σ by $\hat{\sigma}$, the $t(n-2)$ law follows exactly as in Step 4 of the previous proof (SSE is independent of $(\bar{Y}, \hat{\beta})$). ■

Definition (Inference on the mean response $E(Y \mid X = x) = \alpha + \beta x$)

(i) $100(1-\alpha)\%$ CI: $(\hat{\alpha} + \hat{\beta} x) \pm t_{\alpha/2}(n-2)\, \hat{\sigma} \sqrt{n^{-1} + (x - \bar{x})^2 / S_{(xx)}}$.

(ii) For $H_0 : \alpha + \beta x = \mu_0$: $T = \dfrac{(\hat{\alpha} + \hat{\beta} x) - \mu_0}{\hat{\sigma} \sqrt{n^{-1} + (x - \bar{x})^2 / S_{(xx)}}} \sim t(n-2)$, with the usual one- or two-sided P-values and critical regions $t \geq t_\alpha(n-2)$, $t \leq -t_\alpha(n-2)$, $|t| \geq t_{\alpha/2}(n-2)$.

Example 8-7 (Mean sales at given temperatures)

For the hot-drink data ($n = 12$, $\bar{x} = 17.525$, $S_{(xx)} = 1141.162$, $\hat{y} = 69.739 - 0.939x$, $\hat{\sigma}^2 = 15.517$):
(a) at $x = 25°$C, test at level 1% whether mean sales exceed 400,000 won;
(b) find 95% CIs for mean sales at $x = 10, 20, 30°$C and plot them.
Solution.
(a) $H_0 : \alpha + 25\beta = 40$ vs. $H_1 : \alpha + 25\beta > 40$; $t = 4.372 \geq t_{0.01}(10) = 2.764$, so H_0 is rejected: mean sales at 25°C clearly exceed 400,000 won.
(b) At $x = 10$:

$$(69.739 - 0.939 \times 10) \pm 2.228 \sqrt{15.517} \sqrt{\tfrac{1}{12} + \tfrac{(10 - 17.525)^2}{1141.162}} = 60.349 \pm 3.200,$$

i.e. (57.149, 63.549). Similarly (48.345, 53.573) at $x = 20$ and (37.455, 45.683) at $x = 30$. Joining the upper and lower limits over all x gives a band, narrowest at $x = \bar{x}$ (a "bow-tie" shape).

The CI for the mean response is shorter when x is close to $\bar{x}$ and when $S_{(xx)}$ is large; so it pays to observe the response at widely spread x values around the x of interest.

8.4.4 Inference on the intercept α

Taking $x = 0$ above:

Definition (Inference on α)

(i) $100(1-\alpha)\%$ CI: $\hat{\alpha} \pm t_{\alpha/2}(n-2)\, \hat{\sigma} \sqrt{n^{-1} + \bar{x}^2 / S_{(xx)}}$.

(ii) For $H_0 : \alpha = \alpha_0$: $T = (\hat{\alpha} - \alpha_0) / (\hat{\sigma} \sqrt{n^{-1} + \bar{x}^2 / S_{(xx)}}) \sim t(n-2)$, with the usual P-values and critical regions.

Example 8-8 (CI for the intercept)

Find a 95% CI for α in the hot-drink model.
Solution. $\hat{\alpha} \pm t_{0.025}(10) \hat{\sigma} \sqrt{1/12 + 17.525^2 / 1141.162} = 69.739 \pm 5.210$, i.e. (64.529, 74.949).

8.5 Residual analysis in simple linear regression

Motivation. All of §8.4 is valid only if the model assumptions hold. Residuals are the observable proxies for the errors, and plotting them is the standard way to check the assumptions.

The full model is $Y_i = \alpha + \beta x_i + e_i$ with (i) linearity $E(e_i) = 0$;

(ii) equal variance $\mathrm{Var}(e_i) = \sigma^2$;

(iii) independence of $e_1, \ldots, e_n$;

(iv) normality $e_i \sim N(0, \sigma^2)$. If they hold, $e_i/\mathrm{SD}(e_i) = (Y_i - (\alpha + \beta x_i))/\sigma$ are independent $N(0,1)$. Replacing e_i by $\hat{e}_i$ and dividing by an estimate of its standard deviation gives the **studentized residuals**

$$\hat{e}_{st,i} = \frac{\hat{e}_i}{\widehat{\mathrm{SD}}(\hat{e}_i)} = \frac{y_i - \hat{y}_i}{\widehat{\mathrm{SD}}(Y_i - \hat{Y}_i)} \qquad (i = 1, \ldots, n),$$

which should look like n independent standard normal observations. (In simple regression $\mathrm{Var}(\hat{e}_i) = \sigma^2\{1 - \frac{1}{n} - (x_i - \bar{x})^2/S_{(xx)}\}$, estimated by plugging in $\hat{\sigma}^2$.) Checking assumptions this way is **residual analysis**, usually via a **residual plot** of the points $(x_i, \hat{e}_{st,i})$.

Definition (Residual analysis)

Compute $\hat{e}_{st,i} = \hat{e}_i/\widehat{\mathrm{SD}}(\hat{e}_i)$ and judge whether they look like n independent $N(0,1)$ observations. The model is adequate if the residuals
(i) are roughly symmetric about 0;
(ii) have constant spread across x;
(iii) show no systematic pattern;
(iv) lie about 95% within ± 2.

The residual plot for the hot-drink data (Figure 8, bottom) satisfies (i)–(iv), so the simple linear regression model is appropriate there and the inferences of §8.4 are meaningful.

Recommended workflow

(i) Draw a scatter plot to see the rough linear trend.
(ii) Fit the least squares line and MSE, then check the studentized residuals (symmetric about 0, mostly within ± 2, no pattern).
(iii) If nothing is wrong, carry out CIs and tests.
Typical violations show up as: a curved pattern (linearity fails); a funnel of growing spread (equal variance fails); runs or waves in order (independence fails); heavy skew or outliers (normality fails). Then a different regression method is needed.

Example 8-9 (Electricity bills and temperature)

Monthly mean temperature x (°C) and the monthly electricity bill y (1,000 won) of an apartment household:

x	y	x	y	x	y	x	y	x	y
−2.2	14.0	0.8	12.2	8.2	12.9	15.2	11.9	21.8	10.3
−1.9	13.1	1.8	13.0	9.2	11.3	16.3	10.9	22.3	9.1
−1.7	14.4	3.9	11.3	14.2	10.8	21.1	9.6	23.6	7.5
−1.3	13.1	7.0	10.5	14.5	10.0	21.1	8.1	23.6	10.5
−0.7	14.8	8.0	9.7	14.9	9.9	21.5	9.2	24.8	10.0

(a) Check by residual analysis whether simple linear regression is appropriate.

(b) Test the significance of the regression line at level 5%.
(c) Give a 95% CI for the mean bill when the mean temperature is 10°C.
Solution. The scatter plot shows a rough decreasing linear relation.
(a) $n = 25$, $\bar{x} = 11.44$, $\bar{y} = 11.124$, $S_{(xx)} = 2210.44$, $S_{(yy)} = 88.8856$, $S_{(xy)} = -374.9945$. So $\hat{\beta} = -0.17$, $\hat{\alpha} = 13.06$, $\hat{y} = 13.06 - 0.17x$, and $\hat{\sigma}^2 = \{S_{(yy)} - S^2_{(xy)}/S_{(xx)}\}/(n-2) = 1.0987$. The studentized residuals are symmetric about 0, have constant spread, show no pattern and all lie within ±2: the model is adequate. Also $r^2 = S^2_{(xy)}/(S_{(xx)}S_{(yy)}) = 0.7157$, so about 72% of the variation is explained.
(b) SST = 88.8856, SSR $= S^2_{(xy)}/S_{(xx)} = 63.6165$, SSE = 25.2691 (d.f. 24, 1, 23); $f = (\text{SSR}/1)/(\text{SSE}/23) = 57.902 \geq F_{0.05}(1, 23) = 4.28$, $P = 0.0000001$: the line is significant.

Source	SS	d.f.	MS	F	P
Regression	63.6165	1	63.6165	57.902	0.0000001
Residual	25.2691	23	1.0987		
Total	88.8856	24			

(c) $\hat{\alpha} + 10\hat{\beta} = 13.06 - 1.7 = 11.36$ with standard error $\hat{\sigma}\sqrt{1/n + (10-\bar{x})^2/S_{(xx)}} = 0.2121$; $t_{0.025}(23) = 2.069$ gives 11.36 ± 0.44, i.e. (10.92, 11.80).

8.6 Multiple regression analysis

Motivation. A response is often affected by several explanatory variables at once; using them all in one model usually explains and predicts better than any single-variable regression.

A regression model with two or more explanatory variables is a **multiple regression model**. With p explanatory variables, the **multiple linear regression model** is

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \quad (i = 1, \ldots, n),$$

with unknown real coefficients $\beta_0, \ldots, \beta_p$ and error variance $\sigma^2 > 0$.

8.6.1 Two explanatory variables

For $Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$, least squares minimizes $\sum\{Y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})\}^2$. Setting the three partial derivatives to zero gives the normal equations

$$\begin{aligned} -2\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{1i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{2i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0. \end{aligned}$$

The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2$; substituting into the other two yields

$$S_{(11)}\hat{\beta}_1 + S_{(12)}\hat{\beta}_2 = S_{(1y)}, \qquad S_{(12)}\hat{\beta}_1 + S_{(22)}\hat{\beta}_2 = S_{(2y)},$$

where $S_{(11)} = S_{(x_1x_1)}$, $S_{(22)} = S_{(x_2x_2)}$, $S_{(12)} = S_{(x_1x_2)}$, $S_{(1y)} = S_{(x_1y)}$, $S_{(2y)} = S_{(x_2y)}$.

Definition (LSE in multiple linear regression with two explanatory variables)

$\hat\beta_0, \hat\beta_1, \hat\beta_2$ solve

$$\begin{cases} S_{(11)}\hat\beta_1 + S_{(12)}\hat\beta_2 = S_{(1y)} \\ S_{(12)}\hat\beta_1 + S_{(22)}\hat\beta_2 = S_{(2y)} \\ \hat\beta_0 = \bar y - \hat\beta_1\bar x_1 - \hat\beta_2\bar x_2 \end{cases}$$

and the regression plane $E(Y \mid x_1, x_2) = \beta_0+\beta_1 x_1+\beta_2 x_2$ is estimated by $\hat y = \hat\beta_0+\hat\beta_1 x_1+\hat\beta_2 x_2$.

Derivation of the reduced equations. [Supplement] Substitute $\hat\beta_0 = \bar y - \hat\beta_1\bar x_1 - \hat\beta_2\bar x_2$ into the second normal equation: $\sum x_{1i}\{(y_i - \bar y) - \hat\beta_1(x_{1i} - \bar x_1) - \hat\beta_2(x_{2i} - \bar x_2)\} = 0$. Since each bracketed deviation sums to zero over i, x_{1i} may be replaced by $x_{1i} - \bar x_1$, giving $S_{(1y)} - \hat\beta_1 S_{(11)} - \hat\beta_2 S_{(12)} = 0$. The third equation gives the other one in the same way. (As in §8.3, the sum of squares is a convex quadratic, so when $S_{(11)}S_{(22)} - S_{(12)}^2 > 0$ this unique stationary point is the minimum.) ■

Example 8-10 (Standing long jump, age and body fat)

For 10 adult men at a fitness centre, Y = standing long jump (cm), X_1 = age, X_2 = body-fat percentage. Fit the multiple linear regression model.

age	43	64	29	22	31	20	20	26	20	19
body fat	22.7	31.9	12.3	18.4	16.4	26.9	16.0	17.5	22.5	14.5
jump	206	120	250	224	219	195	260	190	226	222

Solution. $n = 10$, $\bar x_1 = 29.4$, $\bar x_2 = 19.91$, $\bar y = 211.2$; $S_{(11)} = 1824.4$, $S_{(22)} = 329.989$, $S_{(12)} = 472.56$, $S_{(1y)} = -3809.8$, $S_{(2y)} = -1723.02$. The normal equations

$$\begin{aligned} 1824.4\,\hat\beta_1 + 472.56\,\hat\beta_2 &= -3809.8, \\ 472.56\,\hat\beta_1 + 329.989\,\hat\beta_2 &= -1723.02, \\ \hat\beta_0 &= 211.2 - 29.4\hat\beta_1 - 19.91\hat\beta_2 \end{aligned}$$

give $\hat\beta_0 = 316.198$, $\hat\beta_1 = -1.1696$, $\hat\beta_2 = -3.5465$, so $\hat y = 316.198 - 1.1696x_1 - 3.5465x_2$.

In practice multiple regression is computed by software; understanding the model, the procedure and the concepts matters more than the arithmetic.

8.6.2 ANOVA in multiple regression

As before $y_i - \bar y = (y_i - \hat y_i) + (\hat y_i - \bar y)$ and

$$\underbrace{\sum (y_i - \bar y)^2}_{\text{SST}} = \underbrace{\sum (y_i - \hat y_i)^2}_{\text{SSE}} + \underbrace{\sum (\hat y_i - \bar y)^2}_{\text{SSR}},$$

with degrees of freedom $n-1$, $n-p-1$ and p. ([Supplement] The proof is the same as in §8.3: the normal equations say that the residuals satisfy $\sum \hat e_i = 0$ and $\sum x_{ki}\hat e_i = 0$ for each k, so $\sum \hat e_i(\hat y_i - \bar y) = 0$ because $\hat y_i - \bar y$ is a linear combination of $1, x_{1i}, \dots, x_{pi}$.) Define

$$\text{MSE} = \frac{\text{SSE}}{n-p-1}, \qquad \text{MSR} = \frac{\text{SSR}}{p}, \qquad \hat\sigma^2 = \text{MSE}, \qquad R^2 = \frac{\text{SSR}}{\text{SST}}.$$

(R^2 is the multiple-regression analogue of r^2.) To test the regression function,

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \qquad \text{vs.} \qquad H_1 : \text{not all } \beta_k \text{ are } 0,$$

use $F = \text{MSR}/\text{MSE}$, which is $F(p, n-p-1)$ under H_0; reject at level α if $f \geq F_\alpha(p, n-p-1)$, $P = P(F \geq f)$.

Definition (ANOVA table for multiple regression)

Source	SS	d.f.	MS	F	P-value
Regression	SSR	p	MSR	$f = \text{MSR}/\text{MSE}$	$P(F \geq f)$
Residual	SSE	$n-p-1$	MSE		
Total	SST	$n-1$			

Example 8-11 (Significance of the long-jump model)

For Example 8-10, SST $= 13503.6$ and SSE $= 2936.9$. Test the regression function at level 5% and compute R^2.

Solution. SSR $=$ SST $-$ SSE $= 10566.7$; with $p = 2$ the d.f. are $9, 7, 2$. MSR $= 10566.7/2 = 5283.35$, MSE $= 2936.9/7 = 419.557$, $f = 12.593 \geq F_{0.05}(2,7) = 4.737$ ($P = 0.0048$): the regression function is significant.

Source	SS	d.f.	MS	F	P
Regression	10566.7	2	5283.35	12.593	0.0048
Residual	2936.9	7	419.557		
Total	13503.6	9			

$R^2 = \text{SSR}/\text{SST} = 0.7825$: about 78% of the variation is explained by the regression function.

Inference as in Example 8-11 also requires checking the model first, again with studentized residuals $\hat{e}_{st,i} = (y_i - \hat{y}_i)/\widehat{\text{SD}}(Y_i - \hat{Y}_i)$. Since there are several explanatory variables, the residual plot uses the fitted values on the horizontal axis: the points $(\hat{y}_i, \hat{e}_{st,i})$. For Example 8-10 this plot shows no problem, so the inference of Example 8-11 is trustworthy.

Chapter 9. Analysis of Variance

9.1 What is analysis of variance?

Motivation. We often ask whether a characteristic differs across several conditions — strength across production methods, yield across temperatures, recovery time across drugs. Analysis of variance answers this by splitting the total variation of the data into parts due to each factor and a part due to error.

Such questions are studied by planned experiments. Suppose tensile strengths of metal parts made by several methods are compared and one method looks best. The difference may be due to the method, or to other factors such as the skill of the workers. To separate them we repeat each method several times (several workers, or one worker repeatedly). The observed variation then has a part due to differences between methods and a part due to other sources (worker skill, fluctuations of one worker). If the variation between methods is large relative to the variation from other sources, we conclude that the method affects strength.

Analysis of variance (ANOVA), devised by R. A. Fisher, is the most widely used tool in the **design of experiments** (DOE). It expresses the variation of the characteristic as a sum of squares, decomposes it into parts attributable to the factors of the experiment, and compares each factor's *mean square* (sum of squares divided by its degrees of freedom) with the mean square due to error. In the example, strength is the characteristic and the method is a factor; the total sum of squares splits as

$$\text{total sum of squares} = \text{sum of squares due to the factor} + \text{sum of squares due to error.}$$

The factor part has an identified cause (the method); the error part collects everything not yet explained. ANOVA generalizes the two-sample test of equal means (Chapter 6) to three or more means. This chapter treats *fixed* factors (levels chosen by the experimenter): one-way ANOVA (equal and unequal replication) and two-way ANOVA (without and with replication).

9.2 One-way analysis of variance

Motivation. The simplest design studies a single factor at $k \geq 3$ levels; the question is whether the k treatment means are all equal.

In a **one-way layout** a single factor A has k levels (treatments), and treatment i is replicated n_i times; the n_i need not be equal, and typically $k = 3$–5 and $n_i = 3$–10. The runs must be performed in random order: this is a **completely randomized design**. For example, with methods A, B, C and 5 runs each, number the $3 \times 5 = 15$ experimental units and run them in an order drawn by lot or random numbers. Statistically this amounts to independent random samples from k populations: from the $N = n_1 + \cdots + n_k$ units, n_1 are randomly chosen for treatment 1, n_2 of the rest for treatment 2, and so on. For $k = 2$ it is the comparison of two treatments based on independent samples.

9.2.1 Equal replication

With n replicates per treatment the data are y_{ij} ($i = 1, \ldots, k$ treatments, $j = 1, \ldots, n$), with treatment means and grand mean

$$\bar{y}_{i\cdot} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{kn}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} \qquad (i = 1, \ldots, k).$$

The observations under treatment i are a random sample of size n from $N(\mu_i, \sigma^2)$: $Y_{ij} = \mu_i + e_{ij}$ with $e_{ij} \sim N(0, \sigma^2)$ independent. Let $\mu = \frac{1}{k}\sum \mu_i$ be the overall mean and $\alpha_i = \mu_i - \mu$ the ith **treatment effect**; by definition $\sum \alpha_i = 0$.

Definition (One-way ANOVA model (equal replication))

$$Y_{ij} = \mu + \alpha_i + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{k} \alpha_i = 0,$$

$i = 1, \dots, k$, $j = 1, \dots, n$. Equivalently:
(i) linearity $E(e_{ij}) = 0$, i.e. $E(Y_{ij}) = \mu + \alpha_i$;
(ii) equal variance $\mathrm{Var}(e_{ij}) = \sigma^2 > 0$;
(iii) independence of the e_{ij};
(iv) normality $e_{ij} \sim N(0, \sigma^2)$.

Testing whether the treatment effects differ is the same as testing whether the k population means differ. Decompose the total deviation:

$$y_{ij} - \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{i\cdot}) + (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}). \tag{9.1}$$

Here $y_{ij} - \bar{y}_{i\cdot}$ is the **residual** (experimental uncertainty) and $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ is the **deviation of the treatment effect**. Squaring and summing gives the following decomposition into the **total sum of squares** SST (total variation), the **treatment sum of squares** SS_{tr} (**between sum of squares**) and the **residual sum of squares** SSE (**within sum of squares**).

Proposition (Decomposition of the total sum of squares, one-way)

With $N = kn$ observations,

$$\underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SST},\ N-1} = \underbrace{n\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SS}_{tr},\ k-1} + \underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2}_{\mathrm{SSE},\ N-k}.$$

Proof. [Supplement] Square (9.1) and sum over i, j. The cross term is

$$2\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot}) = 0,$$

because the inner sum is $n\bar{y}_{i\cdot} - n\bar{y}_{i\cdot} = 0$. The middle square does not depend on j, so it contributes $n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$. As for degrees of freedom: SST has $N-1$ (the deviations sum to 0), SS_{tr} has $k-1$ (k treatment means minus one constraint), and SSE has $N-k$ ($n-1$ in each of k groups); indeed $N - 1 = (k-1) + (N-k)$. ∎

A sum of squares divided by its degrees of freedom is a **mean square**:

$$\mathrm{MS}_{tr} = \frac{\mathrm{SS}_{tr}}{k-1}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k}.$$

We test

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \alpha_i \neq 0,$$

which is the same as $H_0 : \mu_1 = \cdots = \mu_k$. If the treatments matter, SS_{tr} takes a large share of SST, so large values of $F = \mathrm{MS}_{tr}/\mathrm{MSE}$ are evidence against H_0.

Proposition (Expected mean squares, one-way)

$E(\text{MSE}) = \sigma^2$ and $E(\text{MS}_{tr}) = \sigma^2 + \dfrac{n}{k-1}\sum_{i=1}^{k}\alpha_i^2$. Under H_0, $F = \text{MS}_{tr}/\text{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. For each i, $Y_{i1}, \ldots, Y_{in}$ is a random sample from $N(\mu_i, \sigma^2)$, so its sample variance has mean σ^2: $E\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = (n-1)\sigma^2$. Summing over i, $E(\text{SSE}) = k(n-1)\sigma^2 = (N-k)\sigma^2$.

Step 2. The treatment means $\bar{Y}_{i\cdot} \sim N(\mu + \alpha_i, \sigma^2/n)$ are independent. For independent Z_i with means m_i and common variance v, $E\sum(Z_i - \bar{Z})^2 = (k-1)v + \sum(m_i - \bar{m})^2$. With $v = \sigma^2/n$, $m_i = \mu + \alpha_i$ and $\bar{m} = \mu$ (as $\sum \alpha_i = 0$),

$$E(\text{SS}_{tr}) = n\Big\{(k-1)\frac{\sigma^2}{n} + \sum \alpha_i^2\Big\} = (k-1)\sigma^2 + n\sum \alpha_i^2.$$

Step 3. Distribution. $\text{SSE}/\sigma^2 = \sum_i (n-1)S_i^2/\sigma^2$ is a sum of k independent $\chi^2(n-1)$ variables, hence $\chi^2(N-k)$. Each S_i^2 is independent of $\bar{Y}_{i\cdot}$ (normal sampling theory), and the groups are independent, so SSE is independent of $(\bar{Y}_{1\cdot}, \ldots, \bar{Y}_{k\cdot})$, hence of SS_{tr}. Under H_0 the $\bar{Y}_{i\cdot}$ are i.i.d. $N(\mu, \sigma^2/n)$, so $\text{SS}_{tr}/\sigma^2 = \sum(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2/(\sigma^2/n) \sim \chi^2(k-1)$. Therefore $F = \dfrac{(\text{SS}_{tr}/\sigma^2)/(k-1)}{(\text{SSE}/\sigma^2)/(N-k)} \sim F(k-1, N-k)$. ■

Thus MSE always estimates σ^2, while MS_{tr} is inflated by treatment effects — which is why a large F indicates H_1.

Definition (Test of treatment effects (one-way, equal replication))

For $H_0 : \alpha_1 = \cdots = \alpha_k = 0$ use $F = \text{MS}_{tr}/\text{MSE}$. With observed value f:
(i) level-α critical region: $f \geq F_\alpha(k-1, N-k)$;
(ii) P-value: $P = P(F \geq f)$, $F \sim F(k-1, N-k)$.

Source	SS	d.f.	MS	F	P
Treatment	$\text{SS}_{tr} = n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	$k-1$	$\text{MS}_{tr} = \frac{\text{SS}_{tr}}{k-1}$	$f = \frac{\text{MS}_{tr}}{\text{MSE}}$	$P(F \geq f)$
Residual	$\text{SSE} = \sum_i\sum_j(y_{ij} - \bar{y}_{i\cdot})^2$	$N-k$	$\text{MSE} = \frac{\text{SSE}}{N-k}$		
Total	$\text{SST} = \sum_i\sum_j(y_{ij} - \bar{y}_{\cdot\cdot})^2$	$N-1$			

Example 9-1 (Concentration of a treatment solution)

To see whether the concentration of a treatment solution affects the tensile strength of a fabric, four concentrations (A_1: 3.0%, A_2: 3.5%, A_3: 4.0%, A_4: 4.5%) were each run 5 times, the 20 runs in random order. Build the ANOVA table and test at level 5%.

	A_1	A_2	A_3	A_4
	47	51	50	22
	58	62	38	23
	51	31	47	28
	61	46	27	42
	46	49	23	25
mean	52.6	47.8	37.0	28.0

grand mean 41.35

Solution. Model $Y_{ij} = \mu + \alpha_i + e_{ij}$ with $k = 4$, $n = 5$; $H_0 : \alpha_1 = \cdots = \alpha_4 = 0$.

$$\begin{aligned}\text{SST} &= (47 - 41.35)^2 + \cdots + (25 - 41.35)^2 = 3334.55,\\ \text{SS}_{tr} &= 5\{(52.6 - 41.35)^2 + \cdots + (28.0 - 41.35)^2\} = 1826.55,\\ \text{SSE} &= (47 - 52.6)^2 + \cdots + (25 - 28.0)^2 = 1508.00,\end{aligned}$$

with d.f. 19, 3, 16. So $\text{MS}_{tr} = 608.85$, $\text{MSE} = 94.25$, $f = 6.46$, and $P = P(F(3,16) \geq 6.46) = 0.0045$ (table: between 0.001 and 0.01). Concentration has a significant effect on strength at level 5%.

Source	SS	d.f.	MS	F	P
Treatment	1826.55	3	608.85	6.46	0.0045
Residual	1508.00	16	94.25		
Total	3334.55	19			

9.2.2 Unequal replication

In practice the number of replicates often varies by treatment (e.g. more runs for a treatment of special interest). The analysis is almost the same. With n_i replicates of treatment i and $N = \sum n_i$,

$$\bar{y}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}, \qquad \mu = \frac{1}{N}\sum_{i=1}^{k} n_i\mu_i,$$

and $\alpha_i = \mu_i - \mu$ now satisfies $\sum n_i\alpha_i = 0$.

Definition (One-way ANOVA model (unequal replication))

$Y_{ij} = \mu + \alpha_i + e_{ij}$, $e_{ij} \sim N(0, \sigma^2)$ independent, $\sum_{i=1}^{k} n_i\alpha_i = 0$ $(i = 1, \ldots, k,\ j = 1, \ldots, n_i)$.

Proposition (Decomposition and expected mean squares, unequal replication)

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_{i=1}^{k} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with d.f. $N-1 = (k-1)+(N-k)$. Moreover $E(\text{MSE}) = \sigma^2$, $E(\text{MS}_{tr}) = \sigma^2 + \sum n_i\alpha_i^2/(k-1)$, and under $H_0 : \alpha_1 = \cdots = \alpha_k = 0$, $F = \text{MS}_{tr}/\text{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. The cross term is $2\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$, exactly as before; the middle square summed over j gives $n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Step 2. $E(\text{SSE}) = \sum(n_i - 1)\sigma^2 = (N - k)\sigma^2$ as before.

Step 3. Write $\text{SS}_{tr} = \sum n_i\bar{Y}_{i\cdot}^2 - N\bar{Y}_{\cdot\cdot}^2$. Since $E\bar{Y}_{i\cdot}^2 = \sigma^2/n_i + \mu_i^2$ and $\bar{Y}_{\cdot\cdot} \sim N(\mu, \sigma^2/N)$,

$$E(\text{SS}_{tr}) = k\sigma^2 + \sum n_i\mu_i^2 - \sigma^2 - N\mu^2 = (k-1)\sigma^2 + \sum n_i(\mu_i - \mu)^2,$$

using $\sum n_i\mu_i = N\mu$. This gives $E(\text{MS}_{tr})$.

Step 4. The independence argument of the equal-replication case is unchanged. Under H_0, $\sqrt{n_i}(\bar{Y}_{i\cdot} - \mu)/\sigma$ are i.i.d. $N(0,1)$, and SS_{tr}/σ^2 is the squared length of the projection of this vector onto the $(k-1)$-dimensional subspace orthogonal to $(\sqrt{n_1}, \ldots, \sqrt{n_k})$, hence $\chi^2(k-1)$. The F law follows. ■

Definition (Test of treatment effects (one-way, unequal replication))

$F = \mathrm{MS}_{tr}/\mathrm{MSE}$ with $\mathrm{MS}_{tr} = \mathrm{SS}_{tr}/(k-1)$, $\mathrm{MSE} = \mathrm{SSE}/(N-k)$, $N = \sum n_i$. Reject H_0 at level α if $f \geq F_\alpha(k-1, N-k)$; $P = P(F \geq f)$. The ANOVA table is as before with $\mathrm{SS}_{tr} = \sum_i n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Example 9-2 (Ratings on three review sites)

A film's ratings (out of 10) were sampled randomly: 4 from site 1, 5 from site 2, 6 from site 3. Test at level 5% whether the ratings differ by site.

	ratings	mean
Site 1	2, 3, 4, 5	3.5
Site 2	4, 5, 6, 4, 3	4.4
Site 3	6, 5, 7, 4, 6, 8	6.0

grand mean 4.8

Solution. $k = 3$, $n_1 = 4$, $n_2 = 5$, $n_3 = 6$; $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$.

$$\mathrm{SST} = (2-4.8)^2 + \cdots + (8-4.8)^2 = 36.4,$$
$$\mathrm{SS}_{tr} = 4(3.5-4.8)^2 + 5(4.4-4.8)^2 + 6(6.0-4.8)^2 = 16.2,$$
$$\mathrm{SSE} = (2-3.5)^2 + \cdots + (8-6.0)^2 = 20.2,$$

with d.f. 14, 2, 12. $\mathrm{MS}_{tr} = 8.1$, $\mathrm{MSE} = 1.68$, $f = 4.82$, $P = P(F(2,12) \geq 4.82) = 0.0292$. The ratings differ significantly by site.

Source	SS	d.f.	MS	F	P
Treatment	16.2	2	8.1	4.82	0.0292
Residual	20.2	12	1.68		
Total	36.4	14			

9.3 Two-way analysis of variance without replication

Motivation. When two factors may influence the response, a two-way layout examines both at once; with one observation per combination we can still test each factor, provided the factors do not interact.

Two-way ANOVA studies the effects of two factors A (levels $A_1, \ldots, A_p$) and B (levels $B_1, \ldots, B_q$). Each combination of one level of A and one level of B is a treatment, so there are pq treatments. In a completely randomized design all pq runs are done in random order; fixing a level of A and running through the levels of B violates randomization. All factors here are fixed. With one observation y_{ij} per cell, define

$$\bar{y}_{i\cdot} = \frac{1}{q}\sum_{j=1}^{q} y_{ij}, \qquad \bar{y}_{\cdot j} = \frac{1}{p}\sum_{i=1}^{p} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{pq}\sum_{i=1}^{p}\sum_{j=1}^{q} y_{ij}.$$

Definition (Two-way ANOVA model (no replication))

$$Y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{p} \alpha_i = 0, \quad \sum_{j=1}^{q} \beta_j = 0,$$

$i = 1, \dots, p$, $j = 1, \dots, q$; α_i is the effect of level A_i and β_j that of B_j.

The hypotheses of interest are

$$H_0 : \alpha_1 = \cdots = \alpha_p = 0 \ \text{ vs. } \ H_1 : \text{some } \alpha_i \neq 0; \qquad H_0 : \beta_1 = \cdots = \beta_q = 0 \ \text{ vs. } \ H_1 : \text{some } \beta_j \neq 0.$$

The total deviation decomposes as

$$y_{ij} - \bar{y}_{\cdot\cdot} = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}),$$

into the deviation due to A_i, that due to B_j, and the residual.

Proposition (Decomposition, two-way without replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SSE}$, where

$$\mathrm{SST} = \textstyle\sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 \ \ (pq-1), \quad \mathrm{SS}_A = q \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 \ \ (p-1),$$
$$\mathrm{SS}_B = p \textstyle\sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 \ \ (q-1), \quad \mathrm{SSE} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \ \ ((p-1)(q-1)),$$

with degrees of freedom in parentheses.

Proof. [Supplement] Write $a_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $b_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Then $\sum_i a_i = \sum_j b_j = 0$, and for each fixed i, $\sum_j r_{ij} = q\bar{y}_{i\cdot} - q\bar{y}_{i\cdot} - q\bar{y}_{\cdot\cdot} + q\bar{y}_{\cdot\cdot} = 0$; likewise $\sum_i r_{ij} = 0$ for each j.

Step 1. $\sum_i \sum_j a_i b_j = (\sum_i a_i)(\sum_j b_j) = 0$.

Step 2. $\sum_i \sum_j a_i r_{ij} = \sum_i a_i \sum_j r_{ij} = 0$ and $\sum_i \sum_j b_j r_{ij} = \sum_j b_j \sum_i r_{ij} = 0$.

Step 3. Squaring $y_{ij} - \bar{y}_{\cdot\cdot} = a_i + b_j + r_{ij}$ and summing, all cross terms vanish, leaving $q\sum a_i^2 + p\sum b_j^2 + \sum\sum r_{ij}^2$. The residuals satisfy $p+q-1$ independent linear constraints, so SSE has $pq - (p+q-1) = (p-1)(q-1)$ d.f.; and $(pq-1) = (p-1) + (q-1) + (p-1)(q-1)$. ∎

Mean squares and F-statistics:

$$\mathrm{MS}_A = \frac{\mathrm{SS}_A}{p-1}, \quad \mathrm{MS}_B = \frac{\mathrm{SS}_B}{q-1}, \quad \mathrm{MSE} = \frac{\mathrm{SSE}}{(p-1)(q-1)}, \qquad F_A = \frac{\mathrm{MS}_A}{\mathrm{MSE}}, \quad F_B = \frac{\mathrm{MS}_B}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way without replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \dfrac{q}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \dfrac{p}{q-1}\sum \beta_j^2$. Under the respective null hypotheses $F_A \sim F(p-1, (p-1)(q-1))$ and $F_B \sim F(q-1, (p-1)(q-1))$.

Proof. [Supplement]

Step 1. $\bar{Y}_{i\cdot} = \mu + \alpha_i + \bar{e}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and the $\bar{e}_{i\cdot}$ are independent $N(0, \sigma^2/q)$. As in the one-way proof, $E\sum_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 = (p-1)\sigma^2/q + \sum \alpha_i^2$, so $E(\mathrm{SS}_A) = (p-1)\sigma^2 + q\sum \alpha_i^2$. Symmetrically for B.

Step 2. $\mathrm{SST} = \sum\sum (Y_{ij} - \bar{Y}_{\cdot\cdot})^2$ has expectation $(pq-1)\sigma^2 + \sum\sum (\alpha_i + \beta_j)^2 = (pq-1)\sigma^2 + q\sum \alpha_i^2 + p\sum \beta_j^2$ (the cross term vanishes by the constraints). Subtracting, $E(\mathrm{SSE}) = (pq - 1 - (p-1) - (q-1))\sigma^2 = (p-1)(q-1)\sigma^2$.

Step 3. The F laws follow from Cochran's theorem: $\mathrm{SS}_A, \mathrm{SS}_B, \mathrm{SSE}$ are quadratic forms in the errors with ranks $p-1, q-1, (p-1)(q-1)$ adding to $pq-1$, so under the null hypotheses they are independent $\sigma^2\chi^2$ variables with these d.f. ∎

Definition (Tests of treatment effects (two-way, no replication))

(i) Factor A: $H_0 : \alpha_1 = \cdots = \alpha_p = 0$; $F_A = \mathrm{MS}_A/\mathrm{MSE}$; reject if $f_A \geq F_\alpha(p-1,(p-1)(q-1))$; $P_A = P(F_A \geq f_A)$.
(ii) Factor B: $H_0 : \beta_1 = \cdots = \beta_q = 0$; $F_B = \mathrm{MS}_B/\mathrm{MSE}$; reject if $f_B \geq F_\alpha(q-1,(p-1)(q-1))$; $P_B = P(F_B \geq f_B)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Residual	SSE	$(p-1)(q-1)$	MSE		
Total	SST	$pq-1$			

Example 9-3 (Diet and exercise)

Twenty men in their twenties of similar build were randomly assigned to one of 5 diets $(A_1,\ldots,A_5)$ combined with one of 4 exercises (B_1 running, B_2 weight training, B_3 swimming, B_4 yoga). Weight loss (kg) after 4 weeks:

	A_1	A_2	A_3	A_4	A_5	mean
B_1	7.9	7.2	5.1	5.8	6.8	6.56
B_2	7.5	6.6	4.8	5.6	6.5	6.20
B_3	6.9	6.4	4.4	5.4	6.4	5.90
B_4	6.5	6.2	4.1	4.5	5.8	5.42
mean	7.20	6.60	4.60	5.33	6.38	6.02

Test at level 5% whether diet and exercise affect weight loss.
Solution. $H_0 : \alpha_1 = \cdots = \alpha_5 = 0$ and $H_0 : \beta_1 = \cdots = \beta_4 = 0$.

$$\mathrm{SS}_A = 4\{(7.20-6.02)^2 + \cdots + (6.38-6.02)^2\} = 17.417,$$
$$\mathrm{SS}_B = 5\{(6.56-6.02)^2 + \cdots + (5.42-6.02)^2\} = 3.492,$$
$$\mathrm{SST} = (7.9-6.02)^2 + \cdots + (5.8-6.02)^2 = 21.232,$$
$$\mathrm{SSE} = \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B = 0.323,$$

with d.f. $4, 3, 19, 12$. Then $\mathrm{MS}_A = 4.354$, $\mathrm{MS}_B = 1.164$, $\mathrm{MSE} = 0.027$, $f_A = 161.77$, $f_B = 43.24$, both with $P < 0.000001$. Both the diet and the exercise type have significant effects at level 5%.

Source	SS	d.f.	MS	F	P
Factor A	17.417	4	4.354	161.77	< 0.000001
Factor B	3.492	3	1.164	43.24	< 0.000001
Residual	0.323	12	0.027		
Total	21.232	19			

9.4 Two-way analysis of variance with replication

Motivation. Replicating each factor combination improves reliability and, crucially, lets us separate an *interaction* — an effect of A that changes with the level of B — from pure error.

Here factors A and B have p and q levels and each combination is replicated r times. The design

must be completely randomized: all pqr runs in random order. (Randomizing only the order of the pq combinations and then doing the r runs of each consecutively is not a completely randomized design.)

The main advantage over the unreplicated design is that the effect arising from particular combinations of levels can be estimated separately. This is the **interaction**: the effect of A depends on the level of B. Without replication the model has $E(Y_{ij}) = \mu + \alpha_i + \beta_j$, so the difference between two levels of A is the same at every level of B, e.g.

$$(\mu + \alpha_1 + \beta_j) - (\mu + \alpha_3 + \beta_j) = \alpha_1 - \alpha_3 \quad \text{for all } j.$$

Plotting the mean response against the levels of B, one line per level of A, gives parallel line segments: no interaction (Figure 9, left). If the order of the A-levels changes with B, the lines cross and A and B interact (Figure 9, right); then the unreplicated analysis is inappropriate and replication is needed.

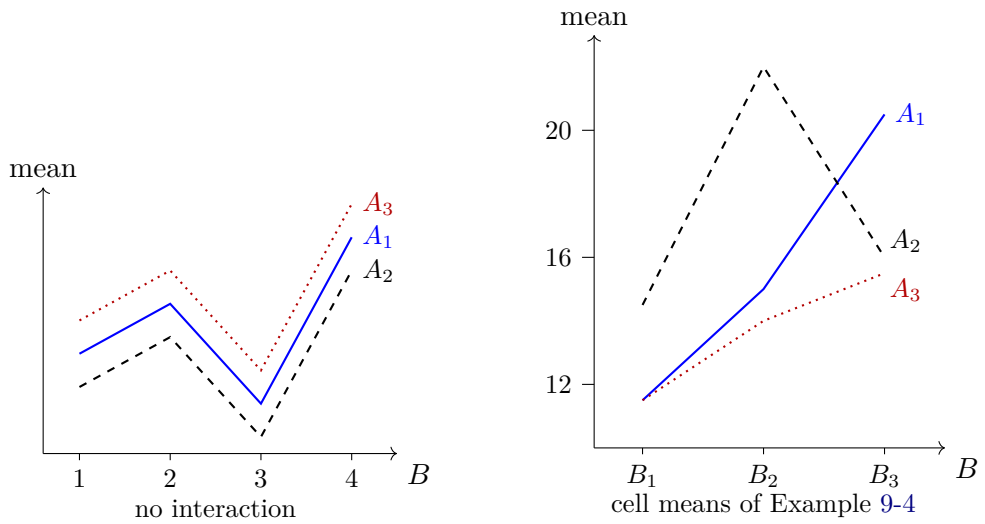


Figure 9: Interaction plots. Left: parallel profiles (no interaction). Right: crossing profiles, here the cell means $\bar{y}_{ij\cdot}$ of Example 9-4.

The data are y_{ijk} $(i = 1, \ldots, p;\ j = 1, \ldots, q;\ k = 1, \ldots, r)$ with means

$$\bar{y}_{ij\cdot} = \frac{1}{r}\sum_k y_{ijk}, \quad \bar{y}_{i\cdot\cdot} = \frac{1}{qr}\sum_j\sum_k y_{ijk}, \quad \bar{y}_{\cdot j\cdot} = \frac{1}{pr}\sum_i\sum_k y_{ijk}, \quad \bar{y}_{\cdots} = \frac{1}{pqr}\sum_i\sum_j\sum_k y_{ijk}.$$

Let γ_{ij} be the interaction effect of A_i and B_j.

Definition (Two-way ANOVA model (with replication))

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad e_{ijk} \sim N(0, \sigma^2) \text{ independent},$$

$$\sum_{i=1}^{p}\alpha_i = 0, \quad \sum_{j=1}^{q}\beta_j = 0, \quad \sum_{i=1}^{p}\gamma_{ij} = \sum_{j=1}^{q}\gamma_{ij} = 0,$$

$i = 1, \ldots, p,\ j = 1, \ldots, q,\ k = 1, \ldots, r.$

Tests of the main effects proceed as in §9.3; the new question is the interaction:

$$H_0 : \gamma_{ij} = 0 \ \ (\text{all } i, j) \qquad \text{vs.} \qquad H_1 : \text{at least one } \gamma_{ij} \neq 0.$$

The total deviation decomposes as

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}),$$

into deviations due to A_i, to B_j, to the interaction, and the residual.

Proposition (Decomposition, two-way with replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B} + \mathrm{SSE}$, where

$$\begin{aligned}
\mathrm{SST} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{\cdots})^2 && (pqr-1),\\
\mathrm{SS}_A &= qr \textstyle\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 && (p-1),\\
\mathrm{SS}_B &= pr \textstyle\sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 && (q-1),\\
\mathrm{SS}_{A\times B} &= r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 && ((p-1)(q-1)),\\
\mathrm{SSE} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2 && (pq(r-1)).
\end{aligned}$$

Proof. [Supplement]

Step 1. Split $y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot})$. This is a one-way layout with the pq cells as treatments, each replicated r times, so by the one-way decomposition

$$\mathrm{SST} = r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 + \mathrm{SSE}.$$

Step 2. The cell means $\bar{y}_{ij\cdot}$ form a $p \times q$ table whose row means are $\bar{y}_{i\cdot\cdot}$, column means $\bar{y}_{\cdot j\cdot}$ and grand mean $\bar{y}_{\cdots}$. Applying the two-way decomposition of §9.3 to this table and multiplying by r,

$$r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B}.$$

Step 3. Combining gives the identity. D.f.: $pq(r-1)$ for SSE ($r-1$ per cell), and $pq - 1 = (p-1) + (q-1) + (p-1)(q-1)$ for the cell-mean part; the total is $pqr - 1$. ■

Mean squares and the interaction test statistic:

$$\mathrm{MS}_{A\times B} = \frac{\mathrm{SS}_{A\times B}}{(p-1)(q-1)}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{pq(r-1)}, \qquad F = \frac{\mathrm{MS}_{A\times B}}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way with replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \dfrac{qr}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \dfrac{pr}{q-1}\sum \beta_j^2$, and $E(\mathrm{MS}_{A\times B}) = \sigma^2 + \dfrac{r}{(p-1)(q-1)}\sum\sum \gamma_{ij}^2$. Under $H_0 : \gamma_{ij} = 0$, $F = \mathrm{MS}_{A\times B}/\mathrm{MSE} \sim F((p-1)(q-1), pq(r-1))$.

Proof. [Supplement]

Step 1. Within each cell, $Y_{ij1}, \ldots, Y_{ijr}$ is a sample from $N(\mu_{ij}, \sigma^2)$ with $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, so $E(\mathrm{SSE}) = pq(r-1)\sigma^2$ and $\mathrm{SSE}/\sigma^2 \sim \chi^2(pq(r-1))$, independent of all cell means (normal sampling theory, cells independent).

Step 2. The cell means $\bar{Y}_{ij\cdot} = \mu_{ij} + \bar{e}_{ij\cdot}$ are independent with variance σ^2/r, i.e. a two-way table without replication with error variance σ^2/r and "means" $\mu+\alpha_i+\beta_j+\gamma_{ij}$. By the computation in §9.3 (now with the extra term γ_{ij}, whose row and column sums vanish), $E(\mathrm{SS}_A) = (p-1)\sigma^2 + qr\sum \alpha_i^2$, $E(\mathrm{SS}_B) = (q-1)\sigma^2 + pr\sum \beta_j^2$ and

$$E(\mathrm{SS}_{A\times B}) = (p-1)(q-1)\sigma^2 + r \textstyle\sum_i \sum_j \gamma_{ij}^2.$$

Step 3. $\mathrm{SS}_{A\times B}$ is a function of the cell means, hence independent of SSE; under H_0 it is $\sigma^2\chi^2((p-1)(q-1))$ (Cochran's theorem applied to the cell-mean table, as in §9.3). The F law follows; the main-effect F-tests are justified the same way. ■

Definition (Test of interaction (two-way with replication))

For $H_0 : \gamma_{ij} = 0$ $(i = 1, \dots, p,\ j = 1, \dots, q)$ use $F = \mathrm{MS}_{A\times B}/\mathrm{MSE}$; reject at level α if $f \geq F_\alpha((p-1)(q-1), pq(r-1))$; $P = P(F \geq f)$. The main effects are tested by $F_A = \mathrm{MS}_A/\mathrm{MSE}$ and $F_B = \mathrm{MS}_B/\mathrm{MSE}$ with denominator d.f. $pq(r-1)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Interaction	$\mathrm{SS}_{A\times B}$	$(p-1)(q-1)$	$\mathrm{MS}_{A\times B}$	$f_{A\times B} = \mathrm{MS}_{A\times B}/\mathrm{MSE}$	$P(F_{A\times B} \geq f_{A\times B})$
Residual	SSE	$pq(r-1)$	MSE		
Total	SST	$pqr-1$			

Example 9-4 (Machines and operators)

Three machines (A_1, A_2, A_3) and three operators (B_1, B_2, B_3) were compared by measuring productivity twice for each combination. Test the main effects and the interaction at level 5%.

	B_1	B_2	B_3	mean
A_1	9, 14	14, 16	19, 22	15.67
A_2	13, 16	18, 26	14, 18	17.5
A_3	11, 12	11, 17	15, 16	13.67
mean	12.5	17	17.33	15.61

Solution. Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$; $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$; $H_0 : \gamma_{ij} = 0$ $(i, j = 1, 2, 3)$.

$$\begin{aligned}
\mathrm{SS}_A &= 3 \times 2\{(15.67 - 15.61)^2 + \cdots + (13.67 - 15.61)^2\} = 44.11,\\
\mathrm{SS}_B &= 3 \times 2\{(12.5 - 15.61)^2 + \cdots + (17.33 - 15.61)^2\} = 87.44,\\
\mathrm{SS}_{A\times B} &= 2\{(11.5 - 15.67 - 12.5 + 15.61)^2 + \cdots\} = 74.22,\\
\mathrm{SST} &= (9 - 15.61)^2 + \cdots + (16 - 15.61)^2 = 288.28,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B - \mathrm{SS}_{A\times B} = 82.51,
\end{aligned}$$

with d.f. $2, 2, 4, 17, 9$. Hence $\mathrm{MS}_A = 22.06$, $\mathrm{MS}_B = 43.72$, $\mathrm{MS}_{A\times B} = 18.56$, $\mathrm{MSE} = 9.17$ and $f_A = 2.41$, $f_B = 4.77$, $f_{A\times B} = 2.02$, with P-values 0.146, 0.039, 0.174. Operators differ significantly in productivity at level 5%, while the machine type and the interaction are not significant.

Source	SS	d.f.	MS	F	P
Factor A	44.11	2	22.06	2.41	0.146
Factor B	87.44	2	43.72	4.77	0.039
Interaction	74.22	4	18.56	2.02	0.174
Residual	82.51	9	9.17		
Total	288.28	17			

Model checking and beyond

As in regression, the error assumptions should be checked before testing, using a residual plot of studentized residuals against the estimated mean responses. For Example 9-4 this plot

shows no problems, so the tests are meaningful. Once a factor is found significant, one would go on to estimate the level means or compare them; this lies beyond the scope of these notes.

Chapter 10. Nonparametric Inference

10.1 Introduction

Motivation. Classical t-based inference assumes normal data. When the data contain outliers or come from a heavy-tailed distribution, those procedures can lose much of their power, and methods built on signs and ranks are a safer alternative.

Most classical procedures assume the population is normal. If $X_1, \ldots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1),$$

and this pivot gives tests and confidence intervals for μ. Real populations are rarely exactly normal. The results still hold approximately because, by the central limit theorem, $\sqrt{n}(\overline{X} - \mu)/\sigma \to N(0,1)$ when n is large.

When the data contain outliers or the population is far from normal, normal-theory methods can fail badly. The actual confidence level can fall below the nominal one, the power of a test can drop sharply, and confidence intervals can become needlessly long.

Definition (Parametric, nonparametric and distribution-free methods)

(a) A **parametric method** assumes a specific parametric family (e.g. normal) for the population and makes inference about its parameters (μ, σ^2).
(b) A **nonparametric method** does not assume a parametric form for the distribution function. Its typical assumptions are only that the population distribution is *continuous*, or *continuous and symmetric*.
(c) A method that is valid whatever the population distribution is called **distribution-free**. For example, a test is distribution-free if the null distribution of its test statistic does not depend on the population distribution.

The two basic tools of nonparametric methods are the **sign** and the **rank**. The sign of an observation is 1 if it is positive and 0 if it is negative. The rank of an observation is its position when the observations are sorted in increasing order, counted from the smallest.

For example, the five observations

$$X_1 = 15, \quad X_2 = 18, \quad X_3 = 11, \quad X_4 = 36, \quad X_5 = 14$$

have ranks

$$R_1 = 3, \quad R_2 = 4, \quad R_3 = 1, \quad R_4 = 5, \quad R_5 = 2.$$

The largest value 36 gets rank 5 however large it is, so working with $R_1, \ldots, R_n$ instead of $X_1, \ldots, X_n$ limits the influence of outliers.

10.2 Wilcoxon rank-sum test

Motivation. Comparing two populations (or two treatments) is the *two-sample problem*. The Wilcoxon rank-sum test compares the locations of two distributions using only the ranks of the pooled sample.

Let the distribution functions of the two populations be $F(x)$ and $G(x) = F(x - \Delta)$. The two distributions have the same shape, and the Y distribution is the X distribution shifted by Δ. We call Δ the **shift parameter**. It can also be read as the difference of the means, $\Delta = \mu_Y - \mu_X$.

Definition (Data structure of the two-sample location problem)

$X_1, X_2, \ldots, X_m$: a random sample from a continuous distribution $F(x)$.
$Y_1, Y_2, \ldots, Y_n$: a random sample from the continuous distribution $F(x - \Delta)$.
The two samples are independent.

$\Delta = 0$ means that X and Y have the same distribution. $\Delta > 0$ means that the center of Y lies to the right of the center of X, and $\Delta < 0$ and $\Delta \neq 0$ are read in the same way.

Definition (Hypotheses on the shift parameter)

(i) $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$.
(ii) $H_0 : \Delta = 0$ versus $H_1 : \Delta < 0$.
(iii) $H_0 : \Delta = 0$ versus $H_1 : \Delta \neq 0$.

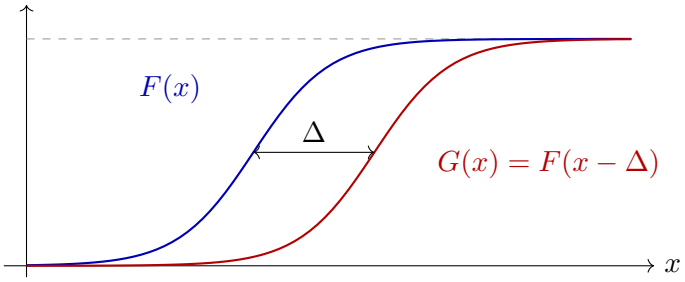


Figure 10.1: The distribution functions F and G under $H_1 : \Delta > 0$.

Pool the m observations X_i and the n observations Y_j into one sample of size $N = m + n$, the **combined sample**. Under $H_1 : \Delta > 0$ the Y's tend to be stochastically larger than the X's, so the Y observations tend to get larger ranks. A large sum of the Y-ranks in the combined sample is therefore evidence for H_1.

Definition 10-1 (Wilcoxon rank-sum statistic)

Let R_j be the rank of Y_j in the combined sample of the X's and Y's. The **Wilcoxon rank-sum statistic** is

$$W = \sum_{j=1}^{n} R_j.$$

10.2.1 Null distribution of W

To test with W we need its distribution under H_0. Take $m = 3$ and $n = 2$. Under $H_0 : \Delta = 0$ the X's and Y's have the same distribution, so $\{X_1, X_2, X_3, Y_1, Y_2\}$ is a random sample of size 5 from F. Assume there are no ties. Then all $\binom{5}{2} = 10$ arrangements of three x's and two y's in increasing order are equally likely.

No.	1	2	3	4	5	6	7	8	9	10
Arrangement	*yyxxx*	*yxyxx*	*yxxyx*	*yxxxy*	*xyyxx*	*xyxyx*	*xyxxy*	*xxyyx*	*xxyxy*	*xxxyy*
W	3	4	5	6	5	6	7	7	8	9

w	3	4	5	6	7	8	9	Total
$P(W = w)$	0.1	0.1	0.2	0.2	0.2	0.1	0.1	1.0

Tables 10-1 and 10-2: the ten arrangements (each with probability 0.1) and the null distribution of W for $m = 3$, $n = 2$.

The null distribution of W does not involve F, so the rank-sum test is distribution-free. For instance, with $m = 3$, $n = 2$ and $\alpha = 0.10$, the rejection region for $H_1 : \Delta > 0$ is $W \geq 9$, whatever F is.

The null distribution of W can be computed for every m and n. Tables of it (Appendix C of the book) or software give the tail probabilities. In the table the smaller sample is taken as the Y sample, and W_s denotes its rank sum. The table lists right-tail probabilities $P(W_s \geq x)$ and left-tail probabilities $P(W_s \leq x^*)$. For example, with smaller sample size $n = 3$ and larger sample size $m = 7$, $x = 25$ and $x^* = 8$ give

$$P(W_s \geq 25) = 0.033 = P(W_s \leq 8).$$

(The null distribution is symmetric about $n(N+1)/2 = 16.5$, so $x^* = n(N+1) - x$.)

Definition (Wilcoxon rank-sum test)

Take the larger sample as the X sample and the smaller one as the Y sample, and let

$$W = \text{sum of the ranks of the } Y \text{ sample in the combined sample.}$$

The rejection regions for $H_0 : \Delta = 0$ at level α are as follows.
(i) For $H_1 : \Delta > 0$: reject if $W \geq c$.
(ii) For $H_1 : \Delta < 0$: reject if $W \leq c^*$.
(iii) For $H_1 : \Delta \neq 0$: reject if $W \geq c_1$ or $W \leq c_1^*$.
The critical values satisfy

$$P(W \geq c) = \alpha, \qquad P(W \leq c^*) = \alpha, \qquad P(W \geq c_1) = \frac{\alpha}{2} = P(W \leq c_1^*).$$

Example 10-1 (Fine dust in two cities)

Cities A and B have similar population density and geography. Their monthly mean PM_{10} concentrations ($\mu g/m^3$) from March to November were recorded.

Month	3	4	5	6	7	8	9	10	11
City A	33	28	23	25	18	17	25.5	27	26
City B	22	18.5	19.5	20	16	24	22.5	21.5	21

At the 5% level, use the rank-sum test to decide whether the two cities differ.
Solution. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, with $X = A$ and $Y = B$ ($m = n = 9$). The ranks in the combined sample of 18 values are listed below.

X_i	33	28	23	25	18	17	25.5	27	26
rank	18	17	11	13	3	2	14	16	15
Y_j	22	18.5	19.5	20	16	24	22.5	21.5	21
rank	9	4	5	6	1	12	10	8	7

Hence $W_s = 9 + 4 + 5 + 6 + 1 + 12 + 10 + 8 + 7 = 62$. From the table with $m = n = 9$,

$$P(W_s \geq 109) = 0.02 = P(W_s \leq 62).$$

The two-sided p-value is $2 \times 0.02 = 0.04 < 0.05$. We reject H_0 and conclude that the PM_{10} levels of the two cities differ.

Example 10-2 (Tire wear)

Tires of company A cost about 20% more than those of company B. Eight tires of each company were driven 10,000 km and their wear rates (%) were measured. One A measurement was lost.

Company A	15	22	27	18	13	19	25	
Company B	20	16	17	26	28	24	30	33

Can we say that tires of A are better than those of B?
Solution. The sample from A is smaller, so A is the Y sample ($n = 7$) and B is the X sample ($m = 8$). Better quality means lower wear, so we test $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$.

X_i (B)	16	17	20	24	26	28	30	33
rank	3	4	7	9	11	13	14	15
Y_j (A)	13	15	18	19	22	25	27	
rank	1	2	5	6	8	10	12	

So $W_s = 44$, and the table with $m = 8$, $n = 7$ gives $P(W_s \le 44) = 0.095$. The p-value 9.5% is above 5% and below 10%. We cannot reject H_0 at the 5% level, but at the 10% level we conclude that tires of A are better.

10.2.2 Ties and the normal approximation

So far there were no ties. Even for a continuous population, real data are rounded, so ties occur. Tied observations get the average of the ranks they occupy (**midranks**), and W_s is computed with these. The tabulated null distribution then holds only approximately, so the test becomes an approximate test.

When m and n are too large for the tables, we use a normal approximation. We standardize W by its null mean and variance and refer the result to $N(0,1)$, which is a Z test.

Theorem 10-1 (Normal approximation of W)

Under $H_0 : \Delta = 0$,

$$E(W) = \frac{n(m+n+1)}{2}, \qquad \mathrm{Var}(W) = \frac{mn(m+n+1)}{12}.$$

When m and n are large,

$$Z_W = \frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Write $N = m + n$ and assume there are no ties.

Step 1. Under H_0 the N observations are i.i.d. from a continuous distribution. Every ordering of them is therefore equally likely, and the vector $(R_1, \dots, R_n)$ of Y-ranks is a random ordered selection of n values drawn without replacement from $\{1, \dots, N\}$. In particular each R_j is uniform on $\{1, \dots, N\}$, and each pair (R_j, R_k) with $j \neq k$ is uniform on the ordered pairs of distinct values.

Step 2. For a uniform R on $\{1, \dots, N\}$,

$$E(R) = \frac{1}{N}\sum_{r=1}^{N} r = \frac{N+1}{2},$$

$$\mathrm{Var}(R) = \frac{1}{N}\sum_{r=1}^{N} r^2 - \Big(\frac{N+1}{2}\Big)^2 = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4} = \frac{N^2-1}{12}.$$

Step 3. By exchangeability $c = \mathrm{Cov}(R_j, R_k)$ is the same for all $j \neq k$. If all N observations are ranked, the ranks sum to the constant $N(N+1)/2$, whose variance is 0:

$$0 = N\,\frac{N^2-1}{12} + N(N-1)\,c, \qquad \text{so} \qquad c = -\frac{N+1}{12}.$$

Step 4. Hence $E(W) = n(N+1)/2 = n(m+n+1)/2$ and

$$\begin{aligned}\mathrm{Var}(W) &= n\,\frac{N^2-1}{12} - n(n-1)\,\frac{N+1}{12} = \frac{n(N+1)}{12}\big[(N-1)-(n-1)\big] \\ &= \frac{n(N+1)(N-n)}{12} = \frac{mn(m+n+1)}{12}.\end{aligned}$$

Step 5. W is the total of a simple random sample of size n drawn without replacement from the finite population $\{1,\dots,N\}$. By the central limit theorem for sampling without replacement (Wald–Wolfowitz, Hájek), such a total is asymptotically normal when $\min(n,m) \to \infty$, because no single population value dominates the population variance. Therefore $Z_W \xrightarrow{d} N(0,1)$. ■

Example 10-3 (Normal approximation for Example 10-2)

Use the normal approximation of W for the data of Example 10-2.
Solution. There $W = 44$ with $m = 8$ and $n = 7$, so

$$E(W) = \frac{7(8+7+1)}{2} = 56, \qquad \mathrm{Var}(W) = \frac{8\times 7\times(8+7+1)}{12} = 74.67,$$
$$Z_W = \frac{44-56}{\sqrt{74.67}} = -1.39.$$

$P(Z \le -1.39) = 0.0823$, so the one-sided p-value is 8.23%. This is very close to the exact result of Example 10-2.

Example 10-4 (New surgical method)

A new surgical method A is compared with the standard method B. Each method was applied to 10 patients, and the recovery times (days) are given below.

A	8	9	11	12	16	17	19	19	21	45
B	13	16	18	18	20	20	21	23	25	72

To decide whether A is more effective than B, carry out
(a) the t test;
(b) the Wilcoxon rank-sum test;
(c) the normal-approximation rank-sum test.
Solution. Shorter recovery means a more effective method. With B as the Y sample, we test $H_0 : \Delta = 0$ against $H_1 : \Delta > 0$.
(a) For A, $\bar{x} = 17.7$ and $\sum(x_i - \bar{x})^2 = 1010.1$. For B, $\bar{y} = 24.6$ and $\sum(y_i - \bar{y})^2 = 2600.4$. Then

$$s_p^2 = \frac{1010.1 + 2600.4}{18} = 200.6, \qquad s_p = 14.16,$$
$$t = \frac{\bar{y} - \bar{x}}{s_p\sqrt{1/m + 1/n}} = \frac{24.6 - 17.7}{14.16\sqrt{1/10 + 1/10}} = 1.09.$$

Since $t_{0.1}(18) = 1.330 > 1.09$, the p-value exceeds 10%, and the t test cannot conclude that A is more effective.
(b) With midranks for the ties, the ranks in the combined sample are as follows.

X_i (A)	8	9	11	12	16	17	19	19	21	45
rank	1	2	3	4	6.5	8	11.5	11.5	15.5	19
Y_j (B)	13	16	18	18	20	20	21	23	25	72
rank	5	6.5	9.5	9.5	13.5	13.5	15.5	17	18	20

So $W_s = 128$. With $m = n = 10$ the table gives $P(W_s \geq 128) = 0.045$. At the 5% level, A is more effective than B.
(c) We have

$$E(W) = \frac{10 \times 21}{2} = 105, \qquad \mathrm{Var}(W) = \frac{10 \times 10 \times 21}{12} = 175,$$
$$Z_W = \frac{128 - 105}{\sqrt{175}} = 1.74,$$

and $P(Z \geq 1.74) = 0.0409$. The p-value is 4.09%, so again A is more effective than B.

Why the t test failed here

In Example 10-4 the rank-sum test gives $p \approx 4$–4.5%, but the t test gives $p > 10\%$. The cause is the two outliers 45 and 72, which inflate s_p and weaken the t test. If these were recording errors or came from exceptional patients and were removed, then

$$\bar{x} = 14.67, \quad \sum(x_i - \bar{x})^2 = 182.00, \qquad \bar{y} = 19.33, \quad \sum(y_i - \bar{y})^2 = 104.00,$$
$$s_p^2 = \frac{182.00 + 104.00}{16} = 17.88, \qquad s_p = 4.23, \qquad t = \frac{19.33 - 14.67}{4.23\sqrt{1/9 + 1/9}} = 4.957.$$

Since $t_{0.0005}(16) = 4.015$, the p-value is below 0.05%, and A is clearly better. With outliers or markedly non-normal data, the rank-sum test is often more efficient than the t test.

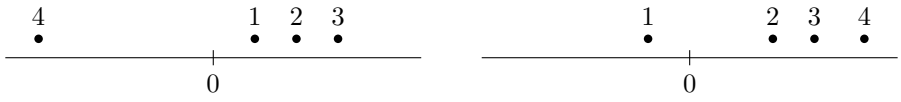
10.3 Wilcoxon signed-rank test

Motivation. For the center of a single population, the t test of Section 6.1 targets the mean. The signed-rank test targets the center of a symmetric distribution using only the signs and the ranks of the absolute values, and it also handles paired comparisons.

Definition (Data structure of the one-sample location problem)

$X_1, X_2, \ldots, X_n$ is a random sample from a distribution $F(x - \theta)$ that is continuous and symmetric about θ. The **location parameter** θ is the center of the distribution, which is both its mean (if it exists) and its median.

Consider two samples of $n = 4$ observations (Figure 10.2). Both have three positive observations. In (a) the negative observation lies far from 0, while in (b) it lies close to 0.



(a) negative value relatively far from 0 (b) negative value relatively close to 0

Figure 10.2: Two samples with $n = 4$; the numbers are the absolute ranks.

The **absolute rank** of an observation is its rank by distance from 0: regardless of sign, values closer to 0 get smaller ranks. To test whether the location is 0, we can use the sum of absolute ranks of the positive observations:

$$\text{(a) } T = 1 + 2 + 3 = 6, \qquad \text{(b) } T = 2 + 3 + 4 = 9.$$

The farther the data are shifted to the right of 0, the larger the ranks or the number of the positive observations, so this sum grows. In general, to test $H_0 : \theta = \theta_0$ we set

$$Z_i = X_i - \theta_0 \qquad (i = 1, \dots, n).$$

Under H_0, $Z_1, \dots, Z_n$ follow a distribution symmetric about 0. Let R_i^+ be the absolute rank of $|Z_i|$ among $|Z_1|, \dots, |Z_n|$. The **Wilcoxon signed-rank statistic** is

$$W^+ = \text{sum of the absolute ranks } R_i^+ \text{ of the positive } Z_i.$$

Under H_0 the Z_i are symmetric about 0, so positive and negative Z_i have the same distribution of absolute ranks (uniform), and each sign is $+$ or $-$ with probability $\frac{1}{2}$. Hence the null distribution of W^+ does not depend on F, and the signed-rank test is distribution-free.

The book's Appendix C tabulates both tails of W^+: right-tail probabilities $P(W^+ \geq x)$ and left-tail probabilities $P(W^+ \leq x^*)$. For $n = 4$, for instance, $P(W^+ \geq 9) = 0.125 = P(W^+ \leq 1)$. Indeed, among the $2^4 = 16$ equally likely sign patterns, only $\{2, 3, 4\}$ and $\{1, 2, 3, 4\}$ have rank sum at least 9.

Definition (Wilcoxon signed-rank test)

(i) Compute $Z_i = X_i - \theta_0$ for $i = 1, \dots, n$.
(ii) Find the absolute rank R_i^+ of $|Z_i|$ among $|Z_1|, \dots, |Z_n|$.
(iii) Compute $W^+ =$ the sum of R_i^+ over the positive Z_i.
The rejection regions for $H_0 : \theta = \theta_0$ are as follows.
For $H_1 : \theta > \theta_0$: reject if $W^+ \geq c$.
For $H_1 : \theta < \theta_0$: reject if $W^+ \leq c^*$.
For $H_1 : \theta \neq \theta_0$: reject if $W^+ \geq c_1$ or $W^+ \leq c_1^*$.
The critical values satisfy

$$P(W^+ \geq c) = \alpha, \quad P(W^+ \leq c^*) = \alpha, \quad P(W^+ \geq c_1) = \frac{\alpha}{2} = P(W^+ \leq c_1^*).$$

Example 10-5 (Smartwatch heart-rate error)

A smartwatch maker claims that the average error of its heart-rate measurement is 5 bpm, where the error is the absolute difference between the true and the measured rates. A consumer group suspects the error exceeds 5 bpm. It tested 10 randomly chosen watches on 10 people of similar build against a medical monitor, obtaining the following errors (bpm).

3.5	2.6	7.8	6.0	6.8	3.1	8.8	8.5	7.7	4.5

Is the consumer group's claim supported?

Solution. Test $H_0 : \theta = 5$ against $H_1 : \theta > 5$. With $\theta_0 = 5$, the values $Z_i = X_i - 5$ are

$$-1.5,\ -2.4,\ 2.8,\ 1.0,\ 1.8,\ -1.9,\ 3.8,\ 3.5,\ 2.7,\ -0.5.$$

negative Z_i	abs. rank	positive Z_i	abs. rank
-0.5	1	1.0	2
-1.5	3	1.8	4
-1.9	5	2.7	7
-2.4	6	2.8	8
		3.5	9
		3.8	10

So $W^+ = 2+4+7+8+9+10 = 40$. With $n = 10$ the table gives $P(W^+ \geq 40) = 0.116$. The p-value 11.6% is above 5%, so the evidence is insufficient to conclude that the error exceeds 5 bpm.

When values of the observations or of $|Z_i|$ are tied, we use average ranks. As with the rank-sum statistic, the null distribution of W^+ then holds only approximately. When n is too large for the table, we use the normal approximation of W^+.

Theorem 10-2 (Normal approximation of W^+)

Under $H_0 : \theta = \theta_0$,

$$E(W^+) = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}.$$

When n is large,

$$Z_{W^+} = \frac{W^+ - E(W^+)}{\sqrt{\mathrm{Var}(W^+)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Assume there are no ties, and let $Z = X - \theta_0$ have a continuous distribution symmetric about 0 under H_0.

Step 1. The sign of Z is independent of $|Z|$. For $a > 0$, symmetry gives

$$P(Z > 0,\ |Z| \leq a) = P(0 < Z \leq a) = \tfrac{1}{2} P(|Z| \leq a) = P(Z > 0)\, P(|Z| \leq a).$$

Step 2. For $k = 1, \ldots, n$ let $I_k = 1$ if the observation with absolute rank k is positive, and $I_k = 0$ otherwise. The absolute ranks depend only on $|Z_1|, \ldots, |Z_n|$, and by Step 1 the signs are i.i.d. fair coin flips independent of these. Hence $I_1, \ldots, I_n$ are i.i.d. Bernoulli($\frac{1}{2}$), and

$$W^+ = \sum_{k=1}^{n} k\, I_k.$$

Step 3. With $E(I_k) = \frac{1}{2}$ and $\mathrm{Var}(I_k) = \frac{1}{4}$,

$$E(W^+) = \frac{1}{2} \sum_{k=1}^{n} k = \frac{n(n+1)}{4},$$

$$\mathrm{Var}(W^+) = \frac{1}{4} \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}.$$

Step 4. The summands $Y_k = k(I_k - \frac{1}{2})$ are independent with mean 0 and $|Y_k| = k/2$. Write $s_n^2 = \mathrm{Var}(W^+)$. Then

$$\frac{\sum_{k=1}^{n} E|Y_k|^3}{s_n^3} = \frac{\frac{1}{8}\sum k^3}{s_n^3} = \frac{n^2(n+1)^2/32}{\big(n(n+1)(2n+1)/24\big)^{3/2}} = O(n^{-1/2}) \to 0.$$

This is Lyapunov's condition, so the Lyapunov CLT gives $Z_{W^+} \xrightarrow{d} N(0,1)$. ■

10.3.1 Paired comparison

Section 6.2 treated paired comparison for the difference of two means. If the experimental units are very heterogeneous, it is better to group them into homogeneous pairs than to split them at random into two groups. With n pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ we form the differences

$$D_i = Y_i - X_i \qquad (i = 1, \dots, n).$$

The hypotheses $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$ on the shift parameter become $H_0 : \theta = 0$ versus $H_1 : \theta > 0$ on the location of the D_i. Applying the signed-rank test to the D_i gives a nonparametric paired comparison.

Example 10-6 (Twins and a new curriculum)

To make statistics easier to learn, a new curriculum A was developed. Researchers recruited 16 pairs of identical twins with equal test scores. One twin of each pair took the old curriculum B and the other took A, and both then sat the same test.

A	70	85	73	75	65	80	71	80
B	65	43	45	80	84	71	52	42
$Z = A - B$	5	42	28	−5	−19	9	19	38
A	51	72	76	79	59	84	56	57
B	78	62	38	80	54	87	38	43
$Z = A - B$	−27	10	38	−1	5	−3	18	14

At the 5% level, is the new curriculum more effective?

Solution. Let θ be the location of Z, and test $H_0 : \theta = 0$ against $H_1 : \theta > 0$. Absolute ranks (with midranks for ties) are as follows.

negative Z_i	−1	−3	−5	−19	−27
abs. rank	1	2	4	10.5	12

positive Z_i	5	5	9	10	14	18	19	28	38	38	42
abs. rank	4	4	6	7	8	9	10.5	13	14.5	14.5	16

So $W^+ = 106.5$. With $n = 16$, Theorem 10-2 gives

$$E(W^+) = \frac{16 \times 17}{4} = 68, \qquad \mathrm{Var}(W^+) = \frac{16 \times 17 \times 33}{24} = 374,$$
$$Z_{W^+} = \frac{106.5 - 68}{\sqrt{374}} = 1.99.$$

$P(Z \ge 1.99) = 0.0233$, so the one-sided p-value 2.33% is below 5%. We reject H_0 and conclude that the new curriculum A is more effective than B.

Chapter 11. Time Series Analysis

11.1 Types of time series data

Motivation. Data recorded yearly, quarterly or monthly are ordered in time and are usually not independent. Before choosing a method we must recognise which components (trend, seasonality, randomness) a series contains.

A **time series** is a sequence of data observed in time order. It is used to find past patterns, forecast the future, or understand dynamic relationships. A **continuous time series** is observable at every instant and consists of values measured over continuous time. A **discrete time series** is observed only at particular times (yearly, monthly, daily, every second, ...). Most series are generated continuously but recorded at fixed intervals, so they appear as discrete series.

An important element of the analysis is the **time lag**, the interval between successive observation times. A time series is written

$$\{Z_t : t = 1, 2, 3, \dots\} \qquad \text{or} \qquad Z_1, Z_2, Z_3, \dots$$

The aims of time series analysis fall into two groups.

(i) **Forecasting**: assuming the past pattern persists, analyse the data so far and forecast the future.

(ii) **Understanding and control**: model the system (stochastic process) that generates the series in order to understand or control it.

The first step is always the **time series plot**, with time t on the horizontal axis and Z_t on the vertical axis. It shows the features of the series at a glance and guides the choice of model.

11.1.1 Components of a time series

The traditional approach regards a series as a sum of components and studies each one. A series has a **systematic component** and a **random** (**irregular**) **component**. The systematic component splits further into the **trend component**, which describes long-term change, and the **seasonal** (or cyclical) **component**, which describes repeated short-term change.

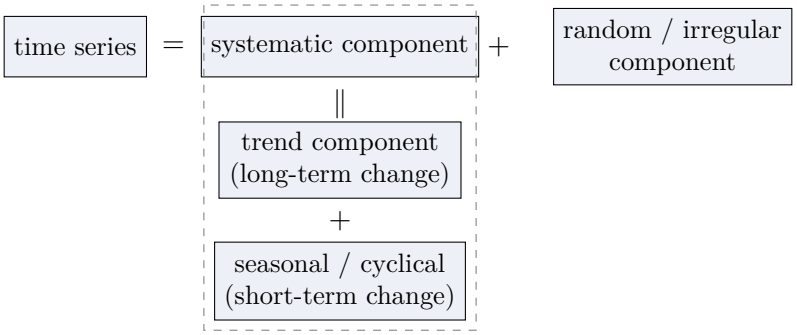


Figure 11.1: Components of a time series.

Definition (Components)

(a) The **random (irregular) component** is variation caused by random factors, unrelated to any regular movement over time. A series made only of a random component around a level is modelled by

$$Z_t = \beta_0 + \epsilon_t. \tag{11.1}$$

(b) The **trend component** is a persistent increase or decrease over time. It need not be linear and may be quadratic or S-shaped. A linear trend gives the **linear trend model**

$$Z_t = \beta_0 + \beta_1 t + \epsilon_t, \tag{11.2}$$

where $\beta_0 + \beta_1 t$ is the trend component and ϵ_t is the irregular component. Curved trends use

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t \quad \text{or} \quad Z_t = \beta_0 + \beta_1 t + \cdots + \beta_k t^k + \epsilon_t.$$

(c) The **seasonal component** is variation that repeats with the week, month or season. Examples are sales of summer goods, rainfall and temperature, which rise in summer and fall in winter.
(d) The **cyclical component** also repeats, but over a long period that is not tied to the seasons. In economics it is the **business cycle**: the Kondratiev cycle (50–60 years), the Juglar cycle (9–10 years) and the Kitchin cycle (about 40 months). A natural example is the roughly 11-year sunspot cycle.

A series whose systematic part is purely seasonal (no trend) can be described with trigonometric or indicator functions. For monthly data with a 12-month seasonal period,

$$Z_t = \beta_0 + \beta_1 \sin\left(\frac{2\pi t}{12}\right) + \beta_2 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.3}$$

The systematic part takes the same value at $t = 1$ and $t = 13$, so it has period 12 months. Many economic series (airline passengers, sales of ice cream or roasted sweet potatoes) have both a trend and seasonality:

$$Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.4}$$

Here $\beta_0 + \beta_1 t$ is the trend, and the sine and cosine terms form the 12-month seasonal component.

In some series the size of the fluctuations grows over time together with the trend, as in many economic indicators. Then one first takes logarithms and fits a model of type (11.4):

$$\log Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t.$$

The log transform stabilises the variability and simplifies the model. Forecasts are then on the log scale, so they must be converted back by the **exponential transformation** $\exp(x) = e^x$.

In real data the components are often not clearly separable. A series may also need two different trend lines in different periods, which indicates a **change point**. A main purpose of the time series plot is to reveal such features.

11.2 Methods of time series analysis

Motivation. No single model fits every series. This section surveys the regression-type trend models and the smoothing methods that the rest of the chapter develops.

A traditional approach is to regress the variable on time and analyse the trend. Expressing Z_t as a polynomial trend in t plus an irregular component gives

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_k t^k + \epsilon_t, \tag{11.5}$$

the **linear polynomial trend model** (or simply **polynomial trend model**). It is a special case of multiple regression,

$$Z_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \epsilon_t,$$

in which the explanatory variables are determined by time: $X_{1t} = t$, $X_{2t} = t^2, \ldots$. If the trend is a nonlinear function of t, we may use a **nonlinear trend model** such as

$$Z_t = \exp(\beta_0 + \beta_1 X_t)\, \epsilon_t.$$

The system generating a series may change over time, so a model that fits well in one period may fit poorly later. Rather than weighting all past observations equally, it is therefore reasonable to give recent observations more weight when forecasting.

If a series has large seasonal swings or much irregular variation, the trend is hard to see. **Smoothing methods** remove the seasonal and irregular components to reveal the overall trend and ease forecasting. The main ones are the **moving average method** and the **exponential smoothing method**.

(a) The moving average gives equal weight to the most recent observations only, not to all observations.

(b) Exponential smoothing gives more weight to recent observations and lets the weights decay exponentially into the past. It is widely used because it adapts flexibly to changes in the generating system and because it is easy to compute without storing much data.

Exponential smoothing is mainly used for forecasting. Moving averages are mainly used to visualise the trend after removing seasonality, which is **seasonal adjustment**. Other approaches include the **decomposition method**, which estimates each component separately and recombines them, and the ARMA/ARIMA models of Box and Jenkins (autoregressive (integrated) moving average). Their three-stage model-building procedure (identification, estimation, diagnosis) is the core of modern time series analysis. Transfer-function models and regression with autocorrelated errors are further options.

This chapter covers trend analysis, moving averages and exponential smoothing. These methods describe the trend in the data. They do not explain why the trend or seasonal variation occurs; they aim only to detect and confirm it. They are empirical, intuitive and convenient, which is why they are widely used.

11.3 Trend analysis

Motivation. When a series fluctuates around a level or follows a clear trend, ordinary least squares on time gives forecasts and prediction intervals. The constant mean model shows the whole workflow in its simplest form.

The choice of model depends on the shape of the series.

(a) A series that moves randomly around a fixed level calls for the constant mean model.

(b) A series with a steady upward or downward trend calls for a linear or a **quadratic trend model**.

(c) A series repeating the same pattern with a fixed period calls for a **seasonal trend model**. If it also has a steady linear trend, we use a **linear seasonal trend model**.

(d) Nonlinear trend models are used mainly for series growing geometrically or moving nonlinearly.

Definition (Three stages of model building)

(i) **Identification**: plot the series, study its features, and choose a tentative model.
(ii) **Estimation**: estimate the parameters by least squares or maximum likelihood.
(iii) **Diagnosis**: analyse the residuals to check the fit. If the fit is inadequate, modify the model or choose another.

The models of trend analysis are fitted exactly as in regression. If the data violate a required assumption (e.g. equal variance), we transform them (e.g. by logs), plot the transformed series again, and choose a tentative model for it.

11.3.1 Constant mean model

A series moving randomly around a fixed level is described by the **constant mean model**

$$Z_t = \beta_0 + \epsilon_t,$$

where the errors ϵ_t are independent with mean 0 and variance σ^2, and β_0 is the unknown constant mean. From the past data $Z_1, \dots, Z_n$ the least squares estimator of β_0 is

$$\hat{\beta}_0 = \frac{1}{n}\sum_{t=1}^{n} Z_t = \overline{Z}.$$

Let $Z_n(l) = E(Z_{n+l} \mid Z_n, Z_{n-1}, \dots)$ be the **l-step-ahead forecast** made at origin n. Under the constant mean model

$$Z_n(l) = \hat{\beta}_0 = \overline{Z} \qquad (l = 1, 2, \dots).$$

Definition (Forecast error, variance estimate and prediction interval)

The mean squared forecast error is

$$E\big[Z_{n+l} - Z_n(l)\big]^2 = \Big(1 + \frac{1}{n}\Big)\sigma^2,$$

and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(Z_t - \overline{Z})^2.$$

If the ϵ_t are normal, a $100(1-\alpha)\%$ **prediction interval** for Z_{n+l} is

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1 + \frac{1}{n}\Big)^{1/2},$$

where $t_{\alpha/2}(n-1)$ is the upper $\alpha/2$ point of the t distribution with $n-1$ degrees of freedom.

[Supplement] Why these hold: $Z_{n+l} - \overline{Z} = \epsilon_{n+l} - \bar{\epsilon}$, and ϵ_{n+l} is independent of $\bar{\epsilon} = \frac{1}{n}\sum_{t\le n}\epsilon_t$. So the error has mean 0 and variance $\sigma^2 + \sigma^2/n$. Under normality it is $N\big(0, (1+\frac{1}{n})\sigma^2\big)$ and independent of $\hat{\sigma}^2$, with $(n-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$. Hence $(Z_{n+l} - \overline{Z})/(\hat{\sigma}\sqrt{1+1/n}) \sim t(n-1)$, which gives the interval.

A prediction interval (L, U) for a future *random* value Z_{n+l} satisfies $P(L < Z_{n+l} < U) \ge 95\%$ when $\alpha = 0.05$. It is not a confidence interval for a parameter.

When a new observation Z_{n+1} arrives, the forecasts from origin $n+1$ are

$$\begin{aligned} Z_{n+1}(l) &= \frac{1}{n+1}(Z_1 + \cdots + Z_n + Z_{n+1}) = \frac{n}{n+1}Z_n(l) + \frac{1}{n+1}Z_{n+1} \\ &= Z_n(l) + \frac{1}{n+1}\big[Z_{n+1} - Z_n(l)\big]. \end{aligned}$$

The new forecast is a linear combination of the old forecast $Z_n(l)$ and the forecast error $e_n(l) = Z_{n+1} - Z_n(l)$. It needs only the latest forecast and the latest error, not all past data, so this updating rule saves computation and storage.

Example 11-1 (Statistics course enrolment)

The (hypothetical) numbers of students taking a statistics course each semester are given below.

Sem.	Students	Sem.	Students	Sem.	Students	Sem.	Students
2012/S1	1,372	2015/S1	1,423	2018/S1	1,420	2021/S1	1,435
2012/S2	1,388	2015/S2	1,337	2018/S2	1,406	2021/S2	1,376
2013/S1	1,478	2016/S1	1,366	2019/S1	1,372	2022/S1	1,347
2013/S2	1,404	2016/S2	1,378	2019/S2	1,489	2022/S2	1,389
2014/S1	1,406	2017/S1	1,461	2020/S1	1,425	2023/S1	1,349
2014/S2	1,486	2017/S2	1,418	2020/S2	1,302	2023/S2	1,364

Using the constant mean model, answer the following.
(a) Find the forecast and the 95% prediction interval for enrolment in the first semester of 2026.
(b) If 1,500 students enrol in the first semester of 2024, find the updated forecast.
Solution. Here $n = 24$, $\overline{Z} = 1399.625$ and $\hat{\sigma} = 47.83$. The first semester of 2026 is $l = 5$ steps ahead, and $Z_n(5) = \overline{Z} = 1399.625$. The forecast for 2024/S1 is also 1399.625.
(a) With $t_{0.025}(23) = 2.069$,

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1+\frac{1}{n}\Big)^{1/2} = 1399.625 \pm 2.069 \times 47.83 \times \Big(1+\frac{1}{24}\Big)^{1/2} = 1399.625 \pm 100.001.$$

The 95% prediction interval is (1298.624, 1500.626), so roughly 1,300 to 1,500 students are expected.
(b) The forecast error is $Z_{n+1} - Z_n(l) = 1500 - 1399.625 = 100.375$. Hence

$$Z_{n+1}(l) = Z_n(l) + \frac{1}{n+1}\big[Z_{n+1} - Z_n(l)\big] = 1399.625 + \frac{100.375}{25} = 1403.64.$$

11.3.2 Linear trend model

A series following $Z_t = \beta_0 + \beta_1 t + \epsilon_t$ is analysed by the regression methods of Chapter 8.

Illustration 11-1: women in senior professional and technical jobs

Consider a (hypothetical) monthly series, January 2018 to January 2024, of the number of women working in senior professional and technical occupations. Its plot shows a clear linear trend. Fitting the linear trend model gives $R^2 = 0.8373$ and the following ANOVA and coefficient tables.

Source	SS	df	MS	F	p-value
Model	2,170,954	1	2,170,954	365.48	$< 2.2 \times 10^{-16}$
Error	421,736	71	5,940		
Total	2,592,690	72			
Variable	**Coefficient**	**SE**	**T**		**p-value**
Intercept	−4,578	262.1	−17.46		$< 2 \times 10^{-16}$
Date	0.2689	0.01407	19.12		$< 2 \times 10^{-16}$

The coefficients are significant and R^2 exceeds 80%, so the trend model fits fairly well. The fitted model is

$$\widehat{\text{Female}} = -4578 + 0.269 \times \text{Date},$$

where Date is the number of days since 1 January 1970 (e.g. 1 January 2018 is day 17,532).

The residuals must still be checked. Their time plot shows runs of residuals with the same sign over stretches of time, so the residuals are positively correlated, not independent. This violates the regression assumption of independent, identically distributed errors. A regression model with autocorrelated errors would be more appropriate.

11.4 Moving averages

Motivation. Instead of averaging all observations, average only the most recent m. This smooths out local fluctuations, tracks a slowly changing level and, with m equal to the seasonal period, removes seasonality.

Consider the model $Z_t = \beta_0 + \epsilon_t$, where now β_0 may drift slowly over time. Locally the series fluctuates around a level, but the level changes. Instead of the least squares estimator $\hat{\beta}_0 = \overline{Z}$, which weights all observations equally, we use the mean of the most recent m observations:

$$M_n = \frac{1}{m} \sum_{t=n-m+1}^{n} Z_t = \frac{1}{m}(Z_{n-m+1} + Z_{n-m+2} + \cdots + Z_n). \tag{11.6}$$

M_n is the **simple moving average**, and it serves as the forecast of Z_{n+l} for every lead l:

$$Z_n(l) = M_n.$$

The choice of m depends on the purpose. A large m smooths strongly, so M_n reacts slowly to local change. A small m follows local change quickly. To smooth out seasonality, m is set equal to the seasonal period. The method has two drawbacks. The first m observations have no forecast, because the first forecast is for observation $m+1$. And all observations in the window get the same weight regardless of their age.

Moving averages are used less for forecasting than for seasonal adjustment: smoothing out the irregular and seasonal components to reveal the trend. For this, instead of (11.6) one uses the **centered moving average** with $m = 2k + 1$:

$$M_n = \frac{1}{2k+1}(Z_{n-k} + \cdots + Z_{n+k}). \tag{11.7}$$

Illustration 11-2: quarterly births in Korea

The national quarterly numbers of births from 2000/Q1 to 2024/Q1 (97 quarters, Statistics Korea) are, row by row:

178245	152324	151913	157607	161437	136920	131907	129670	140157
123005	119727	114022	133304	121051	117707	122974	132259	115630
115593	113476	118216	106313	108074	106104	119423	110753	111082
110501	126767	117315	128434	124306	127560	111416	115866	111050
120664	107525	110135	106525	120550	111369	117792	120460	128997
116100	116491	109677	128802	117652	123317	114779	119432	105518
109651	101854	116004	107103	110808	101520	117952	110126	108263
102079	112366	102337	102192	89348	98529	89532	89638	80072
89760	82040	80480	74542	83030	75395	73683	70568	73595
68263	68900	61579	70125	66129	66551	57757	68339	60149
64175	56523	64468	56142	56743	52618	60474		

Centered moving averages with $m = 3$ ($k = 1$) and $m = 11$ ($k = 5$) give the following (excerpt of Table 11-3; NA means not computable).

t	Year/Q	Actual	$m = 3$	$m = 11$
1	2000/1	178245	NA	NA
2	2000/2	152324	160827.3	NA
3	2000/3	151913	153948	NA
4	2000/4	157607	156985.7	NA
5	2001/1	161437	151988	NA
6	2001/2	136920	143421.3	143901.1
7	2001/3	131907	132832.3	138062.6
8	2001/4	129670	133911.3	136333.5
9	2002/1	140157	130944	133527.9
10	2002/2	123005	127629.7	129900.6
⋮	⋮	⋮	⋮	⋮
89	2022/1	68339	62081.67	62903.36
90	2022/2	60149	64221	62463.73
91	2022/3	64175	60282.33	60872.18
92	2022/4	56523	61722	60358.09
93	2023/1	64468	59044.33	NA
94	2023/2	56142	59117.67	NA
95	2023/3	56743	55167.67	NA
96	2023/4	52618	56611.67	NA
97	2024/1	60474	NA	NA

A larger m gives a smoother curve, and the overall trend is easier to see. With $m = 11$ the smoothed values change little and the trend stands out, while $m = 3$ still follows short-term ups and downs. The cost is data loss: $m = 3$ loses one value at each end, and $m = 11$ loses five at each end. The unsmoothed series is jagged because of the irregular component, and the smoothed series shows the trend clearly.

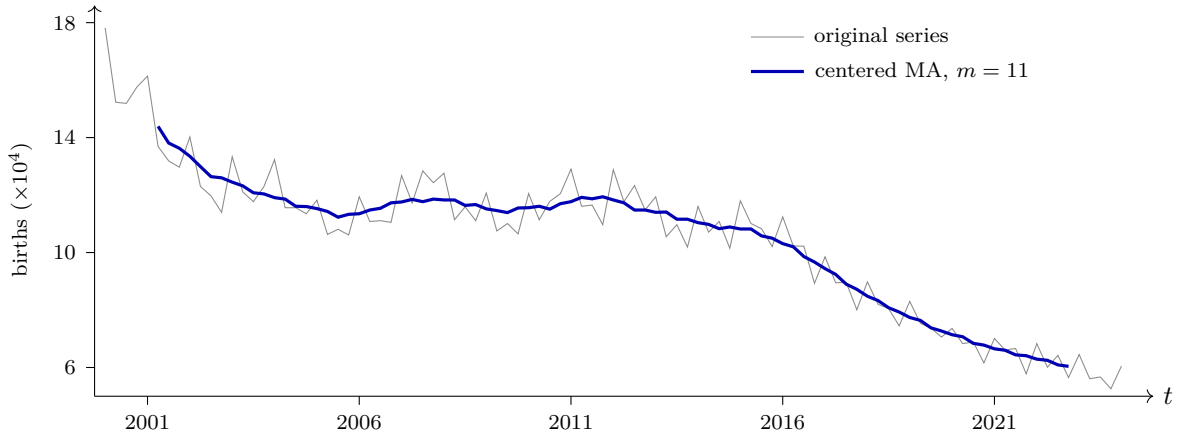


Figure 11.2: Quarterly births and their centered moving average ($m = 11$).

If the series has a linear trend as in (11.2), or a quadratic trend, double or triple moving averages are used instead.

11.5 Exponential smoothing

Motivation. A moving average ignores everything older than m periods and weights the last m equally. Exponential smoothing uses all observations, weights recent ones more, and updates the forecast using just two numbers.

Consider again the model (11.1), whose level may change over time. Given $Z_1, \ldots, Z_n$, exponential

smoothing forecasts Z_{n+1} by weights that decay geometrically into the past:

$$Z_n(1) = c\sum_{t=0}^{n-1}(1-w)^t Z_{n-t} = c\big[Z_n + (1-w)Z_{n-1} + \cdots + (1-w)^{n-1}Z_1\big]. \tag{11.8}$$

The **normalizing constant** c makes the weights sum to 1, i.e. $c\sum_{t=0}^{n-1}(1-w)^t = 1$. As n grows, $c \to w$. So for large n the one-step forecast is

$$Z_n(1) = \sum_{j\geq 0} w(1-w)^j Z_{n-j} = wZ_n + w(1-w)Z_{n-1} + w(1-w)^2 Z_{n-2} + \cdots \tag{11.9}$$

This is a sensible forecast for model (11.1), because it is unbiased:

$$E\big(Z_n(1)\big) = w\sum_{j\geq 0}(1-w)^j E(Z_{n-j}) = w\sum_{j\geq 0}(1-w)^j\beta_0 = \frac{w}{1-(1-w)}\,\beta_0 = \beta_0.$$

Definition (Simple exponential smoothing)

The weight w $(0 < w < 1)$ in (11.8)–(11.9) is the **smoothing constant**, usually between 0.05 and 0.3. The smoothed forecast is the **simple exponentially smoothed value**

$$S_n = Z_n(1),$$

and from origin n every lead has the same forecast: $Z_n(l) = S_n$ for $l = 1, 2, \ldots$. With S_{n-1} the smoothed value at time $n-1$,

$$S_n = wZ_n + (1-w)S_{n-1} = S_{n-1} + w(Z_n - S_{n-1}). \tag{11.10}$$

By (11.10) the new forecast S_n is a linear combination of the previous forecast S_{n-1} and the new observation Z_n. Unlike the moving average, only S_{n-1} and Z_n are needed to produce the updated forecasts $Z_n(l) = S_n$. Computing forecasts recursively by (11.10) is called **simple exponential smoothing**.

11.5.1 Choosing S_0 and w

To start (11.10) we need $S_1 = wZ_1 + (1-w)S_0$, hence an initial value S_0 and a constant w.

(a) *Initial value.* If there is information on data before the sample, it can be used to set S_0; otherwise S_0 must be chosen subjectively. For large n the weight $(1-w)^n$ on S_0 is tiny, so the initial value matters little.

(b) *Smoothing constant.* The choice of w matters a great deal. A small w makes forecasts react slowly to change. A large w makes them follow the latest observations closely. Usually w is treated as fixed over time and taken between 0.05 and 0.3.

A data-based choice uses the **1-step-ahead prediction error** at time $t-1$,

$$e_{t-1}(1) = Z_t - S_{t-1},$$

and picks the w that minimises

$$\mathrm{SSE}(w) = \sum_{t=1}^{n} e_{t-1}^2(1).$$

The choices are linked. If the level of the series changes rapidly, a w close to 1 makes the SSE small, and $S_0 = Z_1$ is appropriate. If the series changes slowly, w is likely to be close to 0, and $S_0 = \overline{Z}$ is a sensible start.

Example 11-2 (KOSPI index)

Monthly closing values of the KOSPI index from January 2018 to December 2024 are given below.

Yr	1	2	3	4	5	6	7	8	9	10	11	12
18	2566.46	2427.36	2445.85	2515.38	2423.01	2326.13	2295.26	2322.88	2343.07	2029.69	2096.86	2041.04
19	2204.85	2195.44	2140.67	2203.59	2041.74	2130.62	2024.55	1967.79	2063.05	2083.48	2087.96	2197.67
20	2119.01	1987.01	1754.64	1947.56	2029.6	2108.33	2249.37	2326.17	2327.89	2267.15	2591.34	2873.47
21	2976.21	3012.95	3061.42	3147.86	3203.92	3296.68	3202.32	3199.27	3068.82	2970.68	2839.01	2977.65
22	2663.34	2699.18	2757.65	2695.05	2685.9	2332.64	2451.5	2472.05	2155.49	2293.61	2472.53	2236.4
23	2425.08	2412.85	2476.86	2501.53	2577.12	2564.28	2632.58	2556.27	2465.07	2277.99	2535.29	2655.28
24	2497.09	2642.36	2746.63	2692.06	2636.52	2797.82	2770.69	2674.31	2593.27	2556.15	2455.91	2399.49

(a) With $S_0 = \overline{Z}$, compute the simple exponentially smoothed values $S_1, \ldots, S_{83}$ for $w = 0.3$ and for $w = 0.1$.
(b) Plot the series with the smoothed values $S_0, \ldots, S_{83}$ for $w = 0.1$ and $w = 0.3$, and describe the effect of w.
Solution.
(a) The series has length $T = 84$, so

$$S_0 = \overline{Z} = \frac{1}{T}\sum_{t=1}^{T} Z_t = 2478.221.$$

For $w = 0.3$, the one-step forecast at $n = 1$, using only Z_1, is

$$S_1 = S_0 + w(Z_1 - S_0) = 2478.221 + 0.3 \times (2566.46 - 2478.221) = 2504.693.$$

Likewise S_2 uses S_1 and Z_2:

$$S_2 = S_1 + w(Z_2 - S_1) = 2504.693 + 0.3 \times (2427.36 - 2504.693) = 2481.493.$$

S_2 uses the data up to time 2 and forecasts Z_3. Continuing gives all of $S_0, \ldots, S_{83}$. Selected values (Table 11-5 lists S_{t-1}, the forecast of Z_t, beside Z_t):

	S_0	S_1	S_2	S_3	$\cdots$	S_{83}
$w = 0.3$	2478.221	2504.693	2481.493	2470.8	$\cdots$	2580.833
$w = 0.1$	2478.221	2487.045	2481.076	2477.554	$\cdots$	2596.527

(b) As with moving averages, the smoothed series moves much more gently than the original (Figure 11.3). With $w = 0.3$ the smoothed series stays fairly close to the original. With $w = 0.1$ it is much smoother and lags further behind the turning points of 2020–2022. [Supplement] For these data the one-step SSE is about 2.75×10^6 for $w = 0.3$ against 6.41×10^6 for $w = 0.1$, so the SSE criterion prefers $w = 0.3$.

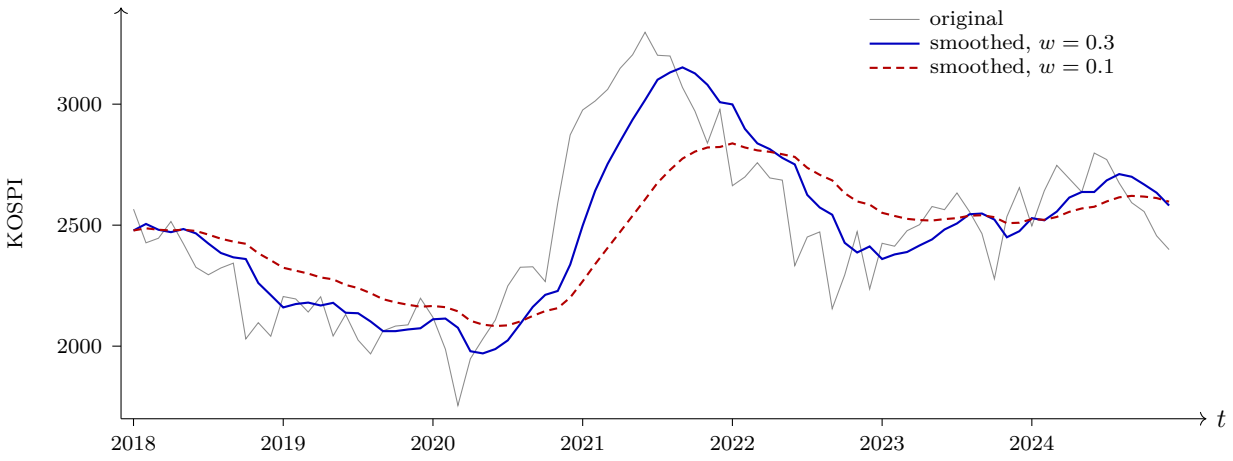


Figure 11.3: KOSPI and its forecasts S_{t-1} by simple exponential smoothing ($w = 0.3$, $w = 0.1$).

Chapter 12. Sample Surveys

12.1 Basic concepts of sample surveys

Motivation. A census is usually too slow, too costly, or even less accurate than a well-designed sample. To use a sample well we must define precisely which population is studied and which errors can arise.

12.1.1 What is a sample survey

To learn about a population we can survey all of it or only part of it and estimate the whole from that part.

Definition (Census and sample survey)

(a) Surveying the entire population is a **census** (complete enumeration).
(b) Drawing a sample from part of the population and inferring population characteristics from it is a **sample survey**.

In theory a census gives exact information. In practice it is often impossible, or it introduces more errors. Surveying the attitudes of every Korean teenager towards the opposite sex, for instance, would take years for one person. The results could be outdated before they appeared, and hiring tens of thousands of interviewers would be very costly. For opinion and attitude surveys a sample survey is the sensible choice.

Definition (Populations, frames and units)

(a) The **target population** is the population we actually want to study. It must be defined precisely. In a survey of "university students", for example, we must decide whether students on leave or graduate students are included.
(b) The **sampling frame** (or simply **frame**) is the list or rule that identifies the units of the target population available for sampling. It is the accessible subset from which the sample is actually drawn.
(c) The **sampled population** is the population actually covered by the frame. If the target is "all adults in Korea" but the frame is "adults listed in the telephone directory", unlisted adults are excluded, so the sampled and target populations differ.
(d) A **sampling unit** is an element of the population that can be selected. In a survey of students' attitudes it is each student; in a survey of farm household incomes it is each household.
(e) The **sampling design** specifies which units are selected and how. It is chosen to suit the parameters to be estimated and the structure of the population, so that estimation is as efficient as possible.

12.1.2 Errors and bias

Definition (Sampling and nonsampling error)

(a) **Sampling error** is the statistical error that arises because we infer about the whole population from part of it.
(b) **Nonsampling error** is every other error: the gap between target and sampled populations, interviewer mistakes, poorly worded questionnaires, and so on.

A sample survey contains both kinds of error. Sampling error is usually designed to shrink at the rate $1/\sqrt{n}$. Nonsampling error does not depend on n and can even be larger in a census. Hence the total error first decreases as n grows, but beyond some point it no longer decreases, because nonsampling error dominates.

The size of the sampling error can be worked out theoretically and controlled through n. Nonsampling error is hard to estimate or predict, so it must be reduced by care in planning, data collection and analysis. Its main sources are the following.

(i) **Faulty questions**: unclear or leading questions distort answers.

(ii) **Wrong answers and interviewer inexperience**: respondents may answer incorrectly or misunderstand the question (**response error**), or the interviewer may convey the purpose wrongly or be careless.

(iii) **Nonresponse error** (nonresponse bias): selected people do not respond. A low response rate makes the sample unrepresentative, especially when nonrespondents have particular tendencies.

(iv) **Coverage error**: some units are missing from (or wrongly included in) the frame, or the frame differs greatly from the target population.

(v) **Processing error**: errors in recording or in computer entry.

For an estimator $\hat{\theta}$ of θ, the difference $E(\hat{\theta}) - \theta$ is its bias. In survey practice the word **bias** is used more broadly: it also covers the systematic distortions caused by the nonsampling errors above. The most serious source of bias is a flawed sampling design.

Case study: the 2015 UK general election polls

The UK Parliament has an elected House of Commons and an appointed or hereditary House of Lords. Real power lies with the Commons, whose majority leader becomes prime minister. Since 1992 the Conservatives had never won a Commons majority. Before the 2015 election, polls taken over five weeks of campaigning predicted that neither party would win a majority and that Labour and the Conservatives would be roughly level. In fact the Conservatives won 331 seats, 91 more than Labour, a clear majority. The British Polling Council's inquiry identified two main causes.
(i) *Wrongly chosen sampled population.* The frames were unrepresentative. Labour supporters were over-represented and Conservative supporters under-represented. Respondents were mostly urban workers who answer surveys easily, while older Conservative voters were hard to reach.
(ii) *Neglect of nonresponse.* Respondents differed from nonrespondents. Conservative supporters likely had a lower response rate, so their real support was underestimated.

Similar errors are common. Examples include surveying national attitudes using only parents at schools in particular apartment complexes, or telephone polls that sample only landline users and exclude mobile users. Such frames produce serious bias, which shows how important the sampling design is.

12.2 Sampling methods

Motivation. Different designs trade off cost, convenience and precision. The basic designs (simple random, systematic, stratified, cluster and stratified cluster sampling) are the building blocks of real surveys.

12.2.1 Simple random sampling

Definition (Simple random sampling)

Drawing n units from a population of N units so that every unit has the same chance of selection, i.e. so that all $\binom{N}{n}$ possible samples are equally likely, is **simple random sampling** (SRS). In practice:
(i) number the units 1 to N;
(ii) select n of the numbers at random.

A random number generator is normally used. For a small population we can also draw lots from shuffled slips of paper.

Theorem (Sample mean under SRS without replacement)

Let a finite population have values $x_1, \dots, x_N$ with mean $\mu = \frac{1}{N}\sum_k x_k$ and variance $\sigma^2 = \frac{1}{N}\sum_k (x_k - \mu)^2$. For an SRS of size n without replacement with sample mean $\overline{X}$ and sample variance S^2,

$$E(\overline{X}) = \mu, \qquad \operatorname{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}, \qquad E(S^2) = \frac{N}{N-1}\sigma^2.$$

Consequently $\left(1 - \frac{n}{N}\right)\frac{S^2}{n}$ is an unbiased estimator of $\operatorname{Var}(\overline{X})$.

Proof. [Supplement]

Step 1. Let $\delta_k = 1$ if unit k is in the sample and $\delta_k = 0$ otherwise, so that $\overline{X} = \frac{1}{n}\sum_{k=1}^{N} x_k \delta_k$. By symmetry of SRS, for $k \neq l$,

$$P(\delta_k = 1) = \frac{n}{N}, \qquad P(\delta_k = \delta_l = 1) = \frac{n(n-1)}{N(N-1)}.$$

Hence $E(\overline{X}) = \frac{1}{n}\sum_k x_k \frac{n}{N} = \mu$.

Step 2. From Step 1,

$$\operatorname{Var}(\delta_k) = \frac{n}{N}\left(1 - \frac{n}{N}\right) = \frac{n(N-n)}{N^2}, \qquad \operatorname{Cov}(\delta_k, \delta_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3. Since $\sum_k \delta_k = n$ is constant, $\sum_k x_k\delta_k$ and $\sum_k d_k\delta_k$ (with $d_k = x_k - \mu$) have the same variance. Using $\sum_k d_k = 0$, so that $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2 = -N\sigma^2$,

$$\begin{aligned}\operatorname{Var}\Big(\sum_k d_k\delta_k\Big) &= \frac{n(N-n)}{N^2}\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big] = \frac{n(N-n)}{N^2}\,N\sigma^2\Big(1 + \frac{1}{N-1}\Big) \\ &= \frac{n(N-n)}{N-1}\,\sigma^2.\end{aligned}$$

Dividing by n^2 gives $\operatorname{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}$.

Step 4. Let $X_1, \dots, X_n$ be the sample values. Each X_i has mean μ and variance σ^2, and

$$E\sum_{i=1}^{n}(X_i - \overline{X})^2 = E\sum_{i=1}^{n}(X_i - \mu)^2 - n\,E(\overline{X} - \mu)^2 = n\sigma^2 - \frac{N-n}{N-1}\sigma^2 = \frac{N(n-1)}{N-1}\sigma^2.$$

So $E(S^2) = \frac{N}{N-1}\sigma^2$. Finally,

$$E\Big[\Big(1 - \frac{n}{N}\Big)\frac{S^2}{n}\Big] = \frac{N-n}{N} \cdot \frac{N}{N-1} \cdot \frac{\sigma^2}{n} = \mathrm{Var}(\overline{X}).$$

■

12.2.2 Systematic sampling

Definition (Systematic sampling)

In **systematic sampling** the units are listed in order and selected at a fixed interval. With population size N and sample size n, the **sampling interval** is $k = N/n$. One unit among the first k is chosen at random, and then every k-th unit after it is taken.

For example, to draw $n = 200$ from $N = 10000$ use $k = 10000/200 = 50$. If the random start is 23, the sample consists of units $23, 73, 123, 173, 223, 273, \dots, 9973$. The method is simpler than SRS and spreads the sample evenly over the population.

Exit polls, which ask voters leaving a polling station how they voted, use systematic sampling. The frame size N and the sample size n are hard to know in advance, so the interval k is fixed beforehand. The joint exit polls of the three Korean broadcasters for the 20th presidential election (2022) and the 22nd National Assembly election (2024) used $k = 5$.

Systematic sampling is convenient and is common in large surveys. If the list has a periodic pattern whose period matches k, however, it can produce serious bias.

12.2.3 Stratified random sampling

Definition (Stratified random sampling)

Divide the population into several groups, the **strata**, draw an SRS within each stratum, and use all of these together as the sample. This is **stratified random sampling** (or simply **stratified sampling**).

Strata are formed so that units are homogeneous within a stratum and heterogeneous across strata. Then every kind of unit appears in the sample, which raises the precision of the estimates. For this reason stratification is regarded as the theoretically most important sampling method.

Suppose, for example, that students' attitudes differ by college and by year. College and year are then natural stratification variables. If students of the same college and year think alike, a small sample from each stratum describes it accurately, and precision for the whole university improves greatly. Strata should therefore be chosen so that units within each are as homogeneous as possible.

12.2.4 Cluster sampling

Definition (Cluster sampling)

In **cluster sampling** the population is divided into subpopulations called **clusters**. Some clusters are chosen by SRS. Either every unit in a chosen cluster is surveyed (**single-stage cluster sampling**) or an SRS of units is drawn within each chosen cluster (**two-stage cluster sampling**).

Because whole clusters (or samples from them) are surveyed, cluster sampling works well when each cluster resembles the population, i.e. when clusters differ little from one another and each represents the whole. Its main advantage is lower cost. If students' attitudes depend on year but not on major, for example, we can take departments as clusters, choose a few departments at random, and survey all or some of their students. This is much cheaper than SRS or stratified sampling over the whole university.

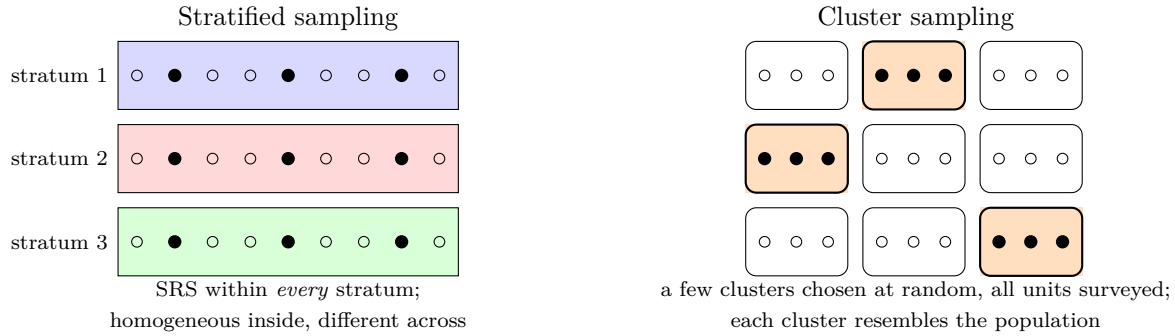


Figure 12.1: Stratified sampling versus (single-stage) cluster sampling; filled dots are sampled units.

12.2.5 Stratified cluster sampling

Definition (Stratified cluster sampling)

In **stratified cluster sampling** the population is first stratified into primary units. Within each stratum, clusters are drawn as primary sampling units, and the chosen clusters are surveyed completely or sampled further. Surveying the chosen clusters completely is **stratified single-stage cluster sampling**. Drawing secondary units from them by SRS is **stratified two-stage cluster sampling**.

Real surveys combine the designs above. Suppose students' attitudes differ considerably between colleges, but little between departments within a college. Take colleges as strata and departments as clusters (primary units). Choose departments at random within each college, and then draw students (secondary units) from the chosen departments by SRS. This is stratified two-stage cluster sampling.

12.3 Estimation under stratified random sampling

Motivation. Stratified sampling is the most important design. Here we derive its estimators of a population mean and a population proportion, with their standard errors and confidence intervals.

A population of N units is divided into h strata of sizes N_i ($\sum_i N_i = N$). The most common way to split the total sample size n among the strata is **proportional allocation**, where each stratum's

sample size is proportional to its size:

$$n_i = n\Big(\frac{N_i}{N}\Big) \qquad (i = 1, 2, \dots, h).$$

For means, let μ_i and σ_i^2 be the mean and variance of stratum i, with $\sigma_i^2 = \frac{1}{N_i}\sum(x - \mu_i)^2$ over the stratum. For proportions, let P_i be the proportion of units in stratum i with the attribute.

Definition 12-1 (Estimation of the population mean)

(i) Estimator:

$$\overline{X}_{\text{st}} = \frac{N_1}{N}\overline{X}_1 + \frac{N_2}{N}\overline{X}_2 + \cdots + \frac{N_h}{N}\overline{X}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \overline{X}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{\sigma_i^2}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\operatorname{SE}}(\overline{X}_{\text{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1 - \frac{n_i}{N_i}\Big)\frac{S_i^2}{n_i}}\,.$$

Here $\overline{X}_i$ and S_i^2 are the sample mean and sample variance of stratum i, and the subscript st stands for "stratified".

Justification. [Supplement] The samples from different strata are independent SRSs without replacement.

Step 1. By the theorem of Section 12.2.1, $E(\overline{X}_i) = \mu_i$. Hence $E(\overline{X}_{\text{st}}) = \frac{1}{N}\sum_i N_i\mu_i = \mu$, so $\overline{X}_{\text{st}}$ is unbiased.

Step 2. By independence across strata and the same theorem,

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_i N_i^2 \operatorname{Var}(\overline{X}_i) = \frac{1}{N^2}\sum_i N_i^2 \frac{N_i - n_i}{N_i - 1}\cdot\frac{\sigma_i^2}{n_i},$$

which is (ii).

Step 3. The same theorem shows that $(1 - n_i/N_i)S_i^2/n_i$ is unbiased for $\operatorname{Var}(\overline{X}_i)$. Summing these estimates with weights N_i^2/N^2 gives an unbiased estimator of $\operatorname{Var}(\overline{X}_{\text{st}})$, and its square root is (iii). ■

Definition 12-2 (Estimation of the population proportion)

(i) Estimator:

$$\hat{P}_{\text{st}} = \frac{N_1}{N}\hat{P}_1 + \frac{N_2}{N}\hat{P}_2 + \cdots + \frac{N_h}{N}\hat{P}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \hat{P}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\hat{P}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{P_i(1 - P_i)}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i}}\,.$$

Here $\hat{P}_i = X_i/n_i$ is the sample proportion of stratum i, where X_i is the number of sampled units in stratum i with the attribute.

Justification. [Supplement] Code each unit as 1 (has the attribute) or 0. Then $\mu_i = P_i$, $\sigma_i^2 = P_i(1-P_i)$ and $\overline{X}_i = \hat{P}_i$, so (i) and (ii) are the special cases of Definition 12-1.

For the sample variance of 0–1 data, $S_i^2 = \frac{n_i}{n_i-1}\hat{P}_i(1-\hat{P}_i)$. The exactly unbiased variance estimate is therefore

$$\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i-1}.$$

Formula (iii) with n_i in the denominator differs from it by the factor $(n_i-1)/n_i \approx 1$. Example 12-2 uses the exact version with $n_i - 1$. ■

Finite population correction and confidence intervals

In part (ii) of Definitions 12-1 and 12-2, the factor $\dfrac{N_i-n_i}{N_i-1}$ appears because the population is finite. It is the **finite population correction factor**. In the estimated standard error it is replaced by $\dfrac{N_i-n_i}{N_i} = 1-\dfrac{n_i}{N_i}$, which makes the variance estimate unbiased (see the theorem of Section 12.2.1). Part (iii) of each definition replaces σ_i^2 and P_i by their estimates S_i^2 and $\hat{P}_i$.

The 95% confidence intervals for the mean and the proportion are

$$\bar{x}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \bar{x}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{s_i^2}{n_i}},$$

$$\hat{p}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}}) = \hat{p}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{p}_i(1-\hat{p}_i)}{n_i}}.$$

The half-widths $z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}})$ and $z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}})$ are called the **margin of error**, also known as the error limit or allowable error.

Example 12-1 (Household living expenses)

To survey monthly living expenses per household in a small city, the city was divided by standard of living into 3 regions (strata), and a stratified sample gave the following data (unit: 100,000 won).

Stratum	N_i	n_i	$\bar{x}_i$	s_i^2
1	2,800	30	18.5	12.2
2	5,600	60	15.2	11.3
3	3,400	30	10.7	10.5
Total	$N = 11{,}800$	$n = 120$		

Estimate the mean monthly living expenses per household and give a 95% confidence interval.

Solution. The estimate is

$$\bar{x}_{\text{st}} = \frac{1}{11800}(2800 \times 18.5 + 5600 \times 15.2 + 3400 \times 10.7) = 14.69.$$

The estimated variance is

$$\widehat{\text{Var}}(\overline{X}_{\text{st}}) = \frac{1}{11800^2}\Big\{2800^2\Big(1 - \frac{30}{2800}\Big)\Big(\frac{12.2}{30}\Big) + 5600^2\Big(1 - \frac{60}{5600}\Big)\Big(\frac{11.3}{60}\Big) + 3400^2\Big(1 - \frac{30}{3400}\Big)\Big(\frac{10.5}{30}\Big)\Big\} = 0.09342.$$

So $\widehat{\text{SE}}(\overline{X}_{\text{st}}) = \sqrt{0.09342} = 0.3056$, and

$$\bar{x}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\overline{X}_{\text{st}}) = 14.69 \pm 1.96 \times 0.3056 = 14.69 \pm 0.60.$$

The 95% confidence interval is $(14.09,\ 15.29)$, i.e. about 1.41 to 1.53 million won.

Example 12-2 (Support for a university policy)

A university grouped similar colleges into 3 strata A, B, C and surveyed students on leave about a university policy by stratified sampling.

College (stratum)	N_i	n_i	$\hat{p}_i$
A	3,500	35	0.56
B	4,500	45	0.60
C	2,000	20	0.40

Estimate the support rate and give a 95% confidence interval.
Solution. The estimate is

$$\hat{p}_{\text{st}} = \frac{1}{10000}(3500 \times 0.56 + 4500 \times 0.60 + 2000 \times 0.40) = 0.546.$$

Using $n_i - 1$ in the denominators,

$$\widehat{\text{Var}}(\hat{P}_{\text{st}}) = \frac{1}{10000^2}\Big\{3500^2\Big(1 - \frac{35}{3500}\Big)\Big(\frac{0.56 \times 0.44}{35 - 1}\Big) + 4500^2\Big(1 - \frac{45}{4500}\Big)\Big(\frac{0.60 \times 0.40}{45 - 1}\Big) + 2000^2\Big(1 - \frac{20}{2000}\Big)\Big(\frac{0.40 \times 0.60}{20 - 1}\Big)\Big\} = 0.00247.$$

So $\widehat{\text{SE}}(\hat{P}_{\text{st}}) = \sqrt{0.00247} = 0.0497$, and

$$\hat{p}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}}) = 0.546 \pm 1.96 \times 0.0497 = 0.546 \pm 0.097.$$

The 95% confidence interval is $(0.449,\ 0.643)$, i.e. 44.9% to 64.3%. (The book centres the interval at 0.564 by a transposition slip and reports $(0.467,\ 0.661)$; the margin of error 0.097 is the same.)

12.4 Problems in practical surveys

Motivation. A published "margin of error" covers only sampling error. Misunderstanding randomness and writing poor questionnaires create nonsampling errors that no formula reports.

Media reports of opinion polls often contain statements like this one. *The survey used automated wireless calls to virtual (safe) mobile numbers, with a sample of 1,003, a response rate of* 2.3%,

and a sampling error of ±3.1 *percentage points at the* 95% *confidence level.* The "sampling error" 3.1% is the margin of error of Section 12.3. It describes only the error from surveying part of the population instead of all of it, i.e. the size of the sampling error. Nonsampling error is not included. Professional survey firms design their samples on sound theory and experience, so their nonsampling errors are comparatively small, and the stated margin of error is then a good indicator of reliability. When non-experts run surveys, however, nonsampling error can grow sharply. Two typical sources are misunderstanding or ignoring randomness, and poorly worded questions.

12.4.1 The concept of randomness

Drawing a sample at random does not mean drawing it any old way. In random sampling, units are selected with probabilities fixed in advance. As noted in Section 12.1.2, if the sample is not drawn from the whole population according to statistical theory, bias and nonsampling error can become severe.

Suppose a graduate student studies "generational differences in attitudes to the division of household chores and gender equality". A sample drawn with all Korean households as the sampled population would be beyond one person's time and budget, so shortcuts get used. For instance, questionnaires might be handed to high-school students in one apartment complex and filled in by their parents. The sampled population is then restricted to families in that complex with a high-school child. Childless families and the elderly are almost excluded, so a "generational" analysis becomes difficult. Moreover, households in one complex share social and economic characteristics, so the sampled population differs greatly from the target population.

The 2015 UK polls of Section 12.1 failed in the same way. Such errors come from reading "random" as "haphazard", or from choosing convenient methods to save cost or effort. Random sampling means fixing each unit's selection probability in advance and then selecting by chance according to those probabilities. Ignoring this can make nonsampling error and bias severe.

12.4.2 Writing the questionnaire

Questions must neither lead respondents to a particular answer nor confuse them with vague wording.

(a) *Leading questions.* "Do you think ancestral rites should be kept in the traditional way, to honour the ancestors and teach tradition to our children?" and "Do you think ancestral rites should be simplified to fit social development and changing values?" both invite a "yes". If asked separately, the proportion favouring keeping rites and the proportion favouring simplifying them may well sum to more than 100%.

(b) *Ambiguous questions.* "Are you satisfied with your current life?" intends to measure life satisfaction, but "current life" is undefined, so respondents may be confused. To measure satisfaction properly, ask about several aspects (work, marriage, family, leisure, friendships) and combine them. "Do you think you have many friends?" depends on how "many" is read: one person regards 3 friends as many, another regards 10 as few.

(c) *Campaigning disguised as a poll.* Around elections, questions such as "Do you know that candidate X devoted his life to the democracy movement?" or "Do you know that candidate X is an economic expert and former minister?" clearly aim to promote a candidate. Such use of a poll abuses statistical surveys.

(d) *Pretesting.* Problems are best found and fixed in a **pilot survey**. Consider "What is your monthly leisure budget? (1) under 100,000 won (2) 100–500 thousand (3) 500 thousand–1 million (4) over 1 million won". Given to university students, most will choose (1). The answers then show only that student leisure spending is low and say nothing useful about its distribution. A pilot survey reveals this easily, and the categories can be revised.

(e) *Length.* With too many questions, respondents tire and answer carelessly. Limit the number of items, and offering fair compensation for the respondent's time also helps.

Every item must be reviewed carefully. However sophisticated the sampling design, a badly written questionnaire yields distorted information.